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Lee et al.

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(54) **LAUNDRY TREATING APPARATUS**

(71) Applicant: **LG ELECTRONICS INC.**, Seoul (KR)

(72) Inventors: **Yunhi Lee**, Seoul (KR); **Jaehyun Kim**, Seoul (KR); **Hyunsun Yoo**, Seoul (KR)

(73) Assignee: **LG ELECTRONICS INC.**, Seoul (KR)

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See application file for complete search history.

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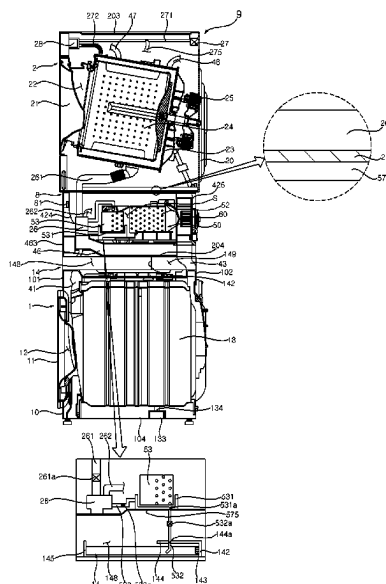
Primary Examiner — Joseph L. Perrin

(74) *Attorney, Agent, or Firm* — Birch, Stewart, Kolasch & Birch, LLP

(57) **ABSTRACT**

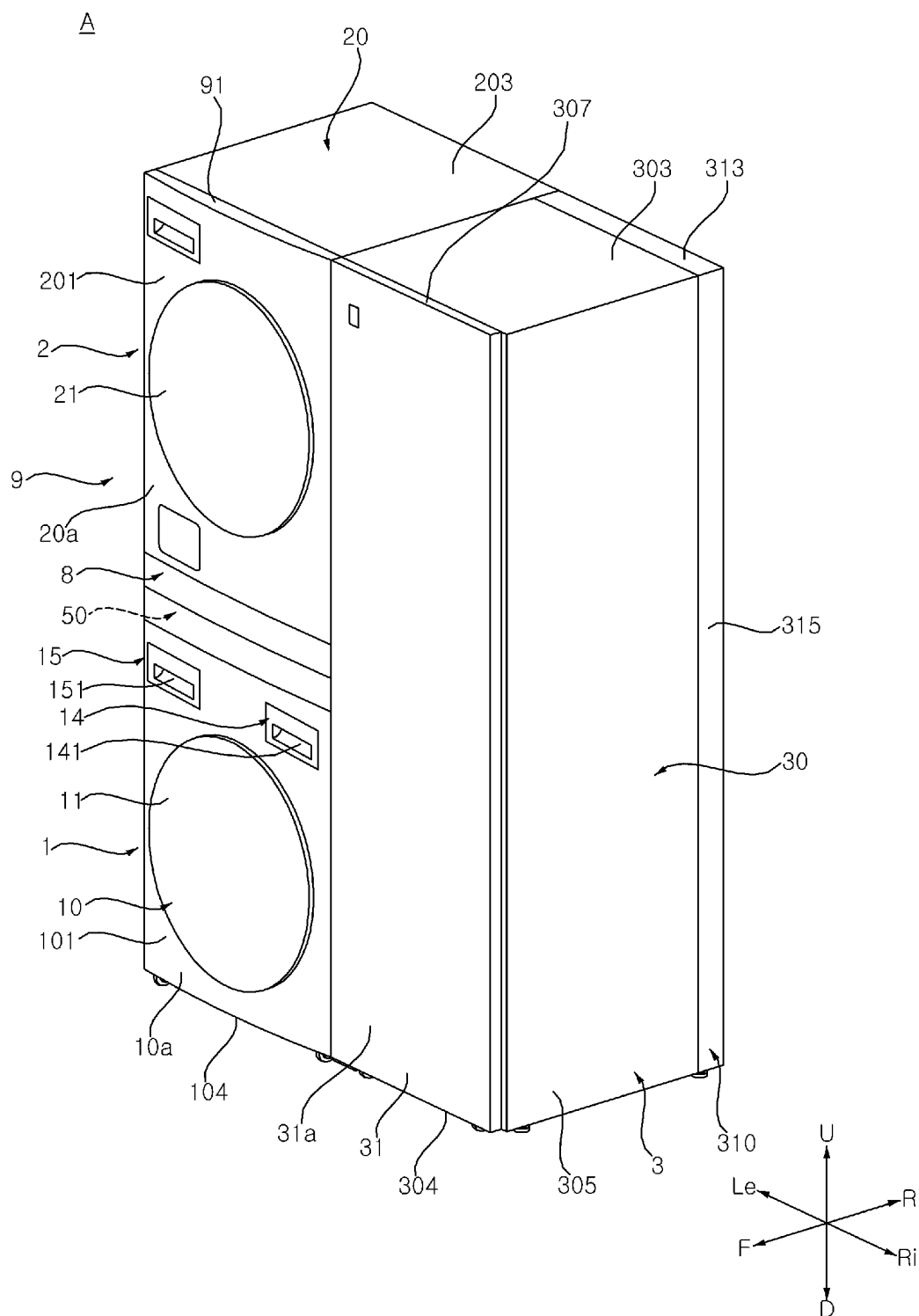
A laundry treating apparatus including a dryer including a first cabinet and a first drum rotatably mounted in the first cabinet; a washing machine including a second cabinet, a tub disposed in the second cabinet, and a second drum rotatably mounted in the tub; a refresher including a third cabinet disposed on one side of the first cabinet and the second cabinet, and having an inner space for receiving laundry; a heating device disposed under the tub and including a heat exchange unit for heating air supplied into the first drum and the inner space of the third cabinet; a drain pan which is disposed under the heat exchange unit and for storing condensate; and a drain pump for discharging water in the tub and the condensate in the drain pan.

14 Claims, 20 Drawing Sheets

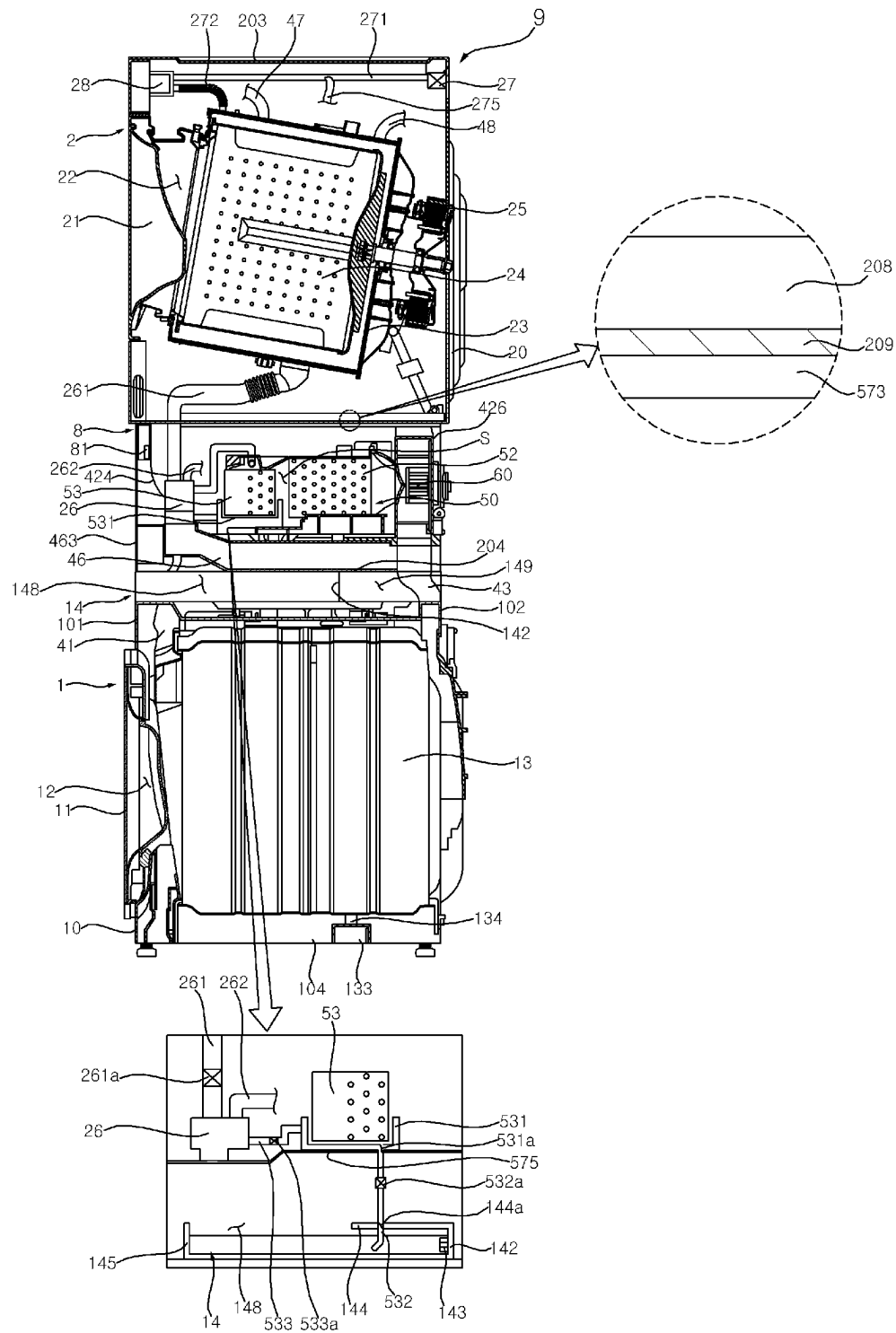


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D06F 73/02 (2006.01)
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D06F 103/32 (2020.01)
D06F 103/50 (2020.01)
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D06F 105/26 (2020.01)
- (52) **U.S. Cl.**
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2103/32 (2020.02); **D06F 2103/50** (2020.02);
D06F 2105/20 (2020.02); **D06F 2105/26**
(2020.02)
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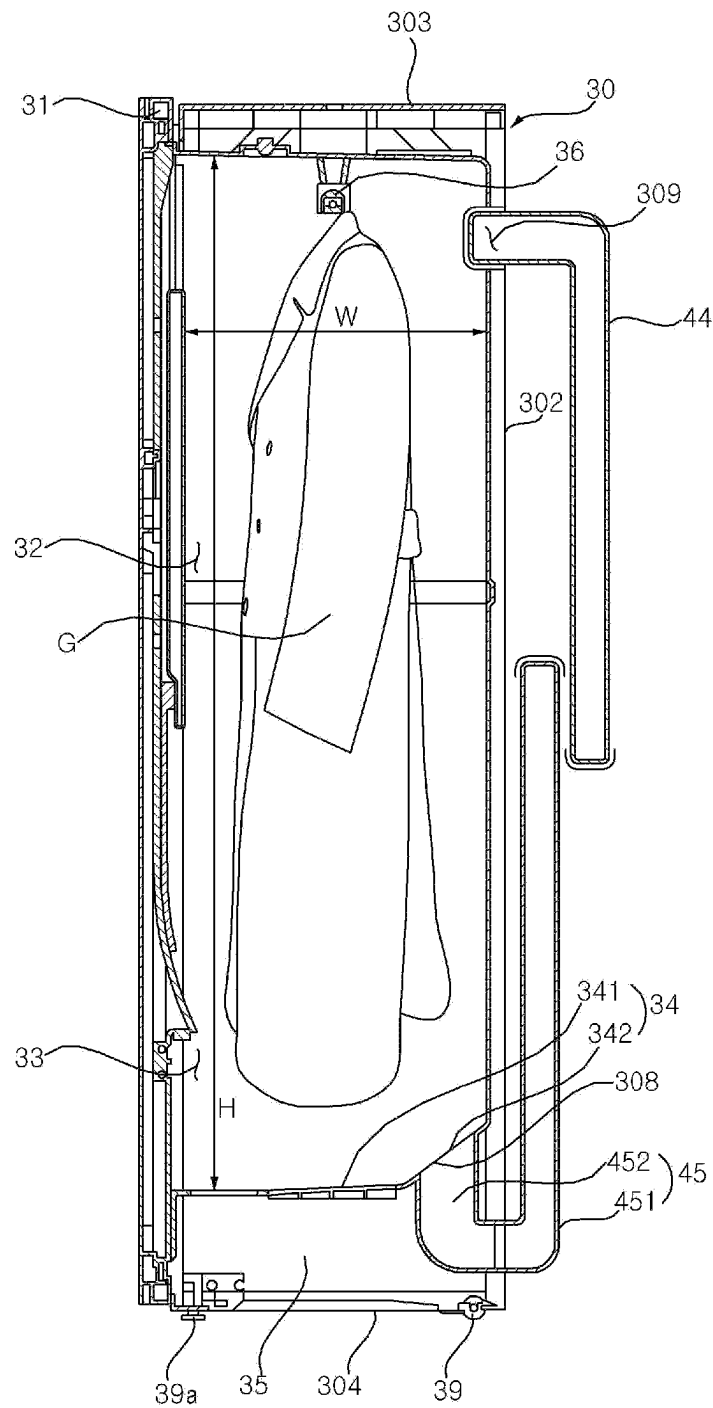
【FIG. 1】



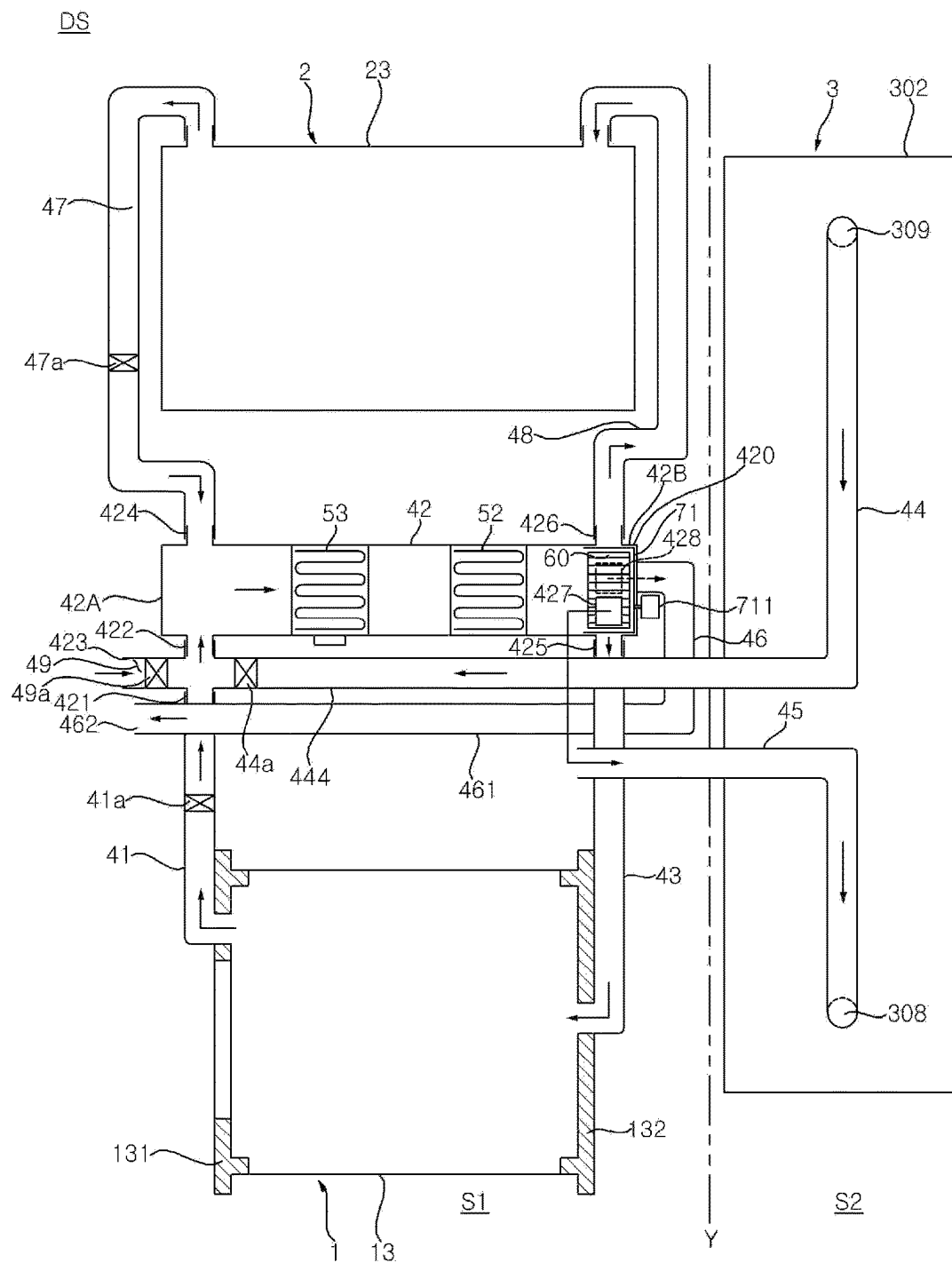
【FIG. 2】



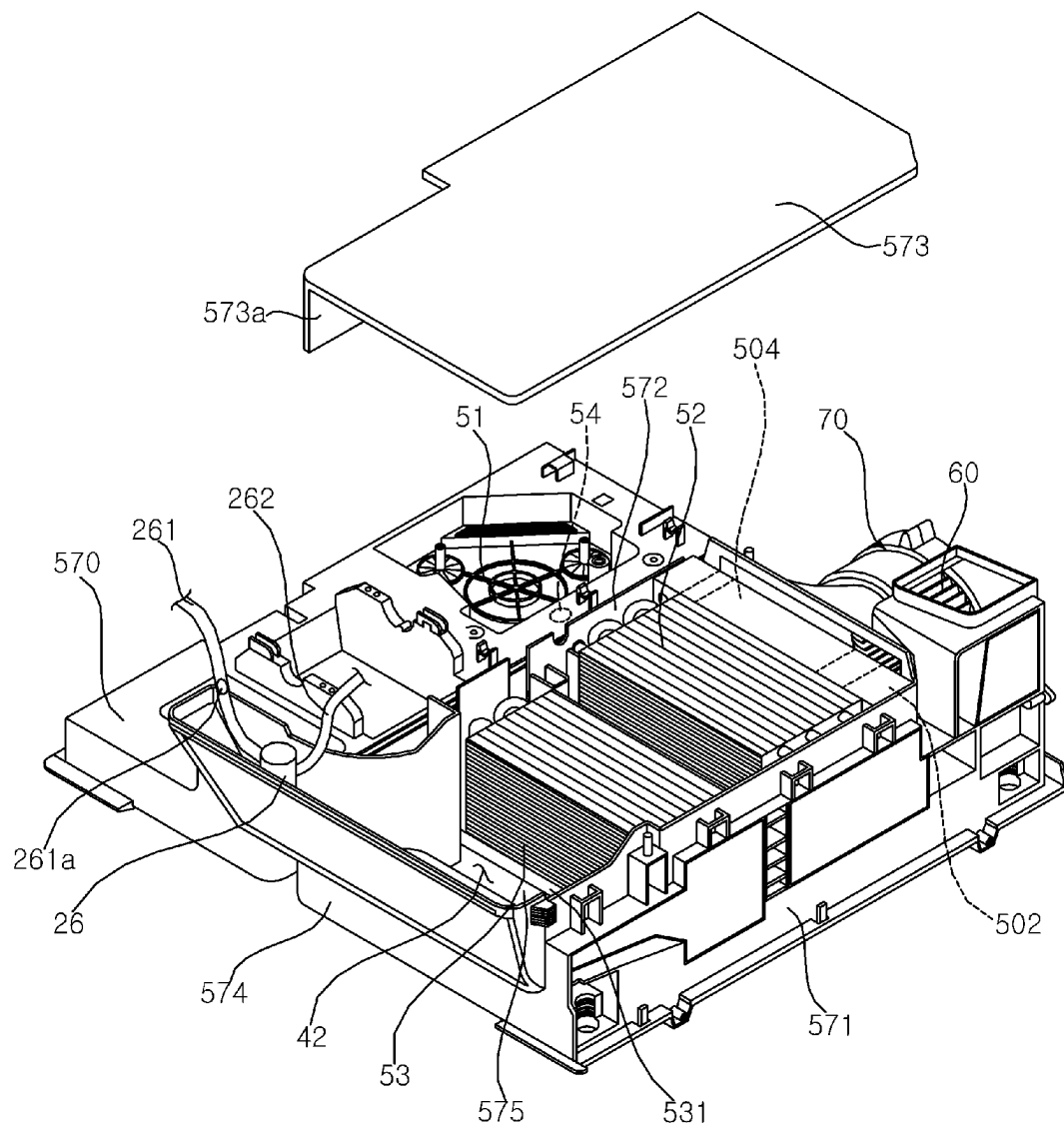
【FIG. 3】



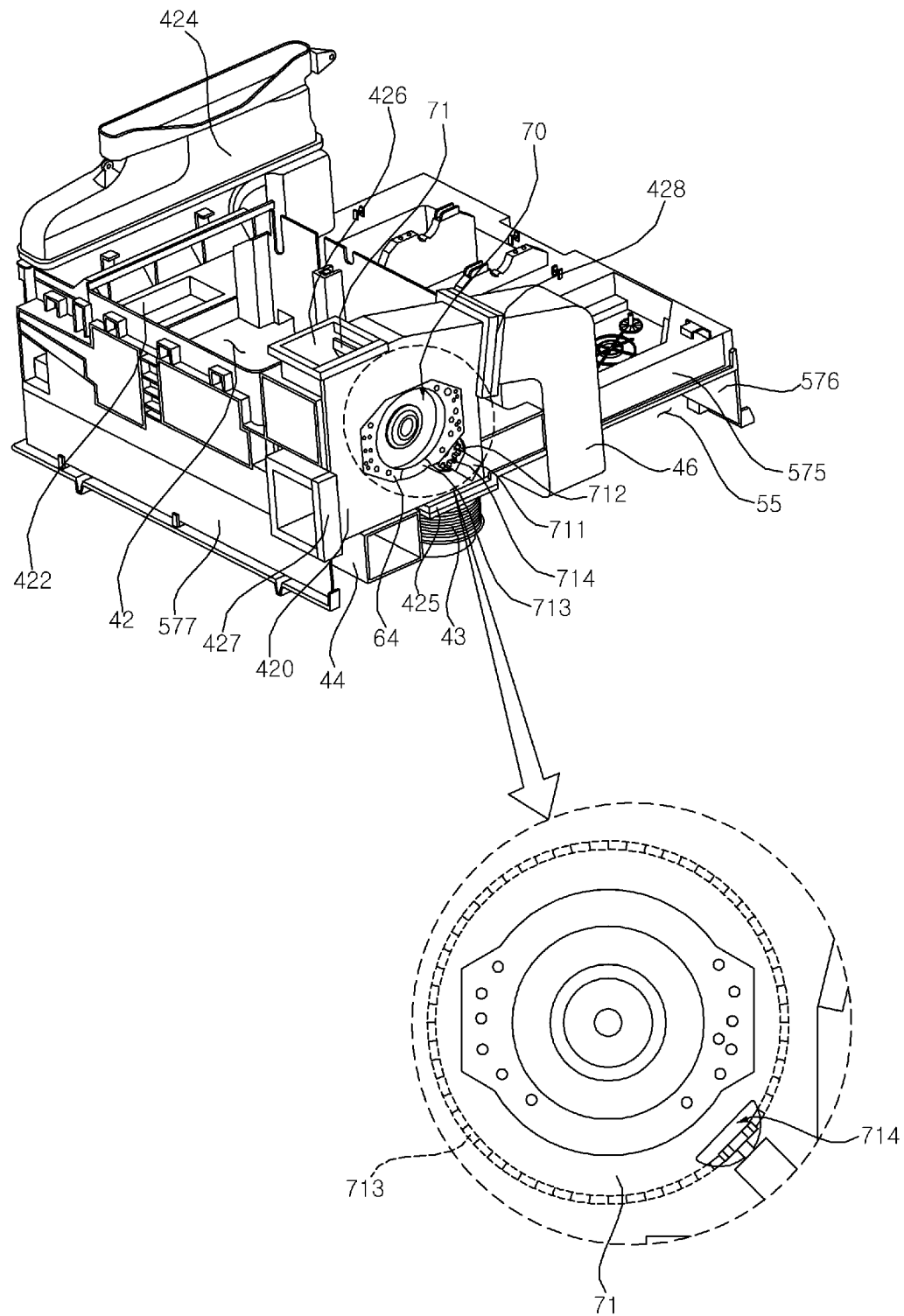
【FIG. 4】



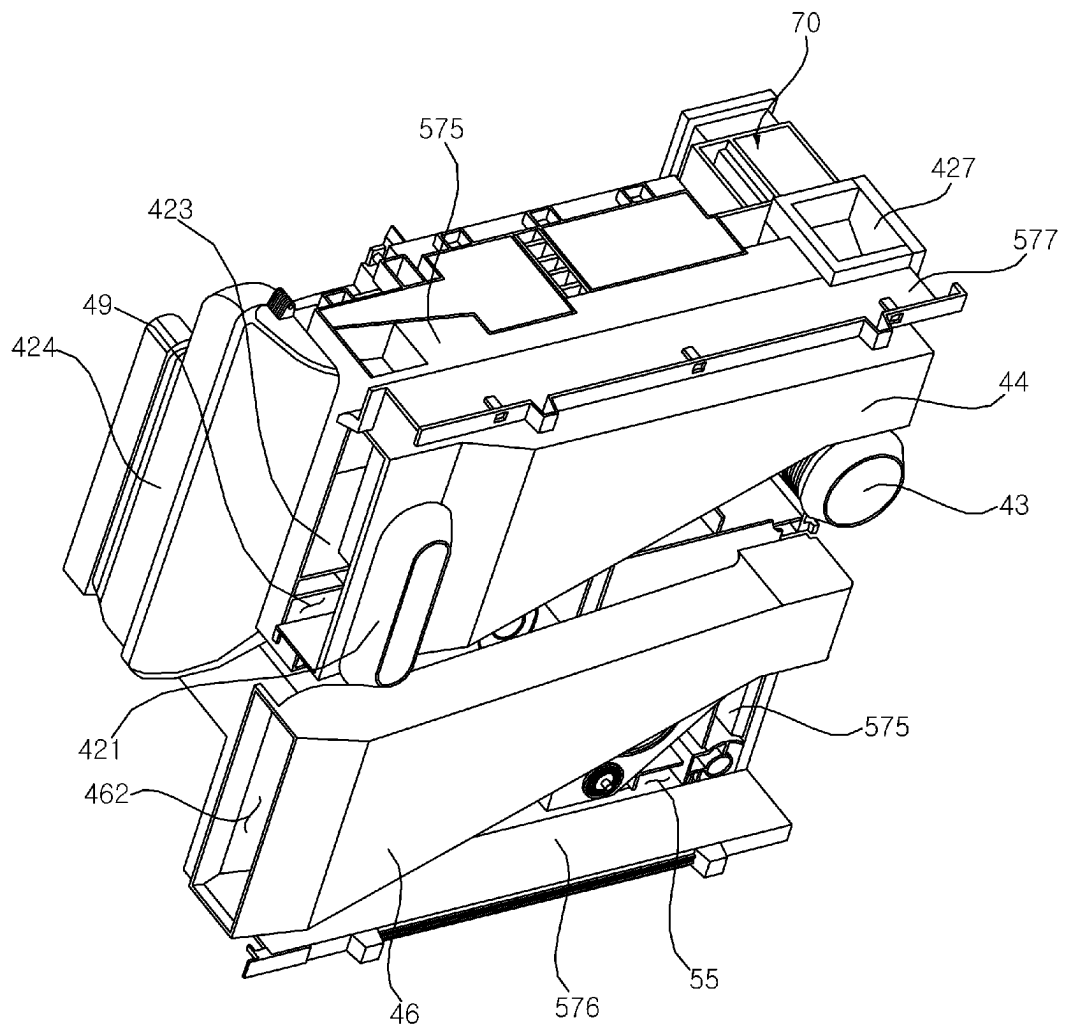
【FIG. 5】



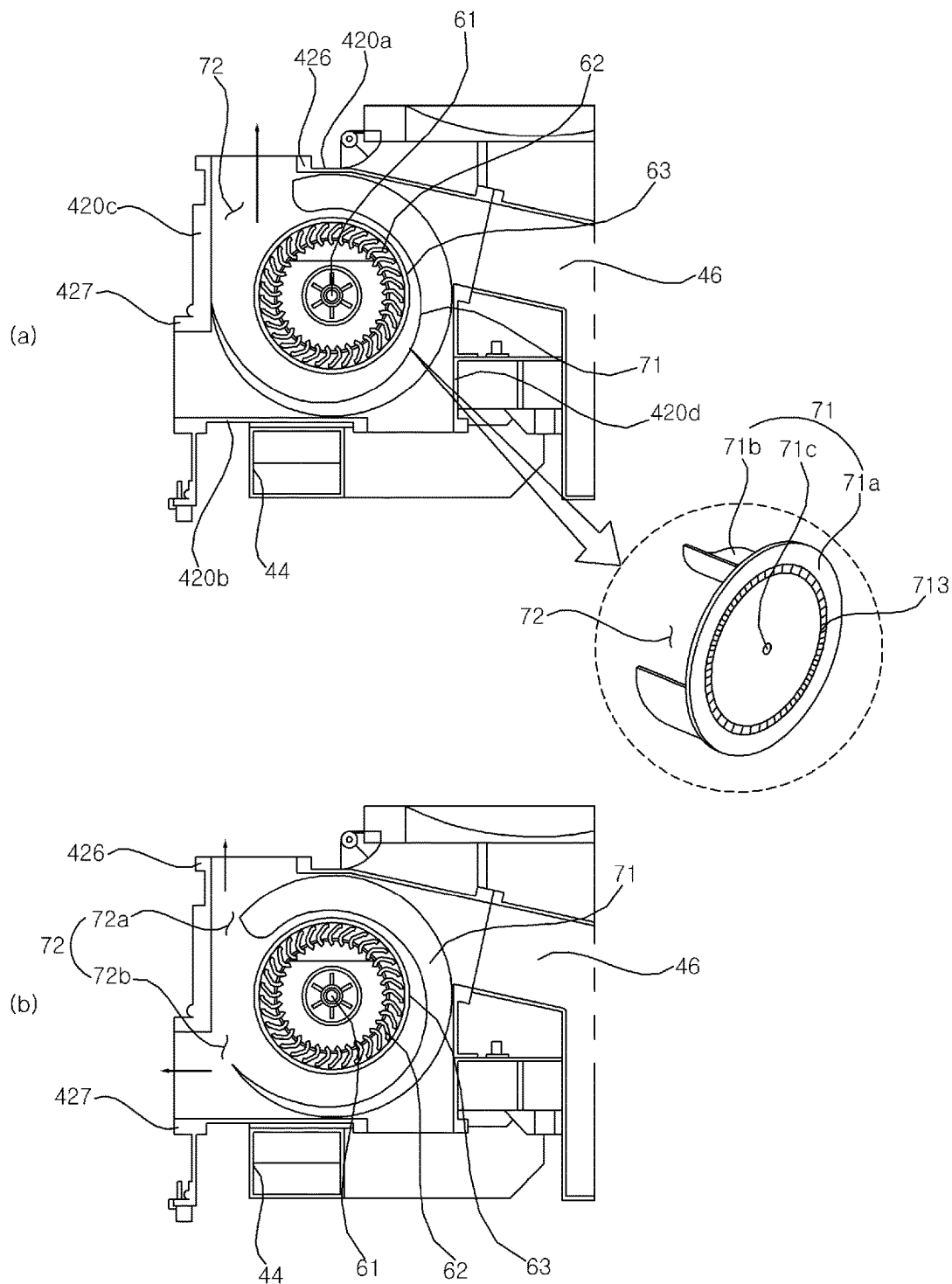
【FIG. 6】



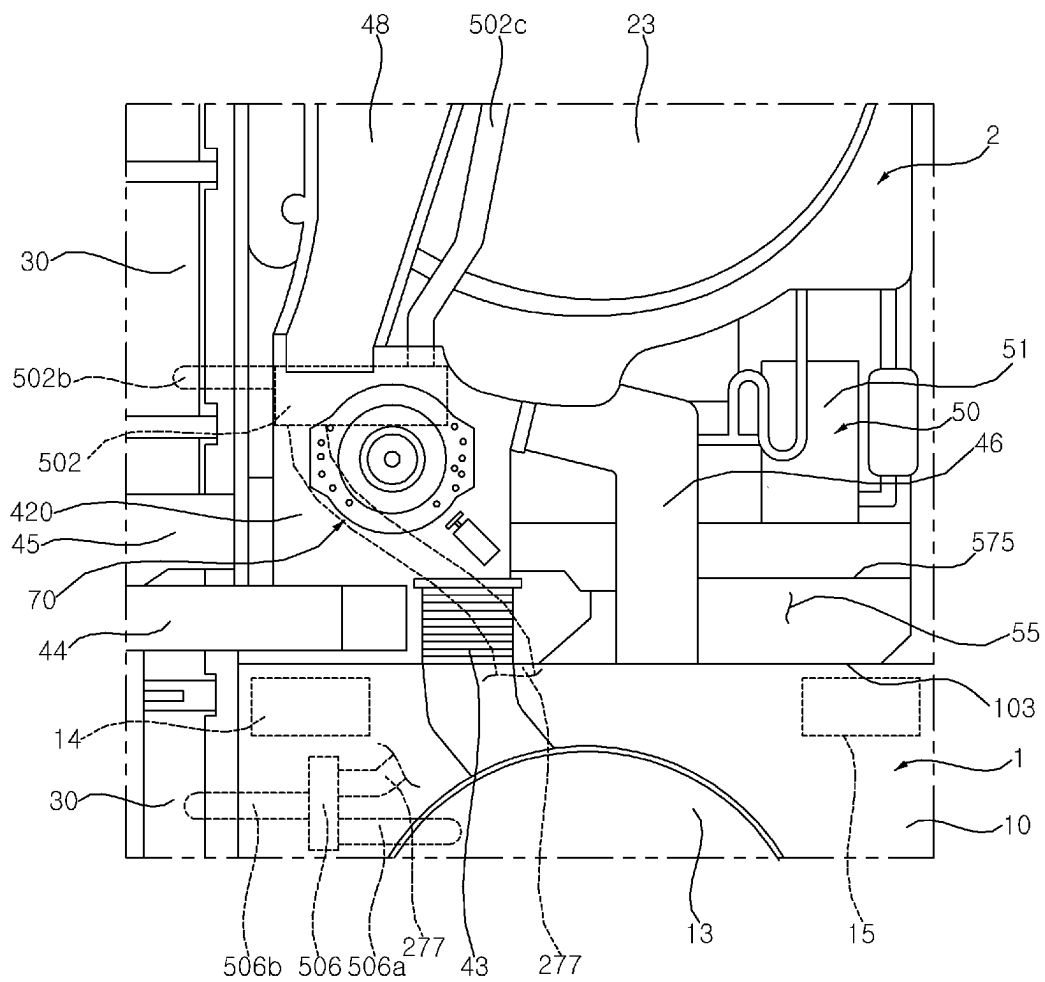
【FIG. 7】



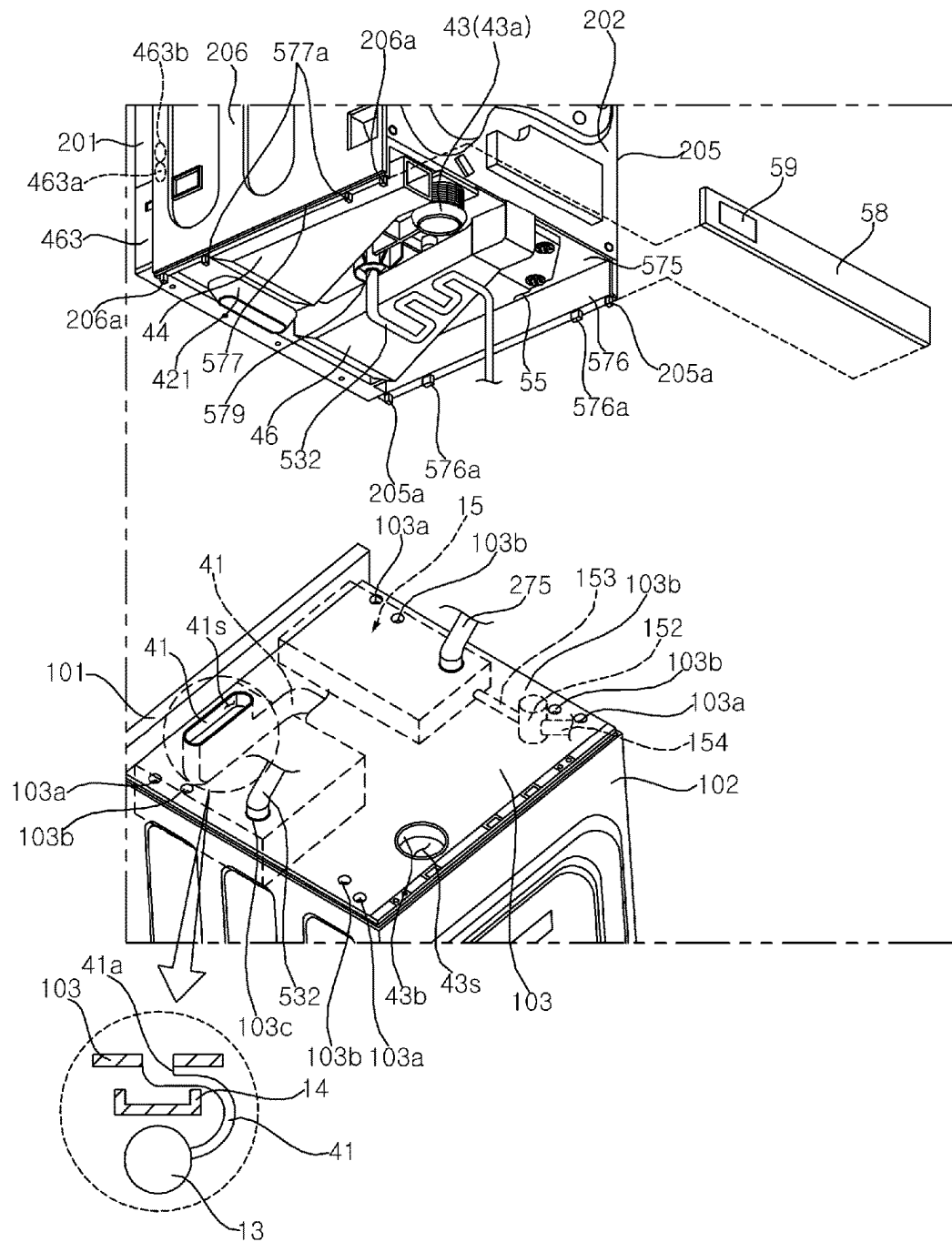
【FIG. 8】



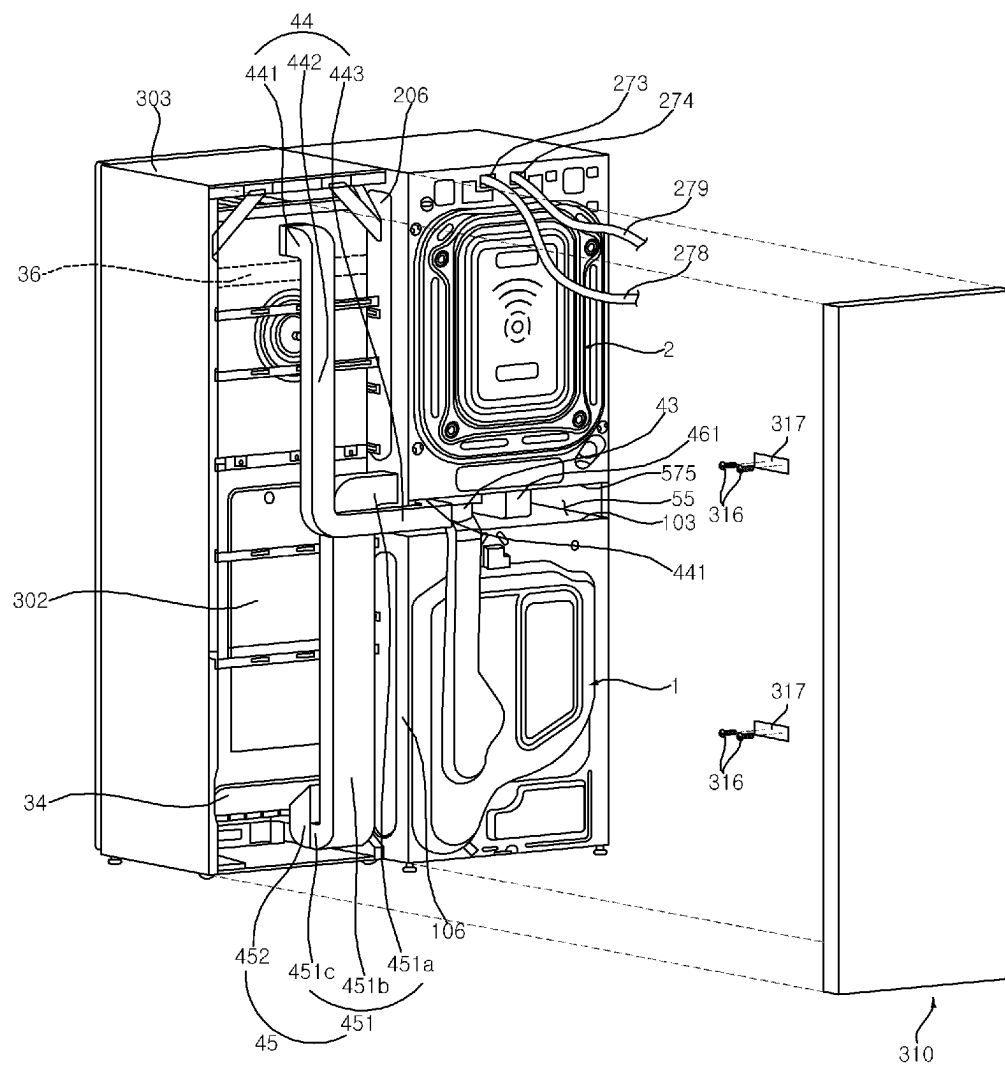
【FIG. 9】



【FIG. 10】

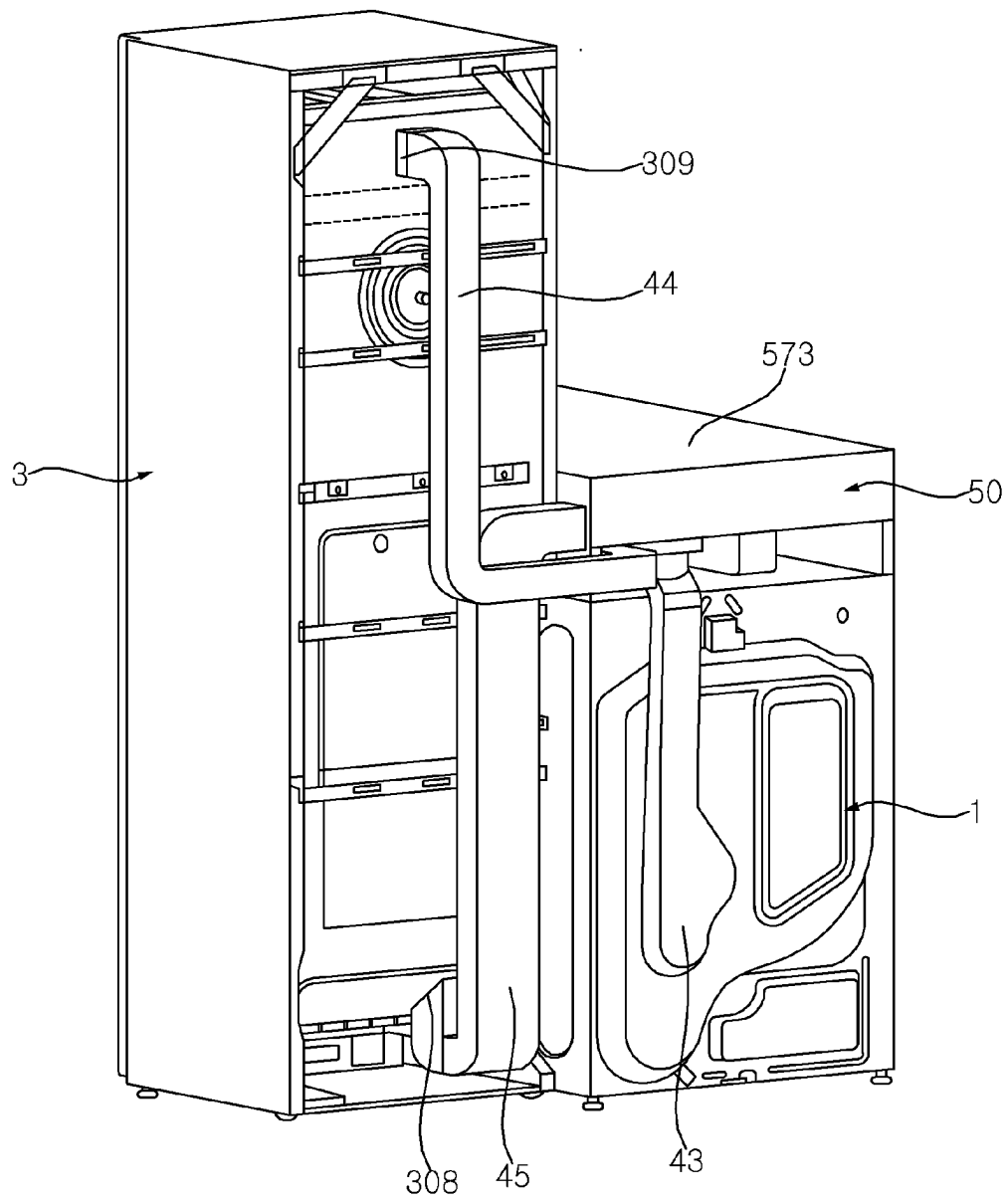


【FIG. 11】

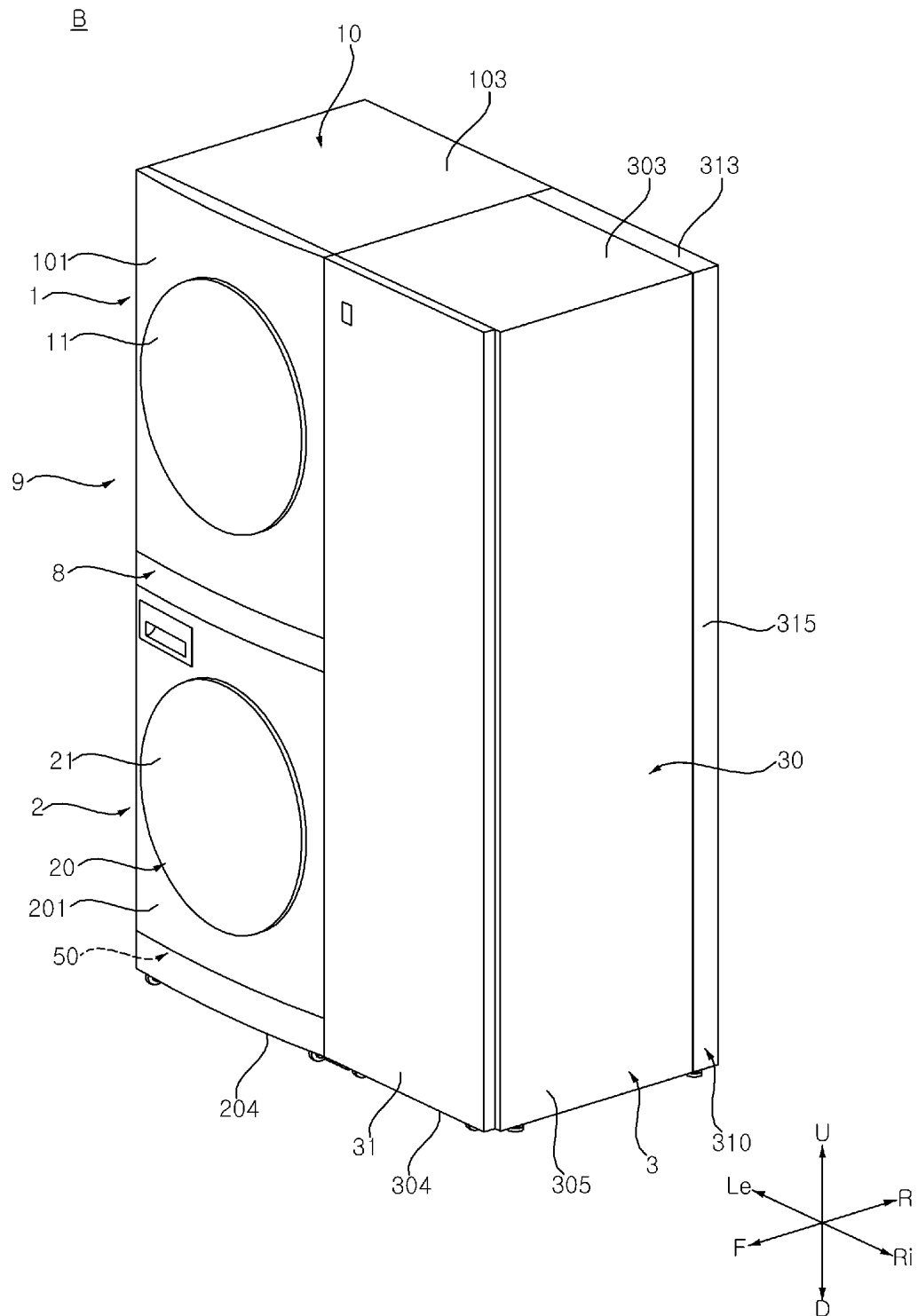


【FIG. 12】

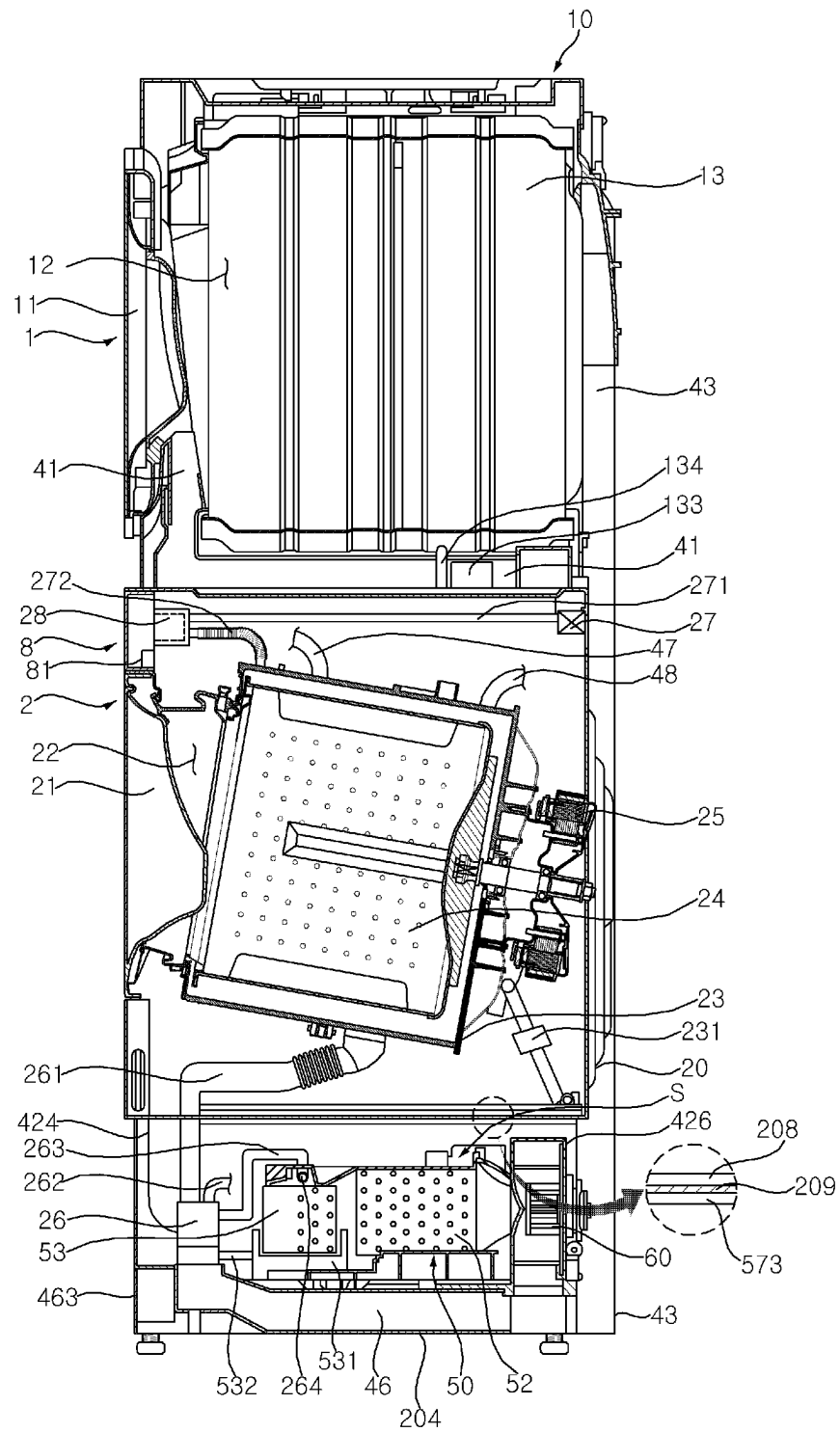
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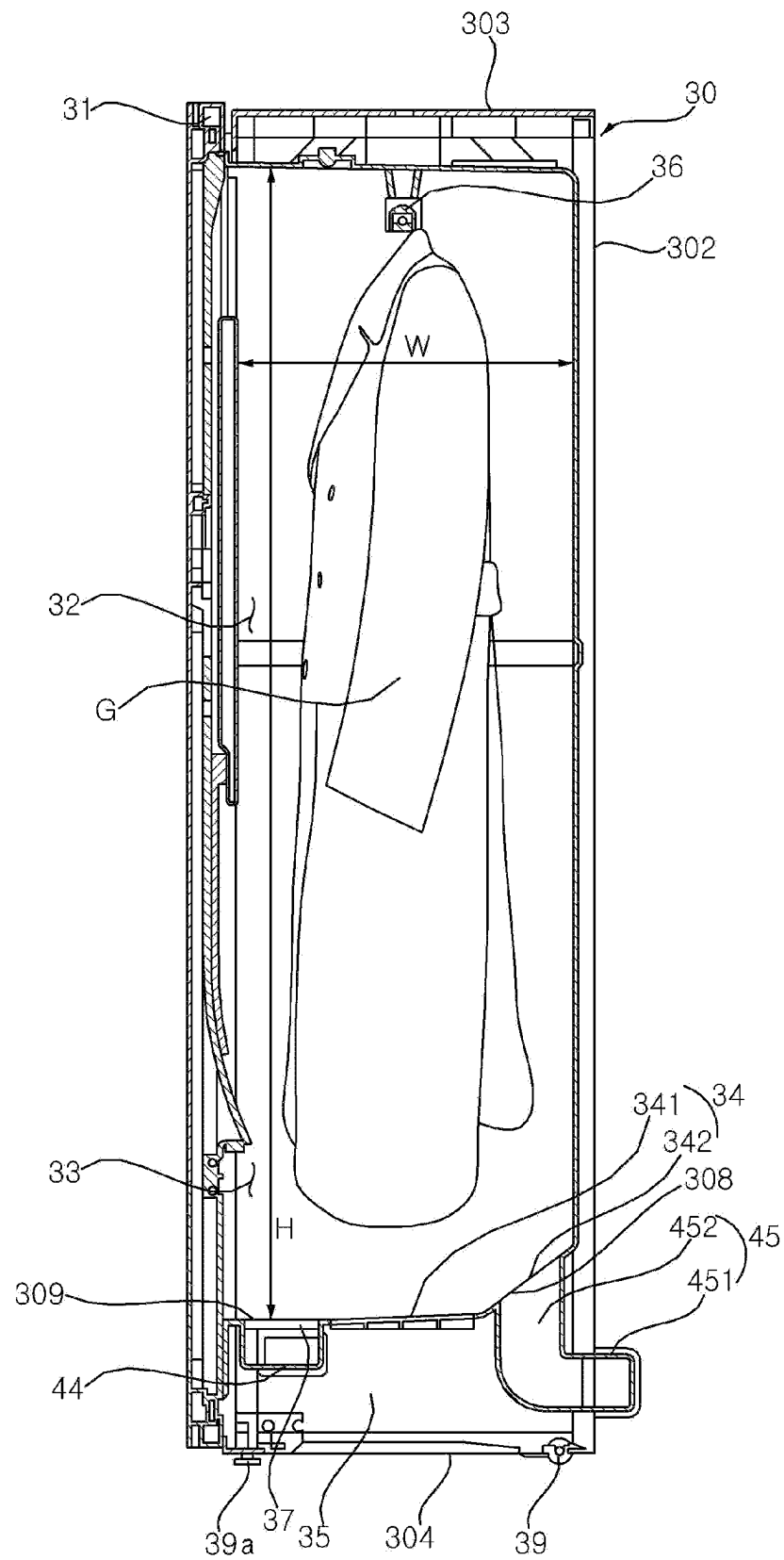
【FIG. 13】



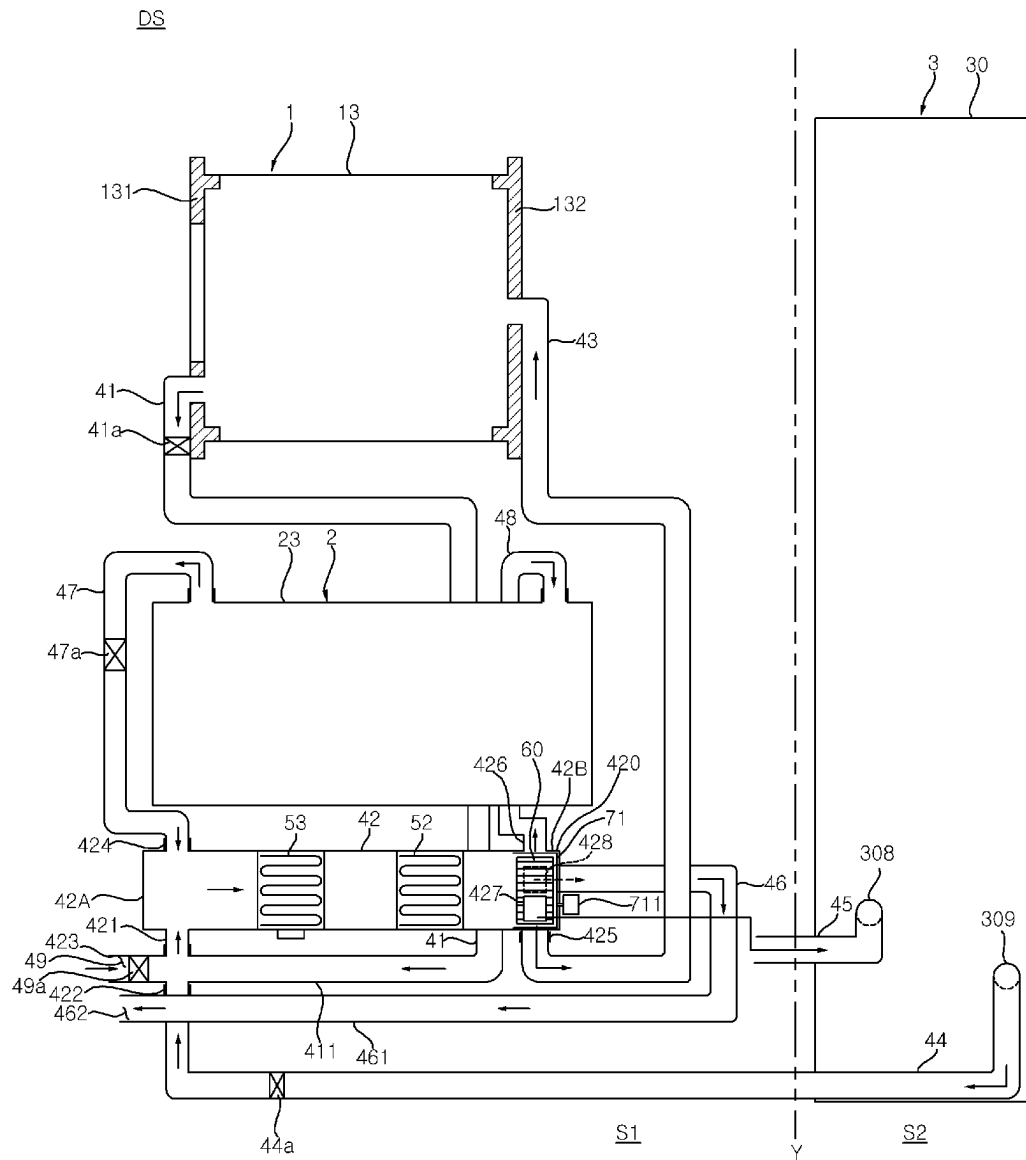
【FIG. 14】



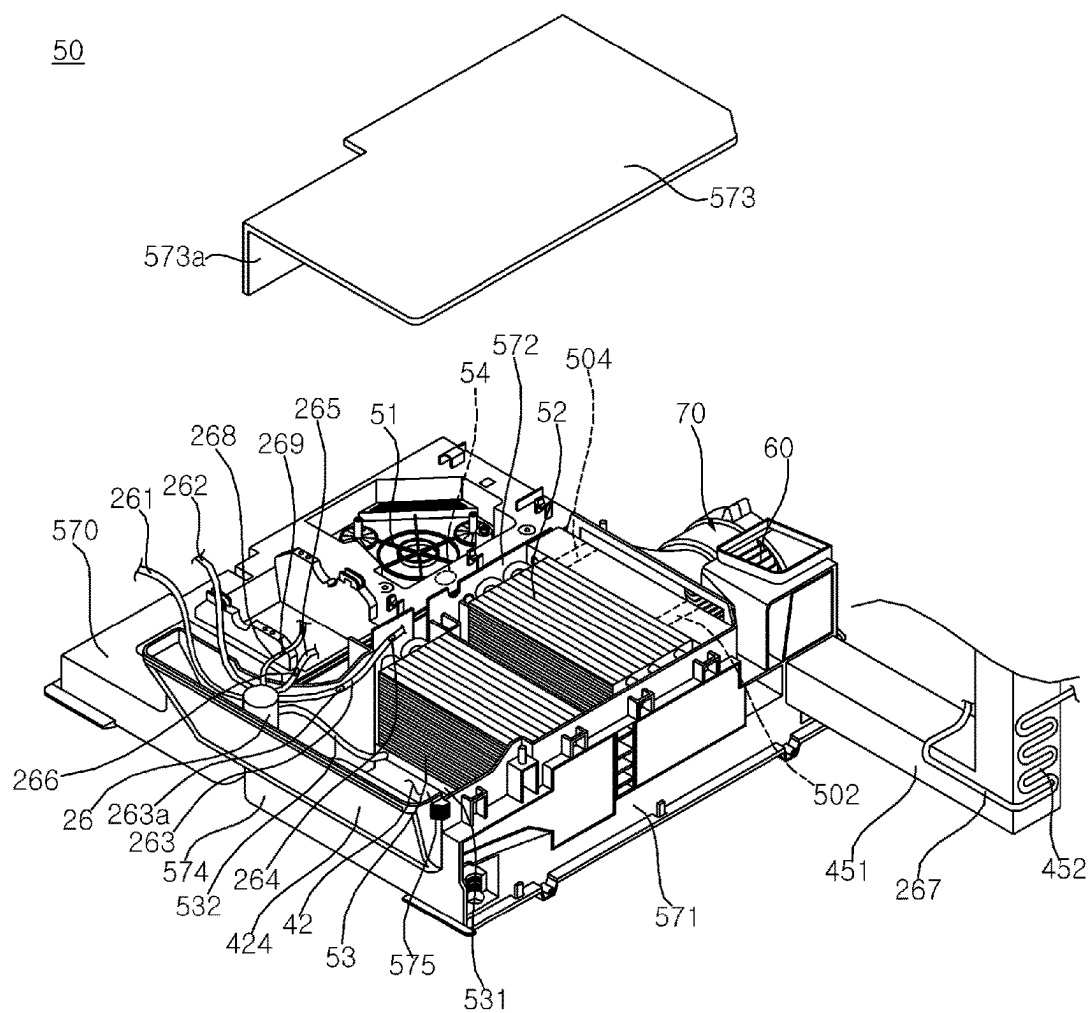
【FIG. 15】



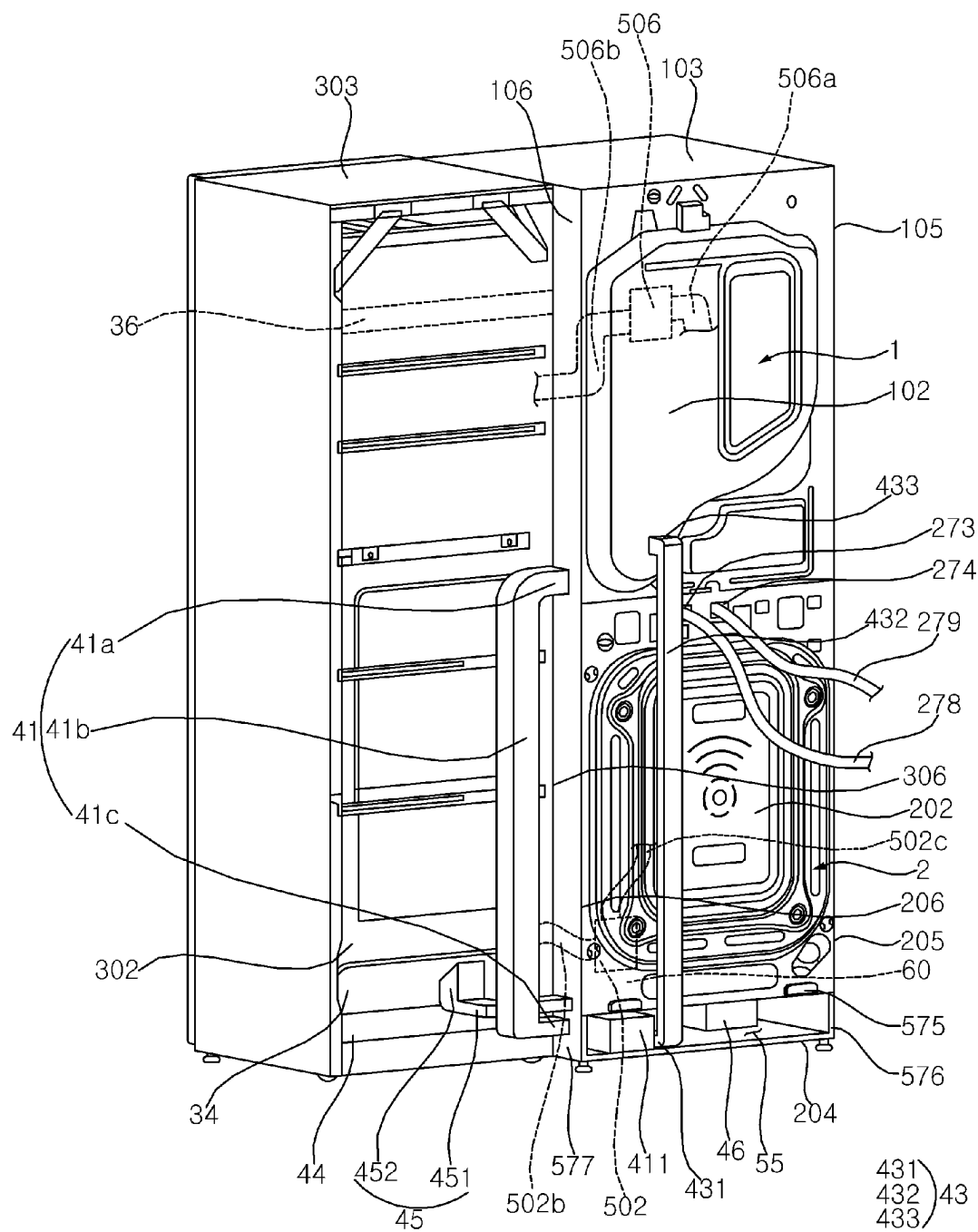
【FIG. 16】



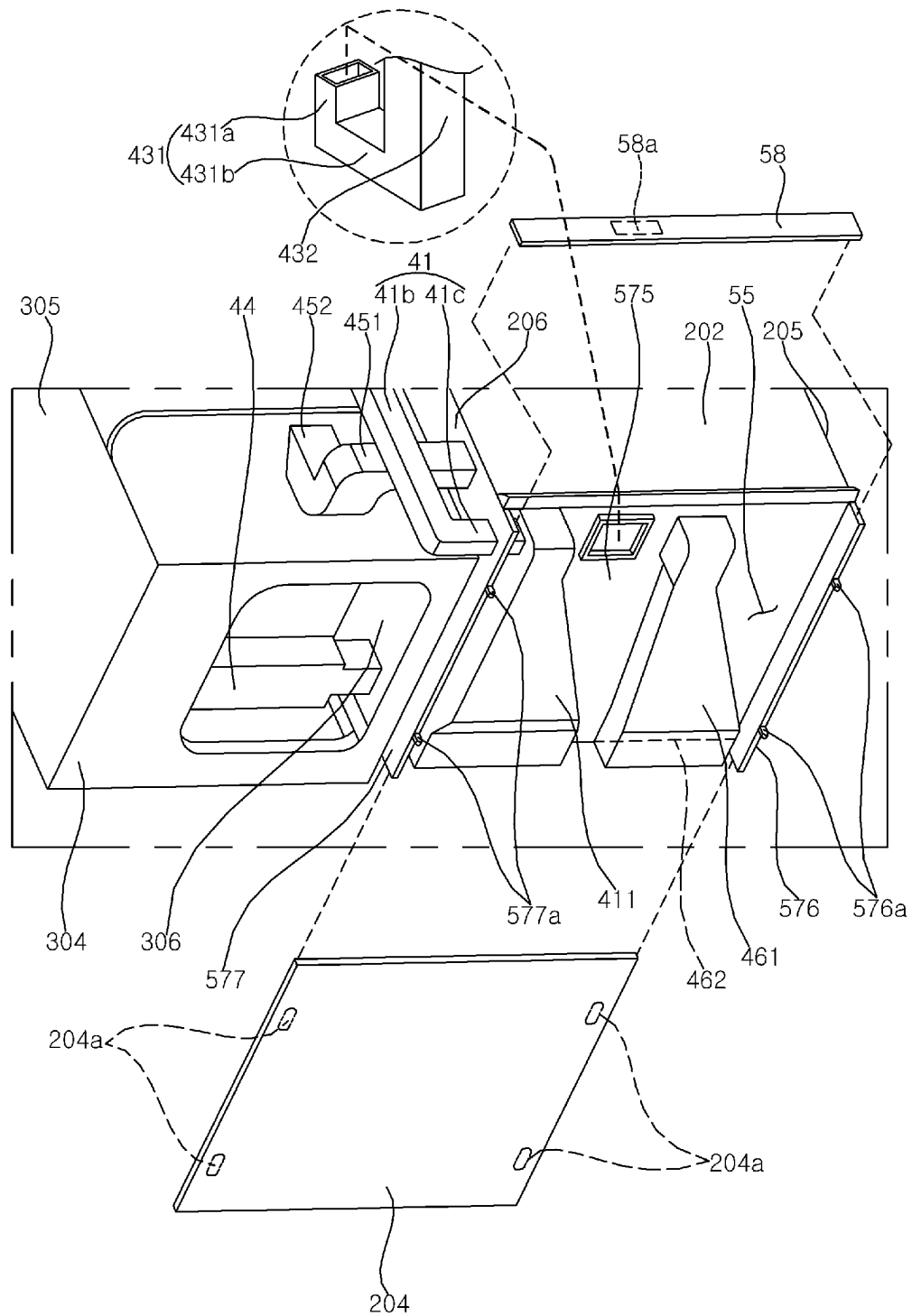
【FIG. 17】



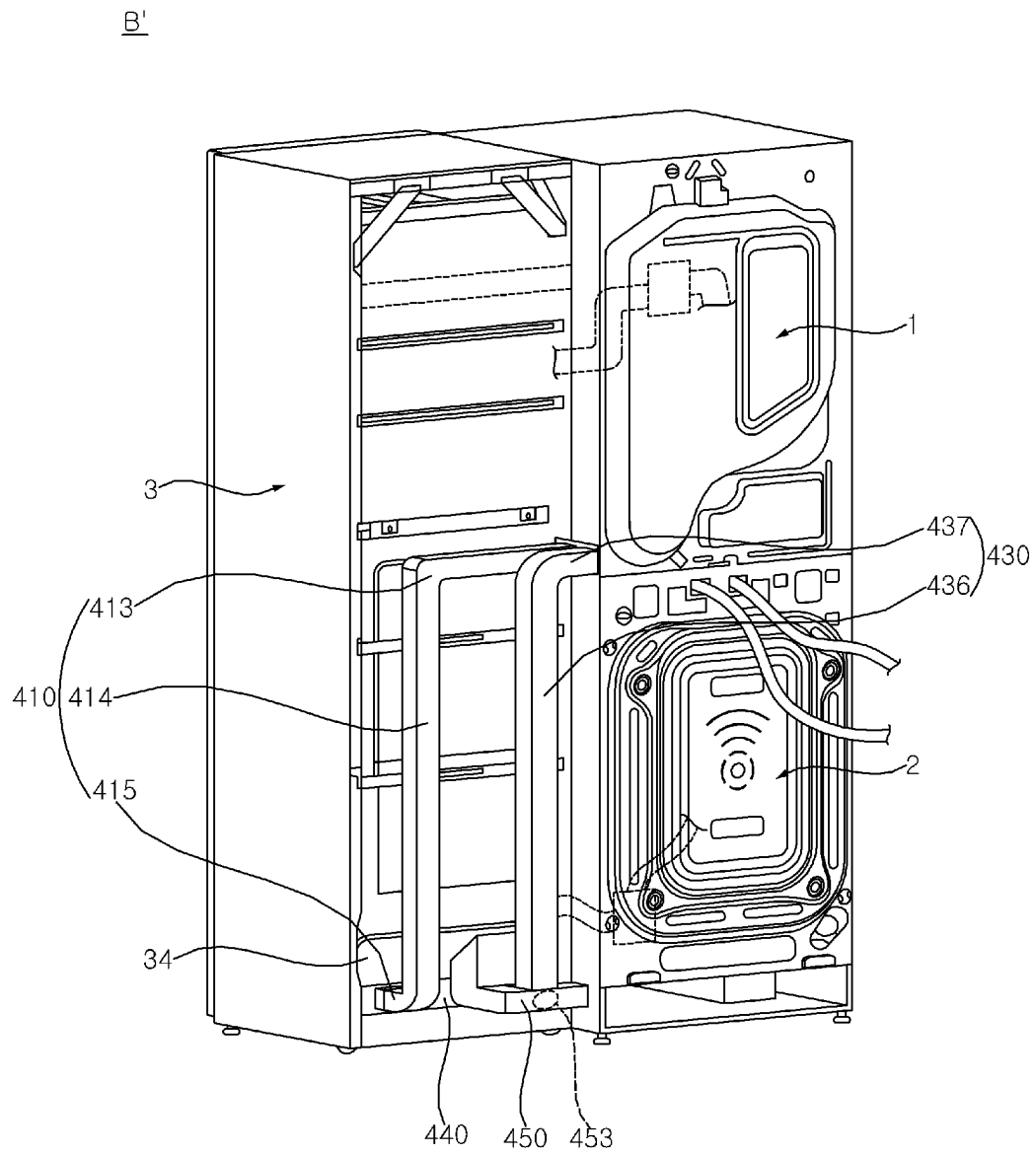
【FIG. 18】



【FIG. 19】



【FIG. 20】



LAUNDRY TREATING APPARATUS**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application claims the priority benefit of the Korean Patent Application No. 10-2021-0069528, filed in the Republic of Korea on May 28, 2021, Korean Patent Application No. 10-2021-0071149, filed in the Republic of Korea on Jun. 1, 2021, Korean Patent Application No. 10-2021-0110913, filed in the Republic of Korea on Aug. 23, 2021, and Korean Patent Application No. 10-2021-0120154, filed in the Republic of Korea on Sep. 9, 2021, which are hereby incorporated by reference in their entirety for all purposes as if fully set forth herein.

BACKGROUND OF THE INVENTION**1. Field of the Invention**

The present disclosure relates to a laundry treating apparatus, and more particularly to a laundry treating apparatus including a plurality of laundry treating machines.

2. Description of the Related Art

A washing machine, a dryer, and refreshers, which are provided for treating clothes, are installed in different spaces, such as a kitchen, a utility room, and an outdoor space, according to usage in a residential space.

Korean Laid-Open Patent Publication No. 10-2020-0109194 discloses a washing machine. The washing machine washes laundry by supplying water and detergent into a tub and a drum and by rotating the drum in which the laundry is placed. In the washing machine, water from an external water source is supplied into the tub or the drum, and a pump is used to circulate the water in the tub to the drum or to discharge the water in the tub to the outside. The washing machine is generally installed in the kitchen or the utility room for the supply and discharge of water.

The washing machine separately includes a heater for heating air, a blower fan, and a steam generator, to dry or sterilize the laundry.

Korean Patent No. 10-2120993 discloses a dryer. The dryer may dry the laundry, placed in the drum, by heating air using a heating means and a blower fan and by supplying the heated air into the drum. The dryer is generally disposed adjacent to the washing machine, so as to dry the washed laundry and to allow a user to easily load the laundry containing moisture into the dryer.

By using the steam generator for generating steam and spraying the steam into the drum, the dryer may sterilize the laundry or may remove wrinkles of the laundry. Water may be supplied into the dryer for generating the steam, and water not changed into steam may be reused or may be discharged to the outside. Further, in order to deodorize the dried laundry, the dryer may further have a deodorization function for filtering the circulated air.

Korean Patent No. 10-2254903 discloses a refresher. The refresher may perform functions, such as drying, deodorizing, de-wrinkling, de-static and/or sterilization (hereinafter referred to as “refreshing”).

The refresher is used for treating clothes, such as suits and coats, which are frequently used by a user after washing. Generally, the user hangs the clothes, such as suits and coats, in a wardrobe. Accordingly, the refresher may be installed adjacent to the wardrobe, or may be installed instead of the

wardrobe, in a dressing room, a living room, or a bedroom. Further, not only for the clothes or garments, the refresher may be used for all washable items, such as shoes, socks, gloves, hats, scarves, etc., which are frequently worn by users, as well as dolls, towels, blankets, etc., which are frequently used by users.

However, the existing laundry treating machines, which are installed in different spaces, have a problem in that a user needs to move the laundry items, which are sorted by a user, to the respective laundry treating machines performing corresponding laundry treating processes, thereby requiring a longer workflow, and making the clothes treating processes uncomfortable.

In addition, as the refresher is installed in a different place from the washing machine and the dryer, it is cumbersome for the user to move the washed wet clothes or the dried clothes to the place where the refresher is installed.

Furthermore, each of the existing laundry treating machines for treating laundry using hot air and steam separately requires water supply equipment, drainage equipment, an air heater, a steam generator, a pump, a blower fan, an air passage, a steam passage, etc., thereby causing a problem in that a space for receiving clothes is reduced. In addition, as each of the machines includes the heater, the steam generator, etc., there is a problem in that costs and energy consumption of the entire system may increase.

Further, there is also a problem in that as the respective laundry treating machines have a hot air passage and a steam passage, the entire length of the passages increases, thus complicating assembly of the passages.

In addition, the existing dryer generally has a heating device for generating hot air which is disposed at a lower portion of the dryer, and a water tank for storing condensate which is disposed at an upper portion of the dryer. Thus, the existing dryer requires a separate pump for moving the condensate generated in the heating device to the water tank, thereby increasing the number of components and making the inner space of the dryer smaller.

SUMMARY OF THE INVENTION

It is an object of the present disclosure to solve the above and other problems.

It is another object of the present disclosure to provide a laundry treating apparatus including a plurality of laundry treating machines.

It is yet another object of the present disclosure to provide a laundry treating apparatus including a washing machine, a dryer, and a refresher.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of performing washing, drying, and refreshing of the laundry in the same space.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of minimizing a user's workflow required for laundry treatment.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of treating laundry which may not be subjected to high temperature drying in a space where washing and drying are performed.

It is still another object of the present disclosure to provide a laundry treating apparatus with improved workability in washing, drying, and refreshing processes of the laundry.

It is still another object of the present disclosure to provide a laundry treating apparatus with reduced vibrations generated in the entire system.

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It is still another object of the present disclosure to provide a laundry treating apparatus capable of improving user convenience in manipulation by lowering the height of components, such as a door, a control panel, etc., which require manipulation by a user.

It is still another object of the present disclosure to provide a laundry treating apparatus having a hot air passage for supplying air to the washing machine, the dryer, and the refresher.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of minimizing a length of a hot air passage for supplying air to the washing machine, the dryer, and the refresher.

It is still another object of the present disclosure to provide a refresher having an extended longitudinal width.

It is still another object of the present disclosure to provide a laundry treating apparatus having a refresher with an extended longitudinal width.

It is still another object of the present disclosure to provide a laundry treating apparatus having an extended vertical height of a laundry receiving space.

It is still another object of the present disclosure to provide a laundry treating apparatus having an extended vertical height of a laundry receiving space in the refresher.

It is still another object of the present disclosure to provide a laundry treating apparatus using a difference between a longitudinal width of the refresher and a longitudinal width of the washing machine/dryer.

It is still another object of the present disclosure to provide a laundry treating apparatus having a steam passage for supplying steam to the washing machine, the dryer, and the refresher.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of minimizing a steam passage for supplying steam to the washing machine, the dryer, and the refresher.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of supplying hot air to the plurality of laundry treating machines using a single heating device.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of supplying steam to the plurality of laundry treating machines using a single heating device.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of facilitating discharge of condensate generated by the heating device.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of facilitating the supply of water for steam generation and discharge of the water.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of minimizing a length of a hot air passage for supplying air to the dryer and the refresher.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of minimizing heat loss of hot air supplied to the refresher.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of minimizing a length of a supply air duct and an exhaust air duct which are connected to the refresher.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of discharging condensate generated by the heating device, without using separate pump equipment.

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It is still another object of the present disclosure to provide a laundry treating apparatus capable of managing condensate by using a drain pump of the washing machine.

It is still another object of the present disclosure to provide a laundry treating apparatus in which temperature of hot air supplied to the refresher is lower than that of the dryer.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of using the condensate generated by the heating device.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of minimizing vibrations generated in the washing machine.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of reducing rocking of cabinets of the washing machine.

It is still another object of the present disclosure to provide a laundry treating apparatus in which the plurality of laundry treating machines are coupled to each other.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of recovering hot air from the upper side of the refresher.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of lowering the height of doors of the washing machine and the dryer.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of lowering the height of a laundry loading opening, through which the laundry is loaded, and the height of a control panel.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of allowing users to easily carry the washed laundry.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of preventing air, heated and dehumidified by the heating device, from being directly discharged to users.

It is still another object of the present disclosure to provide a laundry treating apparatus having a passage formed at a high position for indoor dehumidification.

It is still another object of the present disclosure to provide a laundry treating apparatus having an aligned upper end.

It is still another object of the present disclosure to provide a laundry treating apparatus having an aligned front surface.

It is still another object of the present disclosure to provide a laundry treating apparatus having a height which is constant in the horizontal direction.

It is still another object of the present disclosure to provide a laundry treating apparatus having a constant longitudinal width.

It is still another object of the present disclosure to provide a laundry treating apparatus capable of preventing collision of the laundry treating apparatus with surrounding structures when the laundry treating apparatus is installed.

The objects of the present disclosure are not limited to the aforementioned objects and other objects not described herein will be clearly understood by those skilled in the art from the following description.

In accordance with an aspect of the present disclosure, the above and other objects can be accomplished by providing a laundry treating apparatus may include a dryer including a first cabinet having a first laundry loading opening.

The dryer may have a first door for opening and closing the first laundry loading opening.

The dryer may include a first drum rotatably disposed in the first cabinet.

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The laundry treating apparatus may include a washing machine including a second cabinet vertically disposed with respect to the first cabinet.

The second cabinet may include a second laundry loading opening.

The washing machine may include a second door for opening and closing the second laundry loading opening.

The washing machine may include a tub disposed in the second cabinet.

The washing machine may include a second drum rotatably disposed in the tub.

The laundry treating apparatus may include a refresher disposed on one side of the dryer.

The refresher may be disposed on one side of the washing machine.

The refresher may include a third cabinet having an opening.

The refresher may include a third door for opening and closing the opening.

A space for receiving laundry may be formed in the refresher.

The laundry treating apparatus may include a duct system connected to the dryer and the refresher.

The duct system may include a first supply air duct for supplying air into the first drum.

The duct system may include a second supply air duct for supplying air into an inner space of the refresher.

The duct system may include a heat exchange channel connected to the first supply air duct and the second supply air duct.

The laundry treating apparatus may include a heating device for heating air passing through the heat exchange channel.

The laundry treating apparatus may include a first laundry treating machine including a cabinet having a laundry loading opening, and a door for opening and closing the laundry treating opening.

The first laundry treating machine may include a drum rotatably disposed in the cabinet.

The laundry treating apparatus may include a second laundry treating machine including a cabinet having a laundry loading opening, and a door for opening and closing the laundry treating opening.

A space for receiving laundry may be formed in the second laundry treating machine.

The duct system may include a first supply air duct for supplying air into the drum of the first laundry treating machine.

The duct system may include a second supply air duct for supplying air into the inner space of the second laundry treating machine.

The duct system may include a heat exchange channel connected to the first supply air duct and the second supply air duct.

The heat exchange channel may include an inlet end through which air is introduced, and a discharge end through which the air is discharged.

The laundry treating apparatus may include an integrated frame, having a first laundry loading opening and a second laundry loading opening which are vertically spaced apart from each other, and having a horizontal partition wall disposed at a middle portion thereof.

A drying room may be formed between an upper panel and the horizontal partition wall of the integrated frame.

The first drum may be disposed in the drying room.

A washing room may be formed between a lower panel and the horizontal partition wall of the integrated frame.

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The tub and the second drum may be disposed in the washing room.

The laundry treating apparatus may include a side frame disposed on one side of the integrated frame.

The second supply air duct may supply air into the side frame.

The laundry treating apparatus may include a single frame in which a vertical partition wall is disposed.

The horizontal partition wall may be disposed between the vertical partition wall and the one side panel of the single frame.

The washing room and the drying room may be vertically separated by the horizontal partition wall between the one side panel and the vertical partition wall.

A refreshing room for treating clothes may be formed between the vertical partition wall and the other panel of the single frame.

The dryer may be disposed on one side of the washing machine in a direction horizontal to the washing machine.

The refresher may be disposed over the dryer and the washing machine.

The laundry treating apparatus may include a supply air duct connecting the heating device and the refresher.

The supply air duct may supply the air, heated by the heating device, into the inner space of the refresher.

The heating device may be disposed under the first drum.

The washing machine may be disposed over the dryer.

The heating device may be disposed between the first drum and the tub.

The heating device may include a heat exchange unit for heating air supplied to the refresher.

The laundry treating apparatus may include a drawer disposed below the heat exchange unit.

The drawer may have a space for storing water.

The supply air duct may extend downwardly from the heating device.

The supply air duct may be connected to a supply air hole formed at a lower portion of the refresher.

The laundry treating apparatus may include an exhaust air duct extending upwardly from the heating device.

The exhaust air duct may be connected to an exhaust air hole formed at an upper portion of the refresher.

The heating device may be disposed on an upper side of the dryer.

The supply air duct may extend upwardly from the heating device.

The laundry treating apparatus may include an exhaust air duct extending downwardly from the heating device.

The exhaust air duct may be connected to an exhaust air hole formed at a lower portion of the refresher.

The dryer may be disposed over the washing machine.

The washing may include a drain pump for discharging water in the tub.

The heating device may be disposed under the tub.

The laundry treating apparatus may include a drain pan disposed under the heat exchange unit.

The laundry treating apparatus may include a drain pipe connecting the drain pan and the drain pump.

The washing machine may include an upper panel coupled to a lower portion of the dryer.

The washing machine may include a suspension connecting the cabinet and the tub of the washing machine.

The cabinet of the washing machine may include a side panel disposed between the tub and the cabinet of the refresher.

The side panel may face the cabinet of the refresher.

The side panel may be coupled to the cabinet of the refresher.

The heating device may be disposed over the first drum.

The second drum may be open toward the second laundry loading opening, so that laundry placed therein may be unloaded through the second laundry loading opening.

The first drum may be disposed under the second drum.

The first drum may be open toward the first laundry loading opening, so that the laundry unloaded through the second laundry loading opening, may be loaded through the first laundry loading opening.

The laundry treating apparatus may include a dehumidification duct for discharging the air, heated by the heating device, outside of the laundry treating apparatus.

An upper end of a front surface of the laundry treating apparatus may be aligned horizontally.

An upper end of a front surface of any one, which is located uppermost, among the dryer, the washing machine, and the heating device, may be aligned with an upper end of a front surface of the refresher.

Other detailed matters of the exemplary embodiments are included in the detailed description and the drawings.

Effects of the Invention

According to at least one of the embodiments of the present disclosure, there is provided a laundry treating apparatus including a plurality of laundry treating machines.

According to at least one of the embodiments of the present disclosure, there is provided a laundry treating apparatus including a washing machine, a dryer, and a refresher.

According to at least one of the embodiments of the present disclosure, there is provided a laundry treating apparatus, in which the washing machine, the dryer, and the refresher are disposed in the same space, thereby performing washing, drying, and refreshing in the same space.

According to at least one of the embodiments of the present disclosure, there is provided a laundry treating apparatus, in which laundry may be treated in the same space regardless of the type of laundry.

According to at least one of the embodiments of the present disclosure, there is provided a laundry treating apparatus, in which when washing, drying, and refreshing are performed, a user's load (e.g., amount of work, workflow, etc.) for moving the laundry may be reduced.

According to at least one of the embodiments of the present disclosure, there is provided a laundry treating apparatus capable of treating laundry, which may not be subjected to high temperature drying, in the same space where washing and drying are performed.

According to at least one of the embodiments of the present disclosure, there is provided a laundry treating apparatus, in which the washing machine and the refresher share one surface, thereby reducing vibrations generated in the washing machine.

According to at least one of the embodiments of the present disclosure, there is provided a laundry treating apparatus, in which the heating device of the refresher is disposed between the washing machine and the dryer, thereby extending the height of a laundry receiving space in the refresher.

According to at least one of the embodiments of the present disclosure, a hot air passage is provided in a rear space of the refresher, such that the entire system may have a uniform longitudinal width.

According to at least one of the embodiments of the present disclosure, the heating device is disposed at a middle height of the refresher, thereby minimizing the length of a hot air supply passage for supplying hot air to each of the washing machine, the dryer, and the refresher.

According to at least one of the embodiments of the present disclosure, a steam generator is disposed at a middle height of the refresher, thereby minimizing the length of a steam supply passage for supplying steam to each of the washing machine, the dryer, and the refresher.

According to at least one of the embodiments of the present disclosure, hot air may be supplied to each of the washing machine, the dryer, and the refresher by using a single heating device.

According to at least one of the embodiments of the present disclosure, steam may be supplied to each of the washing machine, the dryer, and the refresher by using a single steam generator.

According to at least one of the embodiments of the present disclosure, condensate may be managed easily by discharging condensate, generated in an evaporator, through a drain pump of the washing machine.

According to at least one of the embodiments of the present disclosure, steam may be generated easily by supplying water, required for generating steam, through a water supply pipe of the washing machine.

According to at least one of the embodiments of the present disclosure, the heating device for supplying hot air to the refresher may be disposed on a lateral side, thereby increasing the height of an inner space of the refresher.

According to at least one of the embodiments of the present disclosure, hot air may be supplied to the refresher by using the heating device of the dryer, thereby increasing a volume of the inner space of the refresher.

According to at least one of the embodiments of the present disclosure, the heating device is disposed at a lower side of the laundry treating apparatus, thereby minimizing the length of the hot air passage connected to the dryer and the refresher.

According to at least one of the embodiments of the present disclosure, a supply air duct, connecting the heating device and the refresher, is reduced in length, thereby minimizing heat loss of the hot air supplied to the refresher.

According to at least one of the embodiments of the present disclosure, condensate generated in the evaporator may be discharged through the drain pump of the washing machine, thereby allowing easy management of the condensate.

According to at least one of the embodiments of the present disclosure, water in a tub and the condensate generated in the heating device are discharged through a single drain pump, thereby reducing the number of components required for drainage.

According to at least one of the embodiments of the present disclosure, a cabinet of the washing machine is coupled to a cabinet of the refresher, thereby reducing vibrations generated in the washing machine.

According to at least one of the embodiments of the present disclosure, as the washing machine is disposed over the dryer, heavy laundry after washing may be dropped to a laundry loading opening of the dryer, thereby allowing a user to easily move the laundry.

According to at least one of the embodiments of the present disclosure, an upper end of the laundry treating apparatus is aligned, such that the plurality of laundry treating machines may be easily coupled to each other.

The effects of the present disclosure are not limited to the aforesaid, and other effects not described herein will be clearly understood by those skilled in the art from the following description of the appended claims.

Further scope of applicability of the invention will become apparent from the detailed description given hereinafter. However, it should be understood that the detailed description and specific examples, while indicating embodiments of the invention, are given by way of illustration only, since various changes and modifications within the scope of the invention will become apparent to those skilled in the art from this detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention will become more fully understood from the detailed description given hereinbelow and the accompanying drawings which are given by way of illustration only, and thus are not limitative of the present invention.

FIG. 1 is a perspective view of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 2 is a longitudinal cross-sectional view of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 3 is a longitudinal cross-sectional view of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 4 is a conceptual diagram illustrating a duct system of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 5 is a perspective view of a heating device according to at least one embodiment of the present disclosure.

FIG. 6 is another perspective view of a heating device according to at least one embodiment of the present disclosure.

FIG. 7 is yet another perspective view of a heating device according to at least one embodiment of the present disclosure.

FIG. 8 is a diagram explaining an operation of a heating device according to at least one embodiment of the present disclosure.

FIG. 9 is a diagram illustrating a portion of a rear surface of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 10 is a diagram illustrating assembly of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 11 is a rear perspective view of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 12 is a rear perspective view of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 13 is a perspective view of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 14 is a longitudinal cross-sectional view of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 15 is a longitudinal cross-sectional view of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 16 is conceptual diagram illustrating a duct system of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 17 is a perspective view of a heating device according to at least one embodiment of the present disclosure.

FIG. 18 is a rear perspective view of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 19 is a view illustrating a lower structure of a laundry treating apparatus according to at least one embodiment of the present disclosure.

FIG. 20 is a rear perspective view of a laundry treating apparatus according to at least one embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Hereinafter, the present disclosure will be described in detail with reference to the accompanying drawings, in which the same reference numerals are used throughout the drawings to designate the same or similar components, and a redundant description thereof will be omitted.

Terms “module” and “unit” for elements used in the following description are given simply in view of the ease of the description, and do not have a distinguishing meaning or role.

It will be noted that a detailed description of known arts will be omitted if it is determined that the detailed description of the known arts can obscure the embodiments of the invention. Further, the accompanying drawings are used to help easily understand various technical features and it should be understood that the embodiments presented herein are not limited by the accompanying drawings. As such, the present disclosure should be construed to extend to any alterations, equivalents and substitutes in addition to those which are particularly set out in the accompanying drawings.

It will be understood that, although the terms first, second, etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another.

It will be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

As used herein, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise.

The laundry treating apparatus according to the embodiments of the present disclosure may include one or more laundry treating machines. The laundry treating machines refer to devices for washing, drying, and/or refreshing objects (e.g., clothes, towels, blankets, etc.).

The laundry treating apparatus may include a plurality of laundry treating machines of the same type. For example, the laundry treating apparatus may include two or more dryers 1. For example, the laundry treating apparatus may include two or more washing machines 2. For example, the laundry treating apparatus may include two or more refreshers 3.

The laundry treating apparatus may include different types of laundry treating machines. For example, the laundry treating apparatus may include the dryer 1, the washing machine 2, and the refresher 3.

The laundry treating apparatus may include laundry treating machines of the same type and laundry treating

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machines of types different from the type. For example, the laundry treating apparatus may include two or more washing machines, one dryer, and one refresher. For example, the laundry treating apparatus may include one washing machine, one dryer, and two or more refreshers.

The laundry treating apparatus of the present disclosure is not limited to the type and number of the laundry treating machines, and may include various combinations thereof. Hereinafter, the laundry treating apparatus including one dryer **1**, one washing machine **2**, and one refresher **3** will be described as an example, but the present disclosure is not limited thereto.

The dryer **1**, the washing machine **2**, and the refresher **3** may be referred to as any one of first to third laundry treating machines, respectively. For example, the dryer **1** may be referred to as the “first laundry treating machine,” the washing machine **2** may be referred to as the “second laundry treating machine,” and the refresher **3** may be referred to as the “third laundry treating machine.”

Hereinafter, it will be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. These elements are not limited to the terms and may include other ordinal numbers. The elements may be used independently without including the ordinal numbers. For example, the elements of the dryer **1** may include the ordinal number “first,” the elements of the washing machine **2** may include the ordinal number “second,” and the elements of the refresher **3** may include the ordinal number “third.” However, the elements of the dryer **1**, the washing machine **2**, and the refresher **3** are not limited to these terms, and may include other ordinal numbers or may be used independently without including the ordinal numbers.

A laundry treating apparatus may include a controller **81** for controlling operation of a dryer **1**, a washing machine **2**, a refresher **3**, and a heating device **50**. The controller **81** may be mounted in a Printed Circuit Board (PCB). The controller **81** may control the dryer **1**, the washing machine **2**, the refresher **3**, and the heating device **50** based on an electric signal received from the control panel **8**. The controller **81** may communicate with the control panel **8**.

The dryer **1** and the washing machine **2** may be vertically disposed. The control panel **8** may be disposed between the dryer **1** and the washing machine **2**. The control panel **8** and the refresher **3** may be vertically disposed.

A user may input a command to the laundry treating apparatus through the control panel **8**. By manipulating the control panel **8**, the user may control the operation of the dryer **1**, the washing machine **2**, and the refresher **3**. By manipulating the control panel **8**, the user may control the operation of the heating device **50**. The control panel **8** may transmit an electrical signal, input by the user's manipulation, to the controller **81**.

A space for providing the PCB may be formed in the control panel **8**. A controller may be mounted in the PCB. The controller may be the controller **81**.

Alternatively, the controller may be disposed on a rear surface of the control panel **8**. The space for providing the PCB may be formed on the rear surface of the control panel **8**. The controller **81** may be mounted in the PCB.

Alternatively, at least some of the dryer **1**, the washing machine **2**, the refresher **3**, and a machine room **S** may include respective controllers for controlling each operation thereof. Even in this case, the respective controllers may be mounted in the PCB. The respective controllers may com-

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municate with each other. In some cases, at least one controller may be an upper level controller that may control other controllers and devices.

The control panel **8** may display an operating state of the laundry treating apparatus. The user may check information on the operating state of the laundry treating apparatus through the control panel **8**.

The laundry treating apparatus may include a heating device **50**. The heating device **50** may supply hot air to the dryer **1**. The heating device **50** may supply hot air to the refresher **3**. The heating device **50** may supply hot air to the washing machine **2**. The hot air, supplied to the respective laundry treating machines, may be drawn into the heating device **50** again for circulation.

However, without circulating the hot air supplied to each of the dryer **1**, the washing machine **2**, and the refresher **3**, the laundry treating apparatus may discharge the hot air to the outside. That is, the laundry treating apparatus may discharge the hot air supplied to a tub **23** of the washing machine **2** to the outside of the washing machine **2**, may discharge the hot air supplied to a first drum **13** of the dryer **1** to the outside of the dryer **1**, and may discharge the hot air supplied to an inner space **33** of the refresher **3** to the outside of the refresher **3**.

The laundry treating apparatus may include the machine room **S** (see FIG. 2). The machine room **S** may provide a space in which the heating device **50** is disposed. The machine room **S** may be a concept that refers to a space in which the heating device **50** and components other than the heating device **50** are disposed. In addition to the heating device **50**, a pump, a blower fan, an air passage, a steam device, a stream passage, a dehumidification device, a controller, a motor, a fan housing, and the like may be disposed in the machine room **S**. The machine room **S** may provide the space in which the aforementioned movable elements (a pump, a blower fan, an air passage, a steam device, a stream passage, a dehumidification device, a controller, a motor, and a fan housing) are disposed.

In the present disclosure, the machine room **S** may refer to the space in which the heating device **50** is disposed, or may be an upper concept that encompasses the heating device **50**.

The machine room **S** may include a separate case. When the machine room **S** includes the separate case, the machine room **S** may be separated from a first cabinet **10** and a second cabinet **20**. Unlike the example, the machine room **S** may be disposed in any one of first to third cabinets **10**, **20**, and **30**. In the case where the machine room **S** is disposed in any one of the first to third cabinets **10**, **20**, and **30**, the machine room **S** may be a concept that refers to a space in which various components including the heating device **50** are disposed. That is, in the case where the machine room **S** is disposed in any one of the first to third cabinets **10**, **20**, and **30**, the machine room **S** may refer to a portion of the inner space of the first to third cabinets **10**, **20**, and **30**.

The machine room **S**, the dryer **1**, and the washing machine **2** may be vertically disposed. The machine room **S** may be disposed on an upper side of the dryer **1** and the washing machine **2**. The machine room **S** may be disposed between the dryer **1** and the washing machine **2**. The machine room **S** may be disposed on a lower side of the dryer **1** and the washing machine **2**. Unlike the example, the machine room **S** and the refresher **3** may be vertically disposed. The machine room **S** may be disposed on the upper side or the lower side of the refresher **3**.

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The heating device **50** may supply hot air to the dryer **1**. The heating device **50** may recover the hot air supplied to the dryer **1**.

The heating device **50** may supply hot air to the washing machine **2**. The heating device **50** may recover the hot air supplied to the washing machine **2**.

The heating device **50** may supply hot air to the refresher **3**. The heating device **50** may recover the hot air supplied to the refresher **3**.

The heating device **50** may supply steam to the dryer **1**. The heating device **50** may supply steam to the washing machine **2**. The heating device **50** may supply steam to the refresher **3**.

The heating device **50** may include a heat pump device. The heat pump device may include a compressor **51**, a condenser **52**, an evaporator **53**, and an expansion device **54** (see FIG. **5**). The heating device **50** may include a fan **60** (see FIG. **5**). The heat pump device may be referred to as a "heat exchange unit."

The heating device **50** may include a heat exchange unit for heating air flowing through a heat exchange channel **42** (see FIG. **5**). The heat exchange unit may be a heat pump device connected by a refrigerant passage. The heat exchange unit may be a heater using electricity as power. The heat exchange unit may be a heating device. The heat exchange unit may include the heater and a cooler. The cooler may be disposed on an upstream side of the heater. The cooler may be disposed in the heat exchange channel **42**. The cooler may dehumidify and cool air passing through the heat exchange channel **42**.

The heating device **50**, the dryer **1**, and the washing machine **2** may be vertically disposed. The heating device **50** may be disposed on the upper side of the dryer **1** and the washing machine **2**, may be disposed between the dryer **1** and the washing machine **2**, or may be disposed under the dryer **1** and the washing machine **2**.

The heating device **50** and the refresher **3** may be vertically disposed. The heating device **50** may be disposed in a rear space of the refresher **3**.

The heating device **50** may be disposed in the first cabinet **10**. The heating device **50** may also be disposed in the second cabinet **20**. The heating device **50** may also be disposed in the third cabinet **30**.

The dryer **1**, the washing machine **2**, and the heating device **50** (or the machine room **S**) may be vertically arranged, and the refresher **3** may be disposed on a lateral side of the washing machine **2**. The heating device **50** may be disposed on a lateral side of the refresher **3**.

Hereinafter, a laundry treating apparatus **A** according to an aspect of the present disclosure will be described below with reference to FIG. **1**.

The laundry treating apparatus **A** illustrated in FIG. **1** may be an example in which the dryer **1** is disposed under the washing machine **2** and the heating device **50** is disposed between the washing machine **2** and the dryer **1**. However, a vertical arrangement of the dryer **1**, the washing machine **2**, and the heating device **50** is not limited thereto.

The refresher **3** may be disposed on a lateral side of the dryer **1** and the washing machine **2**. For example, the refresher **3** may be disposed on the right side of the dryer **1** and the washing machine **2**. However, an arrangement position of the refresher **3** is not limited to the above example, and the refresher **3** may be disposed on the right side of the dryer **1** and the washing machine **2**.

A longitudinal width of the refresher **3** may be smaller than a front-rear width of the dryer **1** and the washing machine **2**.

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A rear case **310**, in which the hot air passage or the steam passage which will be described later is disposed, may be disposed behind the refresher **3**.

The dryer **1** may include the first cabinet **10** having a first laundry loading opening **12** (see FIG. **2**) formed on a front surface thereof, and a first door **11** rotatably coupled to the first cabinet **10**.

By rotating the first door **11**, a user may open the first laundry inlet **12**, and may put the laundry into the first drum **13** through the first laundry inlet **12**.

The washing machine **2** may include a second cabinet **20** having a second laundry loading opening **22** (see FIG. **2**) formed on a front surface, and a second door **21** rotatably coupled to the second cabinet **20**.

By rotating the second door **21**, the user may open the second laundry loading opening **22**, and may load the laundry into the second drum **24** through the second laundry loading opening **22**.

The refresher **3** forms a space in which the laundry is received, and includes a third cabinet **30** having a third laundry loading opening **32** (see FIG. **3**) and a third door **31** rotatably coupled to the cabinet **30**. The third laundry loading opening **32** may be referred to as an "opening."

By rotating the third door **31**, the user may open the third laundry loading opening **32** and may load the laundry into the third cabinet **30** through the third laundry loading opening **32**.

Each of the first cabinet **10**, the second cabinet **20**, and the third cabinet **30** may include an upper panel, a lower panel, a front panel, a rear panel, and a side panel. The upper panel may be referred to as an "upper wall." The lower panel may be referred to as a "lower wall." The front panel may be referred to as a "front wall." The rear panel may be referred to as a "rear wall." The side panel may be referred to as a "side wall."

Each of the first cabinet **10**, the second cabinet **20**, and the third cabinet **30** may include a pair of side panels. The pair of side panels may be horizontally spaced apart from each other. The side panels may be referred to as "side walls." The first cabinet **10** may include a first inner wall **106** facing the refresher **3**, and a first outer wall **105** spaced apart from the first inner wall **106**. The second cabinet **20** may include a second inner wall **206** facing the refresher **3**, and a second outer wall **205** spaced apart from the second inner wall **206**. The third cabinet **30** may include a third inner wall **306** facing the dryer **1** and the washing machine **2**, and a third outer wall **305** spaced apart from the third inner wall **306**.

The first cabinet **10** may include a first upper panel **103**, a first lower panel **104**, a first front panel **101**, a first rear panel **102**, and first side panels **105** and **106**.

The second cabinet **20** may include a second upper panel **203**, a second lower panel **204**, a second front panel **201**, a second rear panel **202**, and second side panels **205** and **206** (second inner and outer walls).

The third cabinet **30** may include a third upper panel **303**, a third lower panel **304**, a third front panel **301**, a third rear panel **302**, and third side panels **305** and **306**. The third cabinet **30** may include the third door **31** instead of the third front panel **301**.

The above description of the first to third cabinets **10**, **20**, and **30** may also be applied to the laundry treating apparatus described with reference to FIGS. **1** to **20**.

A plate may be disposed in a space between the dryer **1** and the refresher **3** and in a space between the washing machine **2** and the refresher **3**. The plate may be a damping material. The dryer **1**, the washing machine **2**, and the refresher **3** may be coupled to each other by the plate.

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The second upper panel **203** of the washing machine **2** and the third upper panel **303** of the refresher **3** may be disposed on the same horizontal plane. The third upper panel **303** of the refresher **3** and the upper wall **313** of the rear case **310** may be disposed on the same horizontal plane.

An upper surface of the washing machine **2** and an upper surface of the refresher **3** may be horizontally aligned. The upper surface of the washing machine **2** and the upper surface of the refresher **3** may be disposed on the same plane.

A front surface **10a** of the dryer **1** may be aligned with a front surface **20a** of the washing machine **2**. The front surface **10a** of the dryer **1** and the front surface **20a** of the washing machine **2** may be disposed on the same plane.

The front surface **10a** of the dryer **1** and a front surface **31a** of the refresher **3** may be horizontally aligned. The front surface **20a** of the washing machine **2** and the front surface **31a** of the refresher **3** may be horizontally aligned. The respective front surfaces of the dryer **1**, the washing machine **2**, and the refresher **3** may be disposed on the same plane.

An upper end **91** of the washing machine **2** may be aligned with an upper end **307** of the refresher **3**. The upper end **91** of the washing machine **2** may be disposed on the same plane as the upper end **307** of the refresher **3**.

An upper end **91** of the washing machine **2** may be aligned with an upper end **307** of the refresher **3**. The upper end **91** of the washing machine **2** may be disposed on the same plane as the upper end **307** of the refresher **3**.

The laundry treating apparatus **A** may include a tower **9** including the dryer **1** and the washing machine **2**. The tower **9** may include the heating device **50**.

The tower **9** may be an assembly in which the dryer **1** and the washing machine **2** are arranged vertically. In the case where the heating device **50** is disposed on one side of the refresher **3**, the tower **9** may be an assembly in which the dryer **1**, the washing machine **2**, and the heating device **50** are arranged vertically.

The dryer **1**, the washing machine **2**, and the heating device **50** may be vertically arranged, so as to form the tower **9**. An arrangement order of the dryer **1**, the washing machine **2**, and the heating device **50** in the tower **9** may not be limited to FIG. **1**. For example, the washing machine **2** may be disposed under the dryer **1**, and the heating device **50** may be disposed above the dryer **1** and the washing machine **2**.

The tower **9** may be disposed on a lateral side of the refresher **3**.

A front surface of the tower **9** may be aligned with the front surface **31a** of the refresher **3**. The front surface of the tower **9** may be disposed on the same plane as the front surface **31a** of the refresher **3**.

An upper end **91** of the front surface of the tower **9** may be aligned with the upper end **307** of the front surface **31a** of the refresher **3**. The upper end **91** of the front surface of the tower **9** may be on the same horizontal plane as the upper end **307** of the front surface **31a** of the refresher **3**. The upper end **91** of the front surface of the tower **9** may be the upper end of the washing machine **2**.

However, in the case where the washing machine **2** is disposed under the dryer **1**, the upper end **91** of the front surface of the tower **9** may refer to the upper end of the front surface of the dryer **1**. Further, in the case where the heating device **50** is disposed on the upper side of the dryer **1** and the washing machine **2**, the upper end **91** of the front surface of the tower **9** may refer to the upper end of the front surface of the heating device **50**.

A third side panel **305** of the refresher **3** and a side wall **315** of the rear case **310** may be disposed on the same plane.

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The front panel **101** of the dryer **1**, the front panel **201** of the washing machine **2**, and the control panel **8** may be disposed on the same plane. The front surface **10a** of the dryer **1**, the front surface **20a** of the washing machine **2**, and the control panel **8** may be disposed on the same plane.

The third door **31** of the refresher **3** may be disposed on the same plane as the front panel **101** of the dryer **1**, the front panel **201** of the washing machine **2**, and the control panel **8**. Alternatively, the third door **31** of the refresher **3** may also be disposed on the same plane as the first door **11** of the dryer **1** and the second door **21** of the washing machine **2**.

The second lower panel **204** of the washing machine **2** may be disposed on the same horizontal plane as the third lower panel **304** of the refresher **3**.

The laundry treating apparatus **A** may include a drawer **14** formed therein which has a space for storing water. The drawer **14** may be disposed in the first cabinet **10**. The drawer **14** may be disposed adjacent to the refresher **3**. The drawer **14** may be disposed at the upper portion of the dryer **1**. The drawer **14** may be disposed above the first door **11**.

The drawer **14** may have a first handle **141**. By holding the first handle **141**, a user may withdraw the drawer **14**. By withdrawing the drawer **14** to the outside of the first cabinet **10**, the user may remove the water stored therein.

The laundry treating apparatus **A** may include a detergent supply device **15** formed therein which has a space for storing detergent. The detergent supply device **15** may be disposed in the first cabinet **10**. The detergent supply device **15** may be disposed farther from the refresher **3** than the drawer **14**. The detergent supply device **15** may be horizontally spaced apart from the drawer **14**. The detergent supply device **15** may be disposed at the upper portion of the dryer **1**. The detergent supply device **15** may be disposed above the first door **11**.

The detergent supply device **15** may have a second handle **151**. By holding the second handle **151**, the user may withdraw the detergent supply device **15**. By withdrawing the detergent supply device **15** to the outside of the first cabinet **10**, the user may fill the detergent supply device **15** with detergent.

The detergent stored in the detergent supply device **15** may be transferred by a detergent pump **152** (see FIG. **28**) to a detergent box **28** disposed at the upper portion of the washing machine **2**.

Hereinafter, an internal structure of the dryer **1**, the washing machine **2**, and the heating device **50** will be described with reference to FIG. **2**. FIG. **2** is a diagram illustrating an internal structure of the dryer **1**, the washing machine **2**, and the heating device **50** which are cut in a longitudinal direction.

The washing machine **2** includes the second cabinet **20**, the second door **21** rotatably coupled to the second cabinet **20**, the tub **23** disposed in the second cabinet **20**, a second drum **24**, which is rotatably disposed in the tub **23** and in which laundry is received, a motor **25** for transmitting power to the second drum **24**, a drain pump **26** for discharging water, generated in the laundry treating apparatus **A** to the outside, a water supply valve **27** connected to an external water source, and a drawer **28** in which detergent is stored. The drawer **28** may be referred to as a "detergent box."

The drain pump **26** may be connected to the tub **23** by a first pipe **261**. The drain pump **26** may discharge water, introduced through the first pipe **261** and a drain pipe **532**, to the outside of the laundry treating apparatus **A** through a second pipe **262**. However, the drain pipe **532** may be directly connected to the second pipe **262**, in which case condensate generated in the evaporator **53** may be dis-

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charged to the outside of the laundry treating apparatus A through the drain pipe 532 and the second pipe 262.

The drain pump 26 may be disposed in the machine room S. The drain pump 26 may be spaced from one side of the heat exchange channel 42. The drain pump 26 may be disposed in the case 570 of the heating device 50.

The water supply valve 27 may open and close water supply pipes 278 and 279 (see FIG. 11) connecting the external water source and the washing machine 2. The water supply valve 27 may control a flow rate of water flowing into the washing machine 2 from the external water source. The water supply valve 27 may be connected to a first water supply pipe 271. The first water supply pipe 271 may be connected to the detergent box 28 in which the detergent is stored. The water flowing into the detergent box 28 through the first water supply pipe 271 may flow into the tub 23 along with the detergent in the detergent box 28 through a second water supply pipe 272.

The first water supply pipe 271 may branch into the third water supply pipe 275. The third water supply pipe 275 branching off from the first water supply pipe 271 may be connected to the detergent supply device 15. The detergent supply device 15 may dissolve the detergent with water supplied through the third water supply pipe 275, and may transfer the dissolved detergent to the detergent box 28 by using the detergent pump 152 (see FIG. 10).

The machine room S may be disposed in the second cabinet 20. The machine room S may refer to a portion of the inner space of the second cabinet 20.

The dryer 1 includes the first cabinet 10, the first door 11 rotatably coupled to the first cabinet 10, and the first drum 13 which is rotatably disposed in the first cabinet 10 and in which the laundry is received.

The dryer 1 may include a motor 133 for rotating the first drum 13. A pulley 134 may be fixed to a rotating shaft of the motor 133. A belt may connect the pulley 134 with a circumferential surface of the drum 13, such that torque of the motor 133 may be transferred to the drum 13 via the pulley 134 and the belt.

The motor 133 may be disposed under the first drum 13. The motor 133 may be disposed in the heating device 50. The motor 133 may be disposed over the first lower panel 104. A space for mounting the motor 133 may be formed between the first drum 13 and the first lower panel 104.

Unlike the example, the motor 133 may be directly coupled to the first drum 13 to rotate the first drum 13.

The heating device 50 may be disposed over the first upper panel 103. The heating device 50 may be disposed in the second cabinet 20.

The washing machine 2 may include a partition wall 208 disposed in the second cabinet 20. The partition wall 208 may extend horizontally. The partition wall 208 may be disposed under the tub 23. The partition wall 208 may be disposed on the upper side of the second lower panel 204.

The machine room S may refer to a space between the partition wall 208 and the second lower panel 204. The drain pump 26 may be disposed under the partition wall 208. The drain pump 26 may be disposed in the machine room S. The first pipe 261 may extend downwardly from the tub 23 to pass through the partition wall 208. The first pipe 261 may pass through the partition wall 208 to extend into the machine room S. The first pipe 261 may be connected to the drain pump 26 in the machine room S.

A buffer material 209 may be disposed between a cover 573 (see FIG. 5) of the heating device 50, which will be described later, and the partition wall 208. The buffer material 209 may be a damping material. The buffer material

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209 may reduce vibrations generated in the washing machine 2. The washing machine 2 may include a suspension 231 connected to the tub 23. The suspension 231 may attenuate vibrations generated in the tub 23. The suspension 231 may be connected to the partition wall 208. The vibrations generated in the tub 23 and transmitted to the second case 20 may be reduced by the suspension 231 and the buffer material 209. The suspension 231 may also be connected to the cover 573.

The heating device 50 includes: a condenser 52 for performing heat exchange between high-temperature refrigerant and air to heat the air; a fan 60 for blowing air heated by passing through the condenser 52; and an evaporator 53 for performing heat exchange between the air flowing into the heating device 50 and low-temperature refrigerant to cool and dehumidify the air. The condenser 52 and the evaporator 53 may be referred to as a "heat exchange unit."

The air heated by passing through the condenser 52 (hereinafter referred to as "hot air") may be blown by the fan 60, and may be supplied to at least one of the dryer 1, the washing machine 2, or the refresher 3.

The heating device 50 may be connected to the first drum 13 by a first supply air duct 43. The hot air generated by the heating device 50 may be blown by the fan 60 to be supplied into the first drum 13 through the first supply air duct 43. The first supply air duct 43 may extend downwardly from the heating device 50.

The heating device 50 may be connected to the tub 23 through a third supply air duct 48. The hot air generated by the heating device 50 may be blown by the fan 60 to be supplied into the tub 23 through the third supply air duct 48. The third supply air duct 48 may extend downwardly from the heating device 50.

The hot air generated by the heating device 50 may be blown by the fan 60 to be supplied to the outside of the laundry treating apparatus A through a dehumidification duct 46. The dehumidification duct 46 may be disposed between a base plate 575 to be described below and the first upper panel 103. The air blown by the fan 60 may be supplied into an indoor space through the dehumidification duct 46. An opening member 463 may be disposed in front of the dehumidification duct 46. The opening member 463 may open and close the front side of the dehumidification duct 46. The opening member 463 may be rotatably coupled to the first cabinet 10 or the second cabinet 20.

The supply air ducts 43, 45, and 48 may be referred to as any one of first to third supply air ducts. For example, the supply air duct connected to the dryer 1 may be referred to as a first supply air duct 43; the supply air duct connected to the refresher 3 may be referred to as a second supply air duct 45, and the supply air duct connected to the washing machine 2 may be referred to as a third supply air duct 48. The supply air ducts 43, 45, and 48 may be referred to as "supply pipes." The respective first, second, and third supply air ducts 43, 45, and 48 may be referred to as any one of the "first to third supply pipes." For example, the first supply air duct 43 may be referred to as a first supply pipe; the second supply air duct 45 may be referred to as a second supply pipe; and the third supply air duct 48 may be referred to as a third supply pipe.

The air supplied to the dryer 1, the washing machine 2, or the refresher 3 may be recovered to the heating device 50.

The heating device 50 may be connected to the first drum 13 by a first exhaust air duct 41. The hot air flowing into the first drum 13 through the first supply air duct 43 may dry the laundry placed in the first drum 13, and then may return to

the heating device 50 through the first exhaust air duct 41. The first exhaust air duct 41 may extend upwardly from the first drum 13.

The heating device 50 may be connected to the tub 23 by a third exhaust air duct 47. The hot air flowing into the tub 23 through the third supply air duct 48 may dry the laundry placed in the second drum 24, and then may return to the heating device 50 through the third exhaust air duct 47.

The exhaust air ducts 41, 44, and 47 may be referred to as any one of first to third exhaust air ducts. For example, the exhaust air duct connected to the dryer 1 may be referred to as a first exhaust air duct 41; the exhaust air duct connected to the refresher 3 may be referred to as a second exhaust air duct 44; and the exhaust air duct connected to the washing machine 2 may be referred to as a third exhaust air duct 47. The respective first, second, and third exhaust air ducts 41, 44, and 47 may be referred to as any one of "first to third return lines." For example, the first exhaust air duct 41 may be referred to as a first return line, the second exhaust air duct 44 may be referred to as a second return line, and a third exhaust air duct 47 may be referred to as a third return line.

The hot air generated by the heating device 50 may be supplied to the dryer 1, the washing machine 2, and the refresher 3, to dry the laundry placed in the respective laundry treating machines 1, 2, and 3, and then may return to the heating device 50. That is, the hot air generated by the heating device 50 may circulate within the laundry treating apparatus A. The hot air, recovered to the heating device 50 after drying the laundry received in each of the plurality of laundry treating machines 1, 2, and 3, may contain a greater amount of moisture than the air drying the laundry placed in one laundry treating machine.

By performing heat exchange between the air returning to the heating device 50 and a refrigerant, the evaporator 53 may reduce the temperature of the returning air and removes moisture contained in the air. The air returning to the heating device 50 may be heat exchanged with the evaporator 53 to generate condensate.

The heating device 50 may include a drain pan 531 disposed on a lower side of the evaporator 53. The drain pan 531 may receive the condensate generated in the evaporator 53.

The drawer 14 may be disposed between the first drum 13 and the tub 23. The drawer 14 may be disposed under the heating device 50. The drawer 14 may be disposed under the evaporator 53. The drawer 14 may be disposed under the drain pan 531.

The drawer 14 may be disposed in the first cabinet 10. The drawer 14 may be disposed at the upper portion of the first cabinet 10. The drawer 14 may be disposed between the first upper panel 103 and the first drum 13.

The drawer 14 may have an inner space 148 formed therein for storing the condensate generated in the evaporator 53.

The drawer 14 may be spaced from the front side of the first rear panel 102. A rear space 149, through which the first supply air duct 43 passes, may be formed between the drawer 14 and the first rear panel 102. The drawer 14 may include a rear wall 142 spaced from the front side of the first rear panel 102 and facing the first rear panel 102. The rear space 149 may be formed between the first rear panel 102 and the rear wall 142.

The condensate generated in the evaporator 53 may be stored in an inner space 148 of the drawer 14. The condensate generated in the evaporator 53 may drop by gravity, to be received in the inner space of the drawer 14.

The drain pan 531 may be disposed between the evaporator 53 and the drawer 14. The condensate generated in the evaporator 53 may be temporarily stored in the drain pan 531, and then may drop into the inner space 148 of the drawer 14. However, the condensate generated in the evaporator 53 may directly drop into the drawer 14, without being stored in the drain pan 531. That is, in the case where the drain pan 531 is not provided, the drawer 14 may be disposed under the evaporator 53 to face the evaporator 53, and the condensate generated in the evaporator 53 may directly drop into the inner space 148 of the drawer 14.

The laundry treating apparatus A may include a drain pipe 532 connecting the drain pan 531 and the drawer 14. The drain pipe 532 may extend downwardly from the drain pan 531 to be connected to the drawer 14.

The drawer 14 may include the aforementioned rear wall 142, a head 145 spaced from the front side of the rear wall 142, and a roof 144 extending forwardly from the rear wall 142.

The first handle 141 may be recessed into the head 145.

The laundry treating apparatus A may include a water level sensor 143 for measuring an amount of water stored in the drawer 14. The water level sensor 143 may be disposed on the rear wall 142 or may be fixed to the rear wall 142.

The roof 144 may be spaced from the rear side of the head 145. The roof 144 may extend forwardly from an upper end of the rear wall 142.

The drain pan 531 may include a first hole 531a which is open downwardly. The drawer 14 may include a second hole 144a which is open upwardly. The first hole 531a and the second hole 144a may communicate with each other. The first hole 531a and the second hole 144a may vertically face each other.

The drain pipe 532 may pass through the first hole 531a and the second hole 144a. The drain pipe 532 may extend downwardly by passing through the first hole 531a, and may extend into the inner space 148 of the drawer 14 by passing through the second hole 144a. The first hole 531a and the second hole 133a may communicate with each other through the drain pipe 532. However, the first hole 531a and the second hole 144a, disposed to vertically face each other, may directly communicate with each other. Specifically, by providing the lower surface of the drain 531 at a position adjacent to the upper surface of the roof 144 so that the first hole 531 and the second hole 144a may vertically face each other, the first hole 531a and the second hole 144a may communicate with each other. In this case, the drain pipe 532 may not be provided, and a separate packing may be provided between the drain pan 531 and the roof 144.

The condensate generated in the evaporator 53 may be temporarily stored in the drain pan 531, and then may be stored in the drawer 14 through the drain pipe 532.

The laundry treating apparatus A may include a second drain pipe 533 connecting the drain pump 26 and the drain pan 531. The drain pipe 532 connecting the drain pan 531 and the drawer 14 may be referred to as a "first drain pipe 532," and the drain pipe 533 connecting the drain pan 531 and the drain pump 26 may be referred to as a "second drain pipe 533."

The drain pump 26 may be disposed on one side of the drain pan 531, and the second drain pipe 533 may extend from the drain pan 531 in the lateral direction to be connected to the drain pump 26. The second drain pipe 533 may transfer the water, stored in the drain pan 531, to the drain pump 26. The drain pump 26 and the drain pan 531 may be disposed on the upper side of the base plate 575. The drain pump 26 may be disposed below the drain pan 531. The

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condensate stored in the drain pan 531 may flow into the drain pump 26 through the second drain pipe 533.

The laundry treating apparatus A may include a first valve 261a for controlling an amount of water flowing in the first pipe 261, and a second valve 533a for controlling an amount of water flowing in the second drain pipe 533.

The controller 81 may be electrically connected to the drain pump 26, the first valve 261a, and the second valve 533a.

The controller 81 may be electrically connected to the water level sensor 143, and may receive information on the amount of water, stored in the drawer 14, from the water level sensor 143. If a value measured by the water level sensor 143 is greater than or equal to a predetermined value, the controller 81 may transmit a signal to the control panel 8. The signal may display information, indicating that it is required to drain the water stored in the drawer 14, on the control panel 8. If the value measured by the water level sensor 143 is greater than or equal to a predetermined value, the control panel 8 may display information, indicating that it is required to drain the water in the drawer 14, on an external display. A user may identify information displayed on the control panel 8 and may remove the water stored in the drawer 14 by withdrawing the drawer 14.

When the washing machine 2 is in operation, and the first valve 261a is open, the controller 81 may open the second valve 533a. That is, the controller 81 may open the second valve 533a so that during the operation of the washing machine 2, the condensate in the drain pan 531 may be discharged to the outside of the laundry treating apparatus A along with the water in the tub 23. The second valve 533a may be opened during a laundry-amount sensing cycle, a washing cycle, a spin-drying cycle, a rinsing cycle, and the like.

When the washing machine 2 is not in operation, and the first valve 261a is closed, the controller 81 may close the second valve 533a. That is, when the washing machine 2 is not in operation, the controller 81 may close the second valve 533a so that the condensate in the drain pan 531 may drop into the drawer 14 without flowing into the drain pump 26.

The amount of condensate stored in the drain pan 531 may be smaller than the amount of water flowing into the drain pump 26 through the first pipe 261. Accordingly, while the washing machine 2 is not in operation such that no water is introduced through the first pipe 261, if the drain pump 26 is operated to discharge the condensate in the drain pan 531, an impeller of the drain pump 26 may be rotated while not being immersed in a sufficient level of water. If the impeller is rotated while not being immersed in a sufficient level of water, the impeller may operate at idle, and an excessive heat may be generated in the impeller, thereby reducing the life span of the drain pump, as well as the energy efficiency. Accordingly, the laundry treating apparatus A of the present disclosure automatically discharges the condensate in the drain pan 531 by using the drain pump 26 only when the washing machine 2 is in operation, and when the washing machine 2 is not in operation, the laundry treating apparatus A allows the condensate to drop into the drawer 14 to store the condensate in the drawer 14.

A third valve 532a for opening and closing the first drain pipe 532 may be disposed in the first drain pipe 532. The controller 81 may be electrically connected to the third valve 532a.

When the washing machine 2 is in operation, the controller 81 may close the third valve 532a. Accordingly, when the

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washing machine 2 is in operation, the condensate in the drain pan 531 may be discharged by the drain pump 26.

When the washing machine 2 is not in operation, the controller 81 may open the third valve 532a. Accordingly, when the washing machine 2 is not in operation, the condensate in the drain pan 531 may drop into the drawer 14.

The dryer 1 may include a drum motor 133 disposed between the first drum 13 and the first lower panel 104, and a pulley 134 rotated by the drum motor 133 and surrounding the first drum 13. The first drum 13 may be rotated by the rotation of the pulley 134.

The drawer 14 may be disposed between the first drum 13 and the first lower panel 104. The drawer 14 may be horizontally spaced apart from the drum motor 133. The drawer 14 may be disposed at the lower portion of the drawer 1. In this case, a large space on the lower side of the first drum 13 may be used for placement of the drawer 14.

Hereinafter, an internal structure of the refresher 3 will be described with reference to FIG. 3. FIG. 3 is a diagram illustrating an internal structure of the refresher 3 of the laundry treating apparatus A which is cut in a longitudinal direction.

The refresher 3 may include the third cabinet 30 forming a space in which the laundry is received.

The refresher 3 may include a hanger 36, on which clothes are hung, and an inner panel 34 defining a lower portion of a space 33 in which clothes are received. The inner panel 34 may be referred to as an "inner plate."

The inner plate 34 may be disposed below the first drum 13. Accordingly, as the height of the position of the inner plate 34 is reduced, a volume of the inner space of the refresher 3 may increase.

The hanger 36 may be disposed between the third upper panel 303 and the third lower panel 304, and may be disposed closer to the third upper panel 303 than the third lower panel 304.

The hanger 36 may be disposed above the tub 23. The hanger 36 may be disposed at a height corresponding to an upper portion of the tub 23. Accordingly, as the height of the position of the hanger 36 increases, a volume of the inner space of the refresher 3 may increase.

The inner panel 34 may be disposed between the third upper panel 303 and the third lower panel 304, and may be disposed closer to the third lower panel 304 than the third upper panel 303.

Garments G received in the inner space 33 of the third cabinet 30 may be located between the hanger 36 and the inner panel 34.

The height H of the inner space 33 may refer to a height between the hanger 36 and the inner panel 34. The height H of the inner space 33 may refer to a vertical gap between the upper panel 303 and the inner panel 34. The height H of the inner space 33 may be extended compared to a case where the heating device 50 is disposed between the inner panel 34 and the third lower panel 304. Accordingly, unlike an existing refresher, the refresher 3 according to the present disclosure may receive garments G, such as a long coat or a suit, which occupy a great height in the vertical direction.

In addition, compared to the existing refresher, the refresher 3 according to the present disclosure may have a longitudinal width which may be extended according to the longitudinal width of the dryer 1 and the washing machine 2. Accordingly, the refresher 3 according to the present disclosure may receive a larger number of garments in the front-rear direction than the existing refresher. The longitu-

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dinal width W of the inner space 33 of the refresher 3 may refer to a width between the third door 31 and the third panel 302.

In the laundry treating apparatus A according to an aspect of the present disclosure, the supply air duct 45 and the exhaust air duct 44 are disposed behind the space 33 in which the laundry is placed, such that the front surface 31a of the refresher 3 and the front surface of the dryer 1 and the washing machine 2 may be aligned.

A storage space 35 may be formed between the inner panel 34 and the third lower panel 304. Garments, such as socks, underwear, hats, scarves, gloves, etc., which take up a relatively small volume, may be placed in the storage space 35.

The inner panel 34 may be connected to the second supply air duct 45. The third cabinet 30 may be connected to the second exhaust air duct 44.

The inner panel 34 may include a first inner panel 341 spaced apart from an upper side of the third lower panel 304, and a second inner panel 342 extending diagonally upward from the first inner panel 341. The second inner panel 342 may extend at an incline upwardly from the first inner panel 341 toward the third rear panel 302.

The second supply air duct 45 may be connected to the second inner panel 342. The second inner panel 342 may have a supply air hole 308 that is open toward the inner space 33. The second supply air duct 45 may be connected to the supply air hole 308. The hot air supplied from the heating device 50 may be discharged to the inner space 33 through the supply air hole 308.

The second supply air duct 45 may include a first hot air duct 451 disposed at a rear side of the third rear panel 302, and a second hot air duct 453 disposed in front of the third rear panel 302.

The first hot air duct 451 may be disposed in the rear case 310. A second hot air duct 452 may be disposed in the storage space 35. The first hot air duct 451 and the second hot air duct 452 may be coupled to the third rear panel 302. The first hot air duct 451 may extend upwardly in the rear case 310 to be connected to the heating device 50. The second hot air duct 452 may extend upwardly to be connected to the second inner panel 342.

The second exhaust air duct 44 may be connected to the third rear panel 302. The third rear panel 302 may have an exhaust air hole 309 that is open toward the inner space 33. The second exhaust air duct 44 may be coupled to the third rear panel 302 so as to correspond to the exhaust air hole 309. The hot air discharged to the inner space 33 of the refresher 3 may dry the garments G, and then may flow into the second exhaust air duct 44 through the exhaust air hole 309.

The supply air hole 308 may be disposed at a lower portion of the refresher 3, and the exhaust air hole 309 may be disposed at an upper portion of the refresher 3. However, the positions of the supply air hole 308 and the exhaust air hole 309 are not limited to the above example. For example, the supply air hole 308 may be disposed at the upper portion of the refresher 3, and the exhaust air hole 309 may be disposed at the lower portion of the refresher 3. In this case, the supply air hole 308 may be disposed above the hanger 36. As the supply air hole 308 is disposed above the hanger 36, dust deposited on the garments G may be removed. Further, any one of the supply air hole 308 or the exhaust air hole 309 is disposed at the upper side, and the other one is disposed at the lower side, air supplied through the supply air hole 308 may be distributed evenly throughout the inner

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space 33, and then may return to the heating device 50 through the exhaust air hole 309.

The second exhaust air duct 44 may pass through the third rear panel 302 to extend into the rear case 310. After passing through the third rear panel 302, the second exhaust air duct 44 may extend downwardly in the rear case 310 to be connected to the heating device 50.

The refresher 3 may include rollers 39 and legs 39a protruding downwardly from the third lower panel 304. A plurality of rollers 39 may be disposed which are spaced apart from each other in the front-rear direction. A roller structure of the refresher 3 may also be applied to the second lower panel 204 of the washing machine 2. The rollers of the washing machine 2 and the refresher 3 may serve to support the weight of the laundry treating apparatus A when the laundry treating apparatus A is moved.

Hereinafter, a duct system of the laundry treating apparatus A according to the present disclosure will be described with reference to FIG. 4. In FIG. 4, a left side of a reference line Y shows a duct system of the dryer 1 and the washing machine 2, and a right side of the reference line Y shows a duct system of the refresher 3. The left side of the reference line Y in FIG. 4 is view conceptually illustrating a duct system of the dryer 1 and the washing machine 2 which are cut by a plane perpendicular to a left and right direction; and the right side of the reference line Y in FIG. 4 is a view conceptually illustrating a rear surface of the refresher 3 to show a duct system. The directions used in the description of FIG. 4 may be the same as those illustrated in FIG. 1.

The left side of the reference line Y in FIG. 4 may be referred to as a first conceptual diagram S1. The right side of the reference line Y in FIG. 4 may be referred to as a second conceptual diagram S2.

The "duct system DS" may be a concept that collectively refers to passages of hot air circulating in the laundry treating apparatus A. The "duct system DS" may be a concept that collectively refers to passages connecting the heating device 50, the dryer 1, the washing machine 2, and the refresher 3. The "duct system DS" may be a concept that collectively refers to passages of hot air heated by the heating device 50.

The heating device 50 heats air and supplies the heated air to each of the dryer 1, the washing machine 2, and the refresher 3.

The duct system DS may include the heat exchange channel 42 in which the air is heated. The duct system DS may include the first supply air duct 43 connecting the heating device 50 and the dryer 1. The duct system DS may include the second supply air duct 45 connecting the heating device 50 and the refresher 3. The duct system DS may include the third supply air duct 48 connecting the heating device 50 and the washing machine 2. The duct system DS may include the dehumidification duct 46 connected to the heating device 50. The air heated by passing through the heat exchange channel 42 is blown by the fan 60 to be supplied to at least any one of the first supply air duct 43, the second supply air duct 45, the third supply air duct 48, and the dehumidification duct 46.

The duct system DS may include the first exhaust air duct 41 connecting the heating device 50 and the dryer 1. The duct system DS may include the second exhaust air duct 44 connecting the heating device 50 and the refresher 3. The duct system DS may include the third exhaust air duct 47 connecting the heating device 50 and the washing machine 2. The heated air supplied to the dryer 1 may flow into the heat exchange channel 42 through the first exhaust air duct 41. The heated air supplied to the refresher 3 may flow into

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the heat exchange channel 42 through the second exhaust air duct 44. The heated air supplied to the washing machine 2 may flow into the heat exchange channel 42 through the third exhaust air duct 47.

The heat exchange channel 42, through the air flows, may be provided in the heating device 50. The heating device 50 may include a heat exchange unit disposed in the heat exchange channel 42. The heat exchange unit may heat the air flowing in the heat exchange channel 42. The heat exchange unit may include the condenser 52 and the evaporator 53.

The laundry treating apparatus A may include the fan 60. The fan 60 may blow the air in the heat exchange channel 42. The fan 60 may be disposed on a downstream side of the condenser 52. The air heated by heat exchange with the evaporator 53 and the condenser 52 may be blown by the fan 60. The fan 60 may be disposed in the heat exchange channel 42.

The air blown by the fan 60 may be supplied to at least any one of the dryer 1, the washing machine 2, the refresher 3, and the dehumidification duct 46.

The laundry treating apparatus A may include a blowing duct 420 surrounding the fan 60. The duct system may include the blowing duct 420. The fan 60 may be disposed in the blowing duct 420.

An inner space of the blowing duct 420 may be a portion of the heat exchange channel 42.

The fan 60 and the blowing duct 420 may be disposed at a discharge end 42B of the heat exchange channel 42. The discharge end 42B may be disposed at a downstream side of the condenser 52 and the evaporator 53.

The blowing duct 420 may include a first discharge port 425 connected to the first supply air duct 43. The blowing duct 420 may include a second discharge port 427 connected to the second supply air duct 45. The blowing duct 420 may include a third discharge port 426 connected to the third supply air duct 48. The blowing duct 420 may include a fourth discharge port 428 connected to the dehumidification duct 46.

The first discharge port 425 may protrude downwardly from the blowing duct 420. The first supply air duct 43 may connect the first drum 13 and the first discharge port 425.

The second discharge port 427 may protrude from the blowing duct 420 to a lateral side. The second supply air duct 45 may connect the third cabinet 30 and the second discharge port 427.

The third discharge port 426 may protrude upwardly from the blowing duct 420. The third supply air duct 48 may connect the tub 23 and the third discharge port 426.

The fourth discharge port 428 may protrude from the blowing duct 420 to the lateral side. The fourth discharge port 428 may protrude in a direction opposite to the second discharge port 427. The dehumidification duct 46 may be connected to the fourth discharge port 428.

A rotating body 71 of a switching device 70 (see FIG. 6) which will be described later may be disposed in the blowing duct 420. The rotating body 71 may be connected to a driving motor 711 to be rotated in the blowing duct 420. The air blown by the fan 60 may flow into at least any one of the first supply air duct 43, the second supply air duct 45, the third supply air duct 48, and the dehumidification duct 46 by the operation of the switching device 70.

The laundry treating apparatus A according to the present disclosure may also include a separate distribution device in addition to the switching device 70 which will be described later. The distribution device may distribute the air blown by the fan 60 to each of the dryer 1, the washing machine 2, the

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refresher 3, and the dehumidification duct 46. That is, the air blown by the fan 60 may be supplied at the same time to each of the dryer 1, the washing machine 2, the refresher 3, and the dehumidification duct 46. The distribution device may be disposed in the blowing duct 420 or may be disposed on a distribution passage connected to the blowing duct 420. The distribution device may be a valve. The distribution device may include an actuator and a switching damper. The laundry treating apparatus A may include both the switching device 70 and the distribution device at the same time, may include only the switching device 70, or may include only the distribution device.

The hot air flowing into the first drum 13 through the first supply air duct 43 may dry the laundry placed in the first drum 13, and then may flow into the heat exchange channel 42 through the first exhaust air duct 41. The first drum 13 may include a front cover 131 disposed on a front side, and a rear cover 132 disposed on a rear side. The first supply air duct 43 may be connected to the rear cover 132, and the first exhaust air duct 41 may be connected to the front cover 131.

The first exhaust air duct 41 may connect the first drum 13 and the heat exchange channel 42. The first exhaust air duct 41 may extend upwardly from the first drum 13 to be connected to the heat exchange channel 42.

A first inlet port 421 connected to the first exhaust air duct 41 may be formed at an inlet end 42A of the heat exchange channel 42. The first inlet port 421 may extend downwardly from the heat exchange channel 42. The first inlet port 421 may extend downwardly from the second exhaust air duct 44. The first inlet port 421 may extend downwardly from a first duct section 444.

The first inlet port 421 may protrude downwardly from a lower surface of the second exhaust air duct 44. The first exhaust air duct 41 may connect the first drum 13 and the first inlet port 421. The first exhaust air duct 41 may connect the first drum 13 and the second exhaust air duct 44. The air in the first exhaust air duct 41 may join the air in the second exhaust air duct 44 to flow into the heat exchange channel 42.

The laundry treating apparatus A may include a first opening and closing valve 41a disposed in the first exhaust air duct 41. The first opening and closing valve 41a may control a flow rate of air in the first exhaust air duct 41. The first opening and closing valve 41a may block an air flow in the first exhaust air duct 41. The first opening and closing valve 41a may block the air flow in the first exhaust air duct 41 when the hot air is not supplied into the first drum 13 through the first exhaust air duct 41.

The hot air flowing into third cabinet 30 through the second supply air duct 45 may dry the laundry placed in the third cabinet 30, and then may flow into the heat exchange channel 42 through the second exhaust air duct 44. The second supply air duct 45 and the second exhaust air duct 44 may be connected to the third rear panel 302 of the third cabinet 30.

The second exhaust air duct 44 may connect the third cabinet 30 and the heat exchange channel 42. The second exhaust air duct 44 may extend downwardly from an upper portion of the third cabinet 30 to be connected to the heat exchange channel 42.

The second exhaust air duct 44 may include a first duct section 444 disposed between the heating device 50 and the second cabinet 20. A first duct section 444 may be disposed between a base plate 575 (see FIG. 6) to be described later and the first upper panel 103 of the first cabinet 10. The first

duct section 444 may extend forwardly and rearwardly in a separation space 55 (see FIG. 6) which will be described later.

The inlet end 42A of the heat exchange channel 42 may have a second inlet port 422 connected to the second exhaust air duct 44. The second inlet port 422 may extend downwardly from the heat exchange channel 42. The second inlet port 422 may extend downwardly from the case 570 of the heating device 50.

The laundry treating apparatus A may include a second opening and closing valve 44a disposed in the second exhaust air duct 44. The second opening and closing valve 44a may control a flow rate of air in the second exhaust air duct 44. The second opening and closing valve 44a may block an air flow in the second exhaust air duct 44. The second opening and closing valve 44a may block the air flow in the second exhaust air duct 44 when the hot air is not supplied into the third drum 30 through the second exhaust air duct 45. The second opening and closing valve 44a may be disposed in the first duct section 444.

The hot air flowing into the tub 23 through the third supply air duct 48 may dry the laundry placed in the tub 23, and then may flow into the heat exchange channel 42 through the third exhaust air duct 47.

The third exhaust air duct 47 may connect the tub 23 and the heat exchange channel 42. The third exhaust air duct 47 may extend downwardly from an upper portion of the tub 23 to be connected to the heat exchange channel 42.

A third inlet port 424 connected to the third exhaust air duct 47 may be formed at the inlet end 42A of the heat exchange channel 42. The third inlet port 424 may extend upwardly from the heat exchange channel 42. The third inlet port 424 may extend upwardly from the case 570 of the heating device 50.

The laundry treating apparatus A may include a third opening and closing valve 47a disposed in the third exhaust air duct 47. The third opening and closing valve 47a may control a flow rate of air in the third exhaust air duct 47. The third opening and closing valve 47a may block an air flow in the third exhaust air duct 47. The third opening and closing valve 47a may block the air flow in the third exhaust air duct 47 when the hot air is not supplied into the tub 23 through the third supply air duct 48.

The hot air flowing through the dehumidification duct 46 may be supplied to the outside of the laundry treating apparatus A through an outlet 462 that is opened forward. The hot air flowing through the dehumidification duct 46 may be supplied to an indoor space through the outlet 462.

The dehumidification duct 46 may include a second duct section 461 disposed between the heating device 50 and the first cabinet 10. The second duct section 461 may be disposed between the base plate 575 (see FIG. 6) to be described later and the first upper panel 103 of the first cabinet 10. The second duct section 461 may extend forwardly and rearwardly in the separation space 55 (see FIG. 6) which will be described later.

An outside air inlet port 423 connected to the heat exchange channel 42 may be formed at the inlet end 42A of the heat exchange channel 42. The outside air inlet port 423 may extend forwardly from the heat exchange channel 42. The outside air inlet port 423 may extend forwardly from the second exhaust air duct 44.

The outside air inlet port 423 may protrude forwardly from one side of the second exhaust air duct 44. The outside air inlet port 423 may allow the indoor space and the second exhaust air duct 44 to communicate with each other. The air in the outside air inlet port 423 may join the air in the second

exhaust air duct 44 to flow into the heat exchange channel 42. The outside air inlet port 423 may have an outside air inlet 49 that is opened forward. The air drawn in through the outside air inlet 49 may join the air in the second exhaust air duct 44 to flow into the heat exchange channel 42.

The laundry treating apparatus A may include a fourth opening and closing valve 49a disposed at the outside air inlet port 423. The fourth opening and closing valve 49a may control a flow rate of air in the outside air inlet port 423. The fourth opening and closing valve 49a may block an air flow in the outside air inlet port 423. The fourth opening and closing valve 49a may block the air flow in the outside air inlet port 423 when the hot air is not supplied to the dehumidification duct 46.

Hereinafter, the heating device 50 will be described with reference to FIGS. 5 to 7. The description of the heating device 50, which will be described below with reference to FIGS. 5 to 7, may be applied commonly to all embodiments of the laundry treating apparatus described with reference to FIGS. 1 to 55. Even in this case, however, depending on relative arrangement positions of the laundry treating machines included in the laundry treating apparatus, a shape of the case 570, a shape of the blowing duct 420, a placement of a drum motor 133, an opening direction of the discharge ports 425, 426, 427, and 428, an opening direction of the inlet ports 421, 422, 423, and 424, and the like may vary.

Referring to FIG. 5, the heating device 50 may include the case 570, the compressor 51, the condenser 52, the evaporator 53, and the expansion device 54.

The compressor 51, the condenser 52, the evaporator 53, and the expansion device 54 may be connected by a refrigerant passage. The heat pump device may include the compressor 51, the condenser 52, the evaporator 53, and the expansion device 54.

The drain pan 531 may be disposed under the evaporator 53.

The case 570 may provide a space in which the compressor 51, the condenser 52, the fan 60, and the evaporator 53 are disposed. The compressor 51, the condenser 52, the fan 60, the evaporator 53, and the switching device 70 may be disposed in the case 570.

The case 570 may include a cover 573, a base plate 575 spaced from a lower side of the cover 573, a front wall 574 disposed in front of the condenser 53, a first side wall 571 disposed on one side of the condenser 53, and a second side wall 572 disposed on the other side of the condenser 53.

The cover 573 may include a shielding wall 573a for separating a space, in which the drain pump 26 is disposed, from the heat exchange channel 42. The shielding wall 573a may extend horizontally. The drain pump 26 may be disposed on one side of the shielding wall 573a. The heat exchange channel 42 may be disposed on the other side of the shielding wall 573a.

The condenser 52, the fan 60, and the evaporator 53 may be disposed on an upper side of the base plate 575.

The heat exchange channel 42 may be a space surrounded by the base plate 575, the front wall 574, the first side wall 571, and the second side wall 572. The cover 573 may not be provided, in which case an upper portion of the heat exchange channel 42 may be open. The heat exchange channel 42 may communicate with the fan 60 disposed at a rear side of the heating device 50. Air in the heat exchange channel 42 may be blown by the fan 60. The heat exchange channel 42 may be referred to as a "heating passage." The condenser 52 and the evaporator 53 may be disposed in the heat exchange channel 42. The heat exchange channel 42 may refer to a partially open space.

The heat exchange channel 42 may be a space surrounded by the cover 573, the base plate 575, the front wall 574, the first side wall 571, and the second side wall 572. The cover 573 may cover the upper portion of the heat exchange channel 42. The heat exchange channel 42 may refer to a space between the cover 573 and the base plate 575. The cover 573, the base plate 575, the front wall 574, the first side wall 571, and the second side wall 572 may form a “heating duct” surrounding the heat exchange channel 42. The heating duct may communicate with the fan 60, and air in the heating duct may be blown by the fan 60. The condenser 52 and the evaporator 53 may be disposed in the heating duct. The heat exchange channel 42 may refer to a duct shielded in all directions.

The heat exchange channel 42 may be formed in the heating device 50. The heat exchange channel 42 may be a portion of the inner space of the heating device 50.

The third inlet port 424 may be disposed in front of the heat exchange channel 42. The third inlet port 424 may cover the front side of the heat exchange channel 42. The third inlet port 424 may be connected to the front wall 574. The third inlet port 424 may be formed in the case 570. The third exhaust air duct 47 may be inserted into the first inlet port 421 and may be fixed thereto.

The heating device 50 may be disposed in the machine room S. The controller 81 may control the operation of components disposed in the machine room S. The machine room S may have a space in which the PCB is disposed, and the controller 81 may be mounted in the PCB.

The heating device 50 may include a steam generator 502 for generating steam and a dehumidifier 504 for removing moisture from air flowing through the heat exchange channel 42.

The steam generator 502 may generate steam by heating water. The steam generator 502 may be disposed between the condenser 52 and the fan 60. The steam generated by the steam generator 502 may be pressurized by the fan 60 to be supplied to each of the dryer 1, the washing machine 2, and the refresher 3. The steam generated by the steam generator 502 may be supplied to each of the first drum 13, the second drum 24, and the inner space 33 of the refresher 3.

The dehumidifier 504 may dehumidify air under room temperature conditions (about 25 degrees Celsius). The dehumidifier 504 may dehumidify air by using desiccant cooling. The dehumidifier 504 may be filled with zeolite. The zeolite filled in the dehumidifier 504 may be replaced periodically.

The dehumidifier 504 may be disposed between the condenser 52 and the fan 60. The dehumidifier 504 may dehumidify the air flowing through the heat exchange channel 42 even when the compressor 51 is not in operation.

The switching device 70 may control the direction of air blown by the fan 60. The switching device 70 may be disposed in the machine room S.

The drain pump 26 may be disposed in the heating device 50. The drain pump 26 may be disposed in the machine room S. The shielding wall 573a may be disposed between the drain pump 26 and the heat exchange channel 42.

Referring to FIG. 6, the switching device 70 may control the supply of hot air to the dryer 1, the washing machine 2, the refresher 3, or the dehumidification duct 46.

The switching device 70 may include the rotating body 71 rotatably mounted in the blowing duct 420. The switching device 70 may include the driving motor 711 that rotates the rotating body 71. The rotating body 71 may be rotated by the driving motor 711 in the blowing duct 420. The fan 60 may be disposed in the rotating body 71.

The driving motor 711 may rotate the rotating body 71. A driving gear 712 may be fixed to the rotating shaft of the driving motor 711. The driving gear 712 may be a pinion gear or a spur gear.

A driven gear 713 may be rotated in engagement with the driving gear 712. The driven gear 713 may be fixed to the rotating body 71 or may be integrally formed with the rotating body 71. The driven gear 713 may be a ring gear. The driven gear 713 may be a ring-shaped rack.

The driving gear 712 and the driven gear 713 may be geared with each other. By the rotation of the driving gear 712, the driven gear 713 may be moved in a circumferential direction with respect to the rotational axis of the fan 60. When the driven gear 713 is moved in a circumferential direction, the rotating body 71 having the driven gear 713 fixed thereto may also be moved in the circumferential direction with respect to the rotational axis of the fan 60.

The driving motor 712 may be disposed outside of the blowing duct 420. One surface (e.g., rear surface) of the blowing duct 420 that faces the rotating body 71 may have a cut-out portion 714. The cut-out portion 714 may be formed at a position corresponding to the driven gear 713. A portion of the driving gear 712 may be inserted into the cut-out portion 714. The driving gear 712 and the driven gear 713 may be geared with each other in the cut-out portion 714.

A rotating shaft of the driving motor 711 may be disposed side by side with the rear surface of the blowing duct 420. Accordingly, a volume occupied by the driving motor 711 and the driving gear 712 in the front-rear direction may be reduced.

The driving motor 711 may be a motor capable of controlling the position, angle, and direction of rotation. For example, the driving motor may be a Brushless Direct Current (BLDC) motor. Alternatively, the driving motor 711 may be a step motor. The driving motor 711 may be electrically connected to the control panel 8 and/or the controller 81. Rotation of the driving motor 711 may be controlled by an electrical signal transmitted from the control panel 8 and/or the controller 81 to the driving motor 711. The control panel 8 and/or the controller 81 may control a hot air supplying direction by controlling the rotation angle of the driving motor 711.

Meanwhile, the driving motor 711 may rotate the rotating body 71 by various known methods used by the motor for rotating the rotating body. For example, the driving motor 711 may rotate the rotating body 71 by using a belt-pulley method, or by using a plurality of gears that are geared with each other, or the rotating shaft of the motor may be rotated together with the rotating body.

The blowing duct 420 may be connected to the first supply air duct 43, the second supply air duct 45, the third supply air duct 48, and the dehumidification duct 46.

The blowing duct 420 may be connected to a motor mount 564, to which a fan motor rotating the fan 60 is fixed. The motor mount 64 may be disposed on a rear surface of the blowing duct 420. The driving motor 711, the driving gear 712, and the driven gear 713 may be disposed radially outwardly from the motor mount 64.

The blowing duct 420 may include the first discharge port 425, the second discharge port 427, the third discharge port 426, and the fourth discharge port 428.

The first supply air duct 43 may be connected to the first discharge port 425. The first supply air duct 43 may be inserted into the first discharge port 425. The first discharge port 425 may be disposed facing downward in the blowing

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duct 420. The first discharge port 425 may provide a supply air hole that is vertically open.

The second supply air duct 45 may be connected to the second discharge port 427. The second supply air duct 45 may be inserted into the second discharge port 427. The second discharge port 427 may be disposed facing toward the lateral side in the blowing duct 420. The second discharge port 427 may provide a supply air hole that is horizontally open.

The third supply air duct 48 may be connected to the third discharge port 426. The third supply air duct 48 may be inserted into the third discharge port 426. The third discharge port 426 may be disposed facing upwardly in the blowing duct 420. The third discharge port 426 may provide a supply air hole that is vertically open.

The dehumidification duct 46 may be connected to the fourth discharge port 428. The dehumidification duct 46 may be inserted into the fourth discharge port 428. The fourth discharge port 428 may be disposed facing a direction opposite to the second discharge port 427. The fourth discharge port 428 may be disposed facing toward the lateral side in the blowing duct 420. The fourth discharge port 428 may provide a supply air hole that is horizontally open.

The heating device 50 may be connected to each of the first exhaust air duct 41, the second exhaust air duct 44, the third exhaust air duct 47, and the outside air inlet port 423. The third exhaust air duct 47 and the outside air inlet port 423 may be connected to the heating device 50 via the second exhaust air duct 44.

The heating device 50 may include the third inlet port 424 connected to the third exhaust air duct 47, and the second inlet port 422 connected to the second exhaust air duct 44. The third inlet port 424 may extend upwardly, and the second inlet port 422 may extend downwardly. The third inlet port 424 and the second inlet port 422 may protrude from the case 570 of the heating device 50.

Air drawn into the heating device 50 through the first exhaust air duct 41, the second exhaust air duct 44, the third exhaust air duct 47, and the outside air inlet port 423 may pass through the heat exchange channel 42 and flow into the fan 60 by the suction force of the fan 60.

Referring to FIG. 7, the air flowing through the first exhaust air duct 41, the second exhaust air duct 44, the third exhaust air duct 47, and the outside air inlet port 423 may meet in the heat exchange channel 42 to flow into the fan 60.

The first inlet port 421 connected to the first exhaust air duct 41 may protrude downwardly from the second exhaust air duct 44. The air in the first exhaust air duct 41 may join the air in the second exhaust air duct 44 to flow into the heat exchange channel 42.

The outside air inlet port 423 communicating with the indoor space may protrude forward from the second exhaust air duct 44. The air flowing into the outside air inlet port 423 through the outside air inlet 49 may join the air in the second exhaust air duct 44 to flow into the heat exchange channel 42.

The air flowing into the heat exchange channel 42 through the second exhaust air duct 44 may join the air flowing into the heat exchange channel 42 through the third exhaust air duct 47 to flow to the fan 60.

The heating device 50 may include a first support plate 576 extending downwardly from the base plate 575, and a second support plate 577 extending downwardly from the base plate 575 and spaced apart from the first support plate 576.

A separation space 55 may be formed between the first support plate 576 and the second support plate 577.

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The second exhaust air duct 44, the first supply air duct 43, and the dehumidification duct 46 may be disposed in the separation space 55. The second exhaust air duct 44, the first supply air duct 43, and the dehumidification duct 46 may be formed between the first support plate 576 and the second support plate 577. The second exhaust air duct 44, the first supply air duct 43, and the dehumidification duct 46 may be disposed under the base plate 575.

The dehumidification duct 42 may be disposed under the base plate 575 and may extend forwardly to discharge the air forwardly through a hot air outlet 462. When an opening member 463 (see FIG. 10) which will be described later opens a front portion of the separation space 55, the air discharged through the hot air outlet 462 may be discharged forwardly from the laundry treating apparatus A.

A portion of the second exhaust air duct 44 disposed under the base plate 575 may be defined as the first duct section 444.

A portion of the dehumidification duct 46 disposed under the base plate 575 may be defined as the second duct section 461.

Hereinafter, a method of controlling a blowing direction of air by the switching device 70 will be described with reference to FIG. 8. In FIG. 8, (a) is a diagram illustrating an example of operation when air is supplied to the third supply air duct 48 by the operation of the switching device 70, and (b) is a diagram illustrating an example of operation when air is supplied to the third supply air duct 48 and the second supply air duct 45 at the same time by the operation of the switching device 70. The description of the switching device 70, which will be described with reference to FIG. 8, may be applied commonly to all embodiments of the laundry treating apparatus described with reference to FIGS. 1 to 20. Even in this case, however, an opening direction of the discharge ports 425, 426, 427, and 428, an arrangement and connection structure of the exhaust air ducts 41, 44, and 47, and the like may vary depending on the relative arrangement of the laundry treating machines included in the laundry treating apparatus.

The fan 60 may be a Sirocco fan. The fan 60 may rotate to blow air in a direction perpendicular to the rotating shaft 61. The fan 60 may blow air in a direction coming into contact with the rotation direction. The air blown by the fan 60 may be concentrated in a predetermined range of angles relative to the rotating shaft 61.

The fan 60 may include the rotating shaft 61 coupled to the fan motor and rotated thereby, a plurality of blades 62 spaced apart in a radially outward direction of the rotating shaft 61, and an outer body 63 coupled to the blades 62 and extending in a rotation direction of the fan 60. The outer body 63 may have an annular shape. The rotating shaft 61 and the blades 62 may be connected by the outer body 63. When the rotating shaft 61 rotates, the outer body 63 and the blades 62 may also be rotated together with the rotating shaft 61.

The rotating body 71 may include a rotating plate 71a having the driven gear 713, a scroll 71b coupled to the rotating plate 71a, and a shaft through hole 71c formed in the rotating plate 71a.

The rotating plate 71a may have a disk shape and may be disposed behind the fan 60. The driven gear 713 may be formed in an annular shape on one side surface of the rotating plate 71a. The driven gear 713 may be formed integrally with the rotating plate 71a. When the driven gear 713 is rotated in engagement with the driving gear 711, the rotating plate 71a may be rotated in the same direction as the rotation direction of the fan 60. The rotating plate 71a may

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have the shaft through hole **71c**, through which the rotating shaft **61** passes. The rotating shaft **61** may pass through the shaft through hole **71c** to be coupled to the fan motor.

The fan **60** may be disposed between the heat exchange channel **42** and the rotating plate **71a**. That is, the rotating plate **71a** may be disposed behind the heat exchange channel **42** and the fan **60**. The rotating plate **71a** may be disposed behind the fan **60** in the blowing duct **420**.

The scroll **71b** may be integrally formed with the rotating plate **71a**. When the rotating plate **71a** is rotated, the scroll **71b** may also be rotated together. The scroll **71b** may extend in the rotation direction of the fan **60**. The scroll **71b** may be disposed to surround the fan **60**. The fan **60** may be disposed in the scroll **71b**.

The scroll **71b** may include a blowing channel **72**. The blowing channel **72** may be a cut-out portion of an outer circumferential surface of the scroll **71b**. The scroll **71b** may cover the outside of the fan **60**, and the blowing channel **72** may be an outer region of the fan **60** which is not covered by the scroll **71b**. The air blown by the fan **60** may be discharged to the outside of the blowing duct **420** through the blowing channel **72**.

When the scroll **71b** is rotated by the rotation of the rotating plate **71a**, the blowing channel **72** may be changed in position. That is, by the rotation of the scroll **71b**, the position of the blowing channel **72** may be changed relative to the rotating shaft **61**. By rotating the scroll **71b** to change the position of the blowing channel **72**, the driving motor **711** may control the direction of air discharged from the blowing duct **420**.

A longitudinal section of the blowing duct **420** may have a square shape. Accordingly, by the rotation of the scroll **71b**, interference between the scroll **71b** and the blowing duct may be avoided.

The blowing duct **420** may include a first wall **420a** disposed on an upper side of the fan **60**, a second wall **420b** disposed on a lower side of the fan **60**, a third wall **420c** disposed on one side of the fan **60**, and a fourth wall **420d** disposed on the other side of the fan **60**.

The first discharge port **425** may protrude upwardly from the second wall **420b**. The second discharge port **427** may protrude toward the lateral side from the third wall **420c**. The third discharge port **426** may protrude upwardly from the first wall **420a**. The fourth discharge port **428** may protrude toward the lateral side from the fourth wall **420d**.

Referring to (a) of FIG. 8, the driving motor **711** may rotate the rotating body **71** by a first angle, and when the rotating body **71** is rotated by the first angle, the air blown by the fan **60** may be supplied to the third supply air duct **48**. In this case, the blowing channel **72** may communicate with only an inner space of the third discharge port **426**. Accordingly, the air blown by the fan **60** may be supplied only to the washing machine **2**.

Referring to (b) of FIG. 8, the driving motor **711** may rotate the rotating body **71** by a second angle, and when the rotating body **71** is rotated by the second angle, the air blown by the fan **60** may be supplied to the third supply air duct **48** and the second supply air duct **45** at the same time. In this case, the blowing channel **72** may communicate with an inner space of the third discharge port **426** and an inner space of the second discharge port **427**. Accordingly, the air blown by the fan **60** may be supplied to the washing machine **2** and the refresher **3** at the same time. The blowing channel **72** may include a first blowing channel **72a** communicating with the inner space of the third discharge port **426**, and a second blowing channel **72b** communicating with the inner space of the second discharge port **427**. The air blown by the

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fan **60** may be supplied to the washing machine **2** through the first blowing channel **72a**. The air blown by the fan **60** may be supplied to the refresher **3** through the second blowing channel **72b**.

A user may control the switching device **70** by inputting a signal to the control panel **8**. Once the signal is input to the control panel **8**, the signal may be transmitted to the driving motor **711**, to control a rotation angle of the driving motor **711**. For example, when the user inputs, to the control panel **8**, a signal for supplying hot air to the washing machine **2**, the driving motor **711** may rotate the rotating body **71** so that the blowing channel **72** may be moved to a position as illustrated in (a) of FIG. 8. For example, when the user inputs, to the control panel **8**, a signal for supplying hot air to the washing machine **2** and the refresher **3** at the same time, the driving motor **711** may rotate the rotating body **71** so that the blowing channel **72** may be moved to a position as illustrated in (b) of FIG. 8.

Hereinafter, a structure for supplying hot air and steam by the heating device **50** to each of laundry treating machines **1**, **2**, and **3** will be described with reference to FIG. 9.

The heating device **50** may be disposed over the first upper panel **103**. The heating device **50** may be disposed in the cabinet **20** of the washing machine **2**.

The switching device **70** may be disposed over the first upper panel **103**. The switching device **70** may be disposed in the cabinet **20** of the washing machine **2**.

The blowing duct **420** may be connected to the first supply air duct **43**, the second supply air duct **45**, and the third supply air duct **48**.

The first supply air duct **43** may extend downwardly from the blowing duct **420**. The first supply air duct **43** may be connected to the first drum **13**. The first supply air duct **43** may pass through the separation space **55** to extend into the first cabinet **10**.

The second supply air duct **45** may extend from the blowing duct **420** to a lateral side. The second supply air duct **45** may be connected to the refresher **3**.

The third supply air duct **48** may extend upwardly from the blowing duct **420**. The third supply air duct **48** may be connected to the tub **23**.

The laundry treating apparatus **A** may include a third steam supply pipe **502c** connecting the steam generator **502** and the tub **23**; and a second steam supply pipe **502b** connecting the steam generator **502** and the refresher **3**.

The steam generator **502** may be disposed in the heating device **50**. The steam generated by the steam generator **502** may be sprayed into the tub **23** through the third steam supply pipe **502c**. The steam generated by the steam generator **502** may be sprayed into the inner space **33** of the refresher **3** through the second steam supply pipe **502b**.

The second steam supply pipe **502b** may extend from the steam generator **502** to the lateral side, to be connected to the refresher **3**.

The third steam supply pipe **502c** may extend upwardly from the steam generator **502** to be connected to the tub **23**.

The steam generated by the steam generator **502** may flow to each of the second drum **24** and the inner space **33** of the refresher **3** by the blowing force of the fan **60**. That is, by pressurizing the steam generated by the steam generator **502**, the fan **60** may cause the steam to flow to the second drum **24** and the inner space **33** of the refresher **3**.

A valve for controlling an amount of steam flow may be provided for each of the second steam supply pipe **502b** and the third steam supply pipe **502c**. By adjusting an opening degree of the valve, the controller **81** may adjust the amount

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of steam supplied to each of the second drum **24** and the inner space **33** of the refresher **3**.

The laundry treating apparatus **A** may include a second steam generator **506** disposed in the dryer **1**. The laundry treating apparatus **A** may include a plurality of steam generators **502** and **506**. The steam generator **502** disposed in the heating device **50** may be referred to as a “first steam generator,” and the steam generator **506** disposed in the dryer **1** may be referred to as a “second steam generator.”

The laundry treating apparatus **A** may include a first steam supply pipe **506a** connecting the second steam generator **506** and the first drum **13**, and a fourth steam supply pipe **506b** connecting the second steam generator **506** and the refresher **3**.

The second steam generator **506** may be disposed in the first cabinet **10**. The steam generated by the second steam generator **506** may be sprayed into the first drum **13** through the first steam supply pipe **506a**. The steam generated by the second steam generator **506** may be sprayed into the inner space **33** of the refresher **3** through the fourth steam supply pipe **506b**.

The fourth steam supply pipe **506b** may extend from the second steam generator **506** in a lateral direction, to be connected to the refresher **3**.

A valve for controlling an amount of steam flow may be provided for each of the first steam supply pipe **506a** and the fourth steam supply pipe **506b**. By adjusting an opening degree of the valve, the controller **81** may adjust the amount of steam supplied to each of the first drum **13** and the inner space **33** of the refresher **3**.

The washing machine **2** may be connected to a cold water supply pipe **278** (see FIG. **11**) for supplying cold water to the tub **23**, and a hot water supply pipe **279** (see FIG. **11**) for supplying hot water to the tub **23**.

The steam generators **502** and **506** may be connected to a third water supply pipe **277** branching off from the cold water supply pipe **278** or the hot water supply pipe **279**. The steam generators **502** and **506** may be connected to the third water supply pipe **277** branching off from the hot water supply pipe **279**.

The steam generators **502** and **506** may be supplied with hot water through the third water supply pipe **277**. The steam generators **502** and **506** may generate steam by heating the water supplied through the third water supply pipe **277**.

Hereinafter, a connection structure of the first cabinet **10** and the second cabinet **20** will be described with reference to FIG. **10**.

The base plate **575** of the heating device **50** may be spaced from the upper side of the first upper panel **103** of the dryer **1**. The separation space **55** may be formed between the base plate **575** and the first upper panel **103**.

The second side panels **205** and **206** of the washing machine **2** may protrude downwardly below the base plate **575**. The second side panels **205** and **206** may be referred to as the “second side walls.”

The second side walls **205** and **206** may include a second outer wall **205** forming one side surface of the laundry treating apparatus **A**, and a second inner wall **206** facing the third cabinet **30**.

The heating device **50** may include the first support plate **576** extending downwardly from the base plate **575**. The first support plate **576** may be disposed inside the second outer wall **205**.

The heating device **50** may include the second support plate **577** extending downwardly from the base plate **575**. The second support plate **577** may be disposed inside the second inner wall **206**.

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The base plate **575** and the support plates **576** and **577** may be integrally formed with each other. The second outer wall **205** and the first support plate **576** may be integrally formed with each other. The second inner wall **206** and the second support plate **577** may be integrally formed with each other.

The second side walls **205** and **206** may include first protrusions **205a** and **206a** protruding downwardly. The second outer wall **205** may include a first outer protrusion **205a** protruding downwardly. The second inner wall **206** may include a first inner protrusion **206a** protruding downwardly. The first protrusions **205a** and **206a** may protrude downwardly from a lower end of the second side walls **205** and **206**.

The first cabinet **10** may include first recesses **103a** into which the first protrusions **205a** and **206a** are inserted. The first recesses **103a** may be formed in the first upper panel **103**. The first recesses **103a** may be formed at positions vertically corresponding to the first protrusions **205a** and **206a**.

The support plates **576** and **577** may include second protrusions **576a** and **577a** protruding downwardly. The first support plate **576** may include a second outer protrusion **576a** protruding downwardly. The second support plate **577** may include a second inner protrusion **577a** protruding downwardly. The second protrusions **576a** and **577a** may protrude downwardly from a lower end of the support plates **576** and **577**.

The first cabinet **10** may include second recesses **103b** into which the second protrusions **576a** and **577a** are inserted. The second recesses **103b** may be formed in the first upper panel **103**. The second recesses **103b** may be formed at positions vertically corresponding to the second protrusions **576a** and **577a**.

The first protrusions **205a** and **206a** and the second protrusions **576a** and **577a** may be referred to as “protrusions.” The first recesses **103a** and the second recesses **103b** may be referred to as “recesses.”

The first cabinet **10** and the second cabinet **20** may be coupled to each other as the protrusions **205a**, **206a**, **576a**, and **577a** are inserted into the recesses **103a** and **103b**.

The machine room **S** may be formed on the upper side of the base plate **575**. Accordingly, the machine room **S** may be spaced from the upper side of the first upper panel **103**.

The separation space **55** may be formed between the first support plate **576** and the second support plate **577**.

The second exhaust air duct **44**, the dehumidification duct **46**, and the first exhaust air duct **41** may be disposed in the separation space **55**. The second exhaust air duct **44**, the dehumidification duct **46**, and the first exhaust air duct **41** may be disposed between the first upper panel **103** and the base plate **575**.

The first supply air duct **43** may protrude downwardly toward the first upper panel **103**. The first supply air duct **43** may include a first connection duct **43a** extending downwardly from the blowing duct **230**, and a second connection duct **43b** connected to the first drum **13**.

The first connection duct **43a** may be disposed in the separation space **55**.

The second connection duct **43b** may extend downwardly from the first upper panel **103** to be connected to the first drum **13**. The second connection duct **43b** may have a first insertion hole **43s**, which is formed on the inside thereof, and into which the first connection duct **43a** is inserted.

The first connection duct **43a** is inserted into the first insertion hole **43s** to be fixed to the second connection duct **43b**.

The first inlet port **421** may be connected to the second exhaust air duct **44**, and may be disposed in the separation space **55**.

The first inlet port **421** may be inserted into a second insertion hole **41s**, formed on the inside of the first exhaust air duct **41**, to be fixed to the first exhaust air duct **41**.

When the first cabinet **10** and the second cabinet **20** are assembled, the first connection duct **43a** and the first inlet port **421** are inserted into the first insertion hole **43s** and the second insertion hole **41s**, respectively, to be fixed thereto, thereby facilitating the assembly and alignment of the first cabinet **10** and the second cabinet **20**.

The laundry treating apparatus **A** may include an auxiliary panel **58** disposed behind the separation space **55**.

The auxiliary panel **58** may shield a rear side of the separation space **55**. The auxiliary panel **58** may be connected to the first rear panel **102** and the second rear panel **202**. The first rear panel **102** and the second rear panel **202** may be connected to each other by the auxiliary panel **58**.

The auxiliary panel **58** may include a duct through hole **59**, through which the second exhaust air duct **44** passes. The second exhaust duct **44** may extend into the separation space **55** by passing through the auxiliary panel **58**.

The auxiliary panel **58** may be coupled to the second side panels **205** and **206** and the support plates **576** and **577**.

The opening member **463** may shield the front side of the separation space **55**. The opening member **463** may be disposed in front of the dehumidification duct **46** and the outside air inlet port **423**.

The laundry treating apparatus **A** may include a hinge **463a** connected to the opening member **463**, and a motor **463b** rotating the hinge **463a**.

The hinge **463a** may be rotatably connected to the second front panel **201**. The hinge **46a** may extend in a left-right direction. The opening member **46** may be rotated in a front-rear direction with the hinge **463a** serving as a rotational axis. The opening member **463** may be integrally formed with the hinge **463a**. When the motor **463b** rotates the hinge **463a**, the opening member **463** may be rotated together with the hinge **463a**. The opening member **463** may rotate forward to open the front side of the separation space **55**. Once the opening member **463** opens the front side of the separation space **55**, outside air of the laundry treating apparatus **A** may flow into the heat exchange channel **42** through the outside air inlet port **423**, and the air passing through the dehumidification duct **46** may be discharged to the outside of the laundry treating apparatus **A**.

The controller **81** may be electrically connected to the motor **463b**. The controller **81** may control the operation of the motor **463b**. The user may open the separation space **55** by manipulating the control panel **8**.

The drawer **14** may be disposed in the first cabinet **10** and may be disposed under the first upper panel **103**.

The drawer **14** may be disposed at a front side of the first rear panel **102** and may be disposed in front of the first supply air duct **43**. As the drawer **14** is disposed in front of the first supply air duct **43**, interference between the first supply air duct **43** and the drawer **14** may be avoided.

The drawer **14** may be disposed below the first exhaust air duct **41**. As the drawer **14** is disposed below the first exhaust air duct **41**, interference between the first exhaust air duct **41** and the drawer **14** may be avoided. The first exhaust air duct **41** may extend upwardly from the first drum **13** and may avoid interference with the drawer **14** by bypassing the drawer **14** to the one side thereof. The first exhaust air duct

41 may curvedly extend along a vertical direction and may extend along the vertical direction from the outside of the drawer **14**.

The first exhaust air duct **41** may include a port insertion part **41a** coupled to the first inlet port **421**. The port insertion part **41** may be connected to the first upper panel **103**. The port insertion part **41a** may be disposed above the drawer **14**.

The detergent supply device **15** may be disposed on one side of the drawer **14**. The detergent supply device **15** may be horizontally spaced apart from the drawer **14**. The detergent supply device **15** may be disposed under the first upper panel **103** and may be disposed in the first cabinet **10**.

The detergent supply device **15** may be connected to the detergent pump **152** by the detergent supply pipe **153**. The detergent supply device **15** may be connected to the third water supply pipe **275** branching off from the first water supply pipe **271**. The detergent stored in the detergent supply device **15** may be dissolved in water supplied through the third water supply pipe **275**. When the detergent pump **152** is operated, the detergent stored in the detergent supply device **15** may be fed into the detergent pump **152** through the detergent supply pipe **153**. The detergent pump **152** may be connected to the detergent box **28** by a detergent transfer pipe **154**. The detergent pump **152** may extrude the detergent, fed into the detergent pump **152**, through the detergent transfer pipe **154** into the detergent box **28**.

The second cabinet **20** may include a first drain hole **579**, through which the drain pipe **532** passes. The first cabinet **10** may include a second drain hole **103c**, through which the drain pipe **532** passes. The first drain hole **579** may be open at the top and bottom in a boss protruding downwardly from the base plate **575**.

The drain pipe **532** may extend downwardly from the drain pan **531** to pass through the first drain hole **579**. The drain pipe **532**, having passed through the first drain hole **579**, may extend downwardly to pass through the second drain hole **103c**. The drain pipe **532**, having passed through the second drain hole **103c**, may extend toward the inner space **148** of the drawer **14**.

The drain pipe **532**, having passed through the first drain hole **579**, may be disposed adjacent to the dehumidification duct **46** for heat exchange with the air flowing in the dehumidification duct **46**. The drain pipe **532** may come into contact with the dehumidification duct **46**. However, it is also possible that the drain pipe **532** is spaced apart from the dehumidification duct **46**, and a separate heat transfer member may be disposed between the drain pipe **532** and the dehumidification duct **46**. After passing through the first drain hole **579**, the drain pipe **532** may come into contact with the dehumidification duct **46**. After coming into contact with the dehumidification duct **46**, the drain pipe **532** may pass through the second drain hole **130c**. The condensate in the drain pipe **532** may pass through the first drain hole **579**, and then may be heat exchanged with the air flowing in the dehumidification duct **46**. The air flowing in the dehumidification duct **46** may be cooled by heat exchange with the condensate in the drain pipe **532**.

Hereinafter, a structure of the second supply air duct **45** and the second exhaust air duct **44** will be described with reference to FIG. **11**.

The second supply air duct **45** and the second exhaust air duct **44** for circulating hot air to the refresher **3** may be disposed at the rear side of the third cabinet **30**.

The second supply air duct **45** and the second exhaust air duct **44** may be disposed behind the third rear panel **302**.

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The second supply air duct **45** may extend downwardly from the heating device **50** to be connected to the supply air hole **308** formed in the inner panel **34**.

The second supply air duct **45** may face the first side panel **106** of the dryer **1**. The second supply air duct **45** may be disposed side by side with the first side panel **106**.

The second supply air duct **45** may include the first hot air duct **451** and the second hot air duct **452**. The first hot air duct **451** may include a first connection part **451a** connected to the heating device **50**, a first extension part **451b** extending downwardly from the first connection part **451a**, and a second connection part **451c** connected to the second hot air duct **452**.

The first connection part **451a** may pass through the second side panel **206** of the second cabinet **10**. The first connection part **451a** may be connected to the blowing duct **420**, and the air blown by the fan **60** may be introduced through the first connection part **451a**. The first connection part **451a** may extend horizontally from the heating device **50**.

The first extension part **451b** may be bent downwardly from the first connection part **451a**. The first extension part **451b** may extend downwardly from one end of the first connection part **451a**. The first extension part **451b** may face the first inner wall **106** of the dryer **1**, and may be disposed side by side with the first inner wall **106**.

The second connection part **451c** may be connected to the third rear panel **302** of the third cabinet **30**. The second connection part **451c** may be connected to the second hot air duct **452**. That is, the third rear panel **302** may be disposed between the second hot air duct **452** and the second connection part **451c**. The second connection part **451c** may extend forwardly and rearwardly from the first extension part **451b**.

The washing machine **2** may include water supply ports **273** and **274** connected to the external water source. The water supply ports **273** and **274** may be connected to the water supply pipes **278** and **279** which are connected to the external water source. The tub **23** may be supplied with water from the external water source through the water supply pipes **278** and **279**.

The water supply ports **273** and **274** may include a first water supply port **273** connected to the cold water supply pipe **278** for supplying cold water, and a second water supply port **274** connected to the hot water supply pipe **279** for supplying hot water. The tub **23** may be supplied with cold water through the cold water supply pipe **278**, and may be supplied with hot water through the hot water supply pipe **279**. The water supply valve **270** may be connected to the cold water supply pipe **278** and the hot water supply pipe **279**, and may control an amount of cold water and hot water flowing into the tub **23**.

The second exhaust air duct **44** may include a third connection part **441** connected to the exhaust air hole **309**, a second extension part **442** extending downwardly from the third connection part **441**, and a fourth connection part **443** connected to the heating device **50**.

The third connection part **441** may be connected to the third rear panel **302** of the third cabinet **30**. The third connection part **441** may communicate with the inner space **33** of the refresher **3**, and air circulating in the refresher **3** may flow to the third connection part **441**. The third connection part **441** may extend rearwardly from the third cabinet **30**. A position where the third connection part **441** and the third cabinet **30** are connected may be between the third upper panel **303** and the hanger **36**.

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The second extension part **442** may be bent downwardly from the third connection part **441**. The second extension part **442** may extend downwardly from the third connection part **441**. The second extension part **442** may extend downwardly from one end of the third connection part **441**. The second connection part **442** may face the second inner wall **206** of the washing machine **2** and may be disposed side by side with the second inner wall **206**.

The fourth connection part **443** may extend into the separation space **55**. The fourth connection part **443** may extend from the second extension part **442** in the left-right direction. The fourth connection part **443** may be connected to the heating device **50** in the separation device **55**.

The rear case **310** may be fixed to the refresher **3** by a fastening member **316** passing through a fastening plate **317** extending in the front-rear direction. There may be a plurality of fastening plates **317** which are vertically spaced apart from each other. The fastening plates **317** may extend in the front-rear direction to be connected to respective side walls of the rear case **310** and the third cabinet **30**. The fastening member **316** may pass through the respective side walls of the rear case **310** and the third cabinet **30** and the fastening plate **317**.

The second supply air duct **45** and the second exhaust air duct **44** may be disposed in the rear case **310**. The rear case **310** may be disposed at the rear side of the third cabinet **30**. A rear surface of the rear case **310** may be aligned horizontally with rear surfaces of the dryer **1** and the washing machine **2**.

Hereinafter, a laundry treating apparatus A' according to another embodiment of the present disclosure will be described with reference to FIG. **12**.

The laundry treating apparatus A' according to another embodiment of the present disclosure may include the dryer **1**, the refresher **3**, and the heating device **50**. The laundry treating apparatus A' according to another embodiment of the present disclosure may not include a washing machine.

The heating device **50** may be disposed over the dryer **1**. The heating device **50** may be disposed on a lateral side of the refresher **3**.

The heating device **50** may include the cover **573** disposed on an upper side of the heat exchange channel **42**. The cover **573** may form an upper surface of the heating device **50**.

The cover **573** may shield an upper portion of the heat exchange channel **42**. The cover **573** may be disposed over the condenser **52** and the evaporator **53**.

A user may remove the cover **573** from the heating device **50**. When the user removes the cover **573**, the condenser **52**, the evaporator **53**, and the fan **60** may be exposed to the outside. The user may clean the condenser **52**, the evaporator **53**, and the fan **60** exposed to the outside. That is, as no structure is disposed on the upper side of the heating device **50**, the user may clean the inner structures of the heating device **50** by removing the cover **573**.

The description of the dryer **1**, the refresher **3**, and the heating device **50** in FIG. **12** may also be applied to the description of the dryer **1**, the refresher **3**, and the heating device **50** of the laundry treating apparatus A (FIGS. **1** to **11**) according to one embodiment of the present disclosure.

The description of the dryer **1**, the washing machine **2**, the refresher **3**, and the heating device **50**, described above with reference to FIGS. **1** to **12**, may also be applied to the following description of laundry treating apparatuses according to embodiments of the present disclosure which will be described below with reference to FIGS. **13** to **20**.

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Further, the description of components illustrated in FIGS. 1 to 12 may also be applied to FIGS. 13 to 20, even when the corresponding components are not illustrated in the description of the laundry treating apparatuses with reference to FIGS. 13 to 20. For example, even when the rear case 310 is not illustrated in FIGS. 13 to 20, the description of the rear case 310 described above with reference to FIGS. 1 to 12 may also be applied to the following description of the rear case 310 which will be described with reference to FIGS. 13 to 20.

Hereinafter, laundry treating apparatuses B and B' according to other embodiments of the present disclosure will be described with reference to FIGS. 13 to 20.

Referring to FIG. 13, in the laundry treating apparatus B, the dryer 1 is disposed over the washing machine 2, and the heating device 50 may be disposed at a lower portion of the washing machine 2. The heating device 50 may be disposed under the tub 23.

The heating device 50 may be disposed in the second cabinet 20. The heating device 50 may be disposed on a lower side of the inner space of the second cabinet 20.

The drain pump 26 may be disposed in the machine room S. The drain pump 26 may be disposed in the case 570 of the heating device 50 (see FIG. 17).

The condensate collected in the drain pan 531 may be extruded by the drain pump 26 to be discharged to the outside of the laundry treating apparatus B through a second pipe 262.

Referring to FIG. 14, the heating device 50 may be disposed under the tub 23.

The heating device 50 may be disposed at the lower portion of the second cabinet 20, and the machine room S, in which the heating device 50 is disposed, may be disposed at the lower portion of the second cabinet 20.

The machine room S may be disposed under the partition wall 208. The partition wall 208 may be disposed between the heating device 50 and the tub 23. The drain pump 26 may be disposed under the partition wall 208. The drain pump 26 may be disposed in the machine room S.

The third supply air duct 48 may extend upwardly from the heating device 50 to be connected to the tub 23.

The third exhaust air duct 47 may extend downwardly from the tub 23 to be connected to the heating device 50.

Referring to FIG. 15, the supply air hole 308 and the exhaust air hole 309 of the refresher 3 may be disposed at the lower portion of the refresher 3.

The second supply air duct 45 may be connected to a lower portion of the third cabinet 30.

The supply air hole 308 connected to the supply air duct 45 may be disposed at the lower portion of the cabinet 30. The supply air hole 308, which is open toward the inner space 33, may be formed in the inner panel 34. The supply air hole 308 may be formed in the second inner panel 342.

The supply air duct 45 may be connected to the supply air hole 308. The supply air duct 45 may be coupled to the third rear panel 302. The supply air duct 45 may pass through the third rear panel 302.

The supply air duct 45 may include the first hot air duct 451 connected to the heating device 50, and the second hot air duct 452 connecting the first hot air duct 451 and the supply air hole 308.

The first hot air duct 451 may extend to the lateral side. The second hot air duct 452 may be integrally formed with the first hot air duct 451. The second hot air duct 452 may extend upwardly. The second hot air duct 452 may extend upwardly from the first hot air duct 451 to be connected to the supply air hole 308.

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The first hot air duct 451 may be disposed behind the rear panel 302. The first hot air duct 451 may extend to the lateral side at the rear side of the rear panel 302. The second hot air duct 452 may pass through the rear panel 302 to be connected to the supply air hole 308.

The supply air duct 45 may extend horizontally at the rear side of the rear panel 302. The supply air duct 45 may extend horizontally at the rear side of the rear panel 302 to be connected to the heating device 50. The first hot air duct 451 may extend horizontally at the rear side of the rear panel 302.

The second exhaust air duct 44 may be disposed at the lower portion of the third cabinet 30.

The exhaust air hole 309 connected to the exhaust air duct 44 may be disposed at the lower portion of the cabinet 30. The exhaust air hole 309 may be disposed below the supply air hole 308.

The exhaust air hole 309 may be open in the inner panel 34. The exhaust air hole 309 may be open in the first inner panel 341. The exhaust air hole 309 may be formed at a front portion of the first inner panel 341. The exhaust air hole 309 may be spaced from the front side of the supply air hole 308.

The exhaust air duct 44 may be connected to the exhaust air hole 309. The exhaust air duct 44 may be coupled to the inner panel 34. The exhaust air duct 44 may be connected to the lower side of the inner panel 34.

The exhaust air duct 44 may extend horizontally from the lower side of the inner panel 34. The exhaust air duct 44 may extend horizontally from the lower side of the inner panel 34 to be connected to the heating device 50.

A filter 37 may be disposed in the exhaust air hole 309. The filter 37 may be disposed on the inner panel 34.

Referring to FIG. 16, the laundry treating apparatus B includes a duct system DS connecting the dryer 1, the washing machine 2, the refresher 3, and the heating device 50.

The blowing duct 420 may include a first discharge port 425 connected to the first supply air duct 43. The blowing duct 420 may include a second discharge port 427 connected to the second supply air duct 45. The blowing duct 420 may include a third discharge port 426 connected to the third supply air duct 48. The blowing duct 420 may include a fourth discharge port 428 connected to the dehumidification duct 46.

The first discharge port 425 may protrude downwardly from the blowing duct 420. The first supply air duct 43 may connect the first drum 13 and the first discharge port 425.

The second discharge port 427 may protrude from the blowing duct 420 to a lateral side. The second supply air duct 45 may connect the supply air hole 308 and the second discharge port 427.

The third discharge port 426 may protrude upwardly from the blowing duct 420. The third supply air duct 48 may connect the tub 23 and the third discharge port 426.

The fourth discharge port 428 may protrude from the blowing duct 420 to a lateral side. The fourth discharge port 428 may protrude in a direction opposite to the second discharge port 427. The dehumidification duct 46 may be connected to the fourth discharge port 428.

The hot air flowing into the first drum 13 through the first supply air duct 43 may dry the laundry placed in the first drum 13, and then may flow into the heat exchange channel 42 through the first exhaust air duct 41. The first drum 13 may include the front cover 131 disposed on a front side, and the rear cover 132 disposed on a rear side. The first supply

air duct 43 may be connected to the rear cover 132, and the first exhaust air duct 41 may be connected to the front cover 131.

The first exhaust air duct 41 may connect the first drum 13 and the heat exchange channel 42. The first exhaust air duct 41 may extend downwardly from the first drum 13 to be connected to the heat exchange channel 42.

The motor 133 rotating the first drum 13 may be disposed on the upper side of the second cabinet 20. The motor 133 may be disposed on one side of the first exhaust air duct 41. The motor 133 may be disposed side by side with the first exhaust air duct 41 in a horizontal direction.

After extending downwardly from the first drum 13, the first exhaust air duct 41 may be bent horizontally. The first exhaust air duct 41 may be disposed under the first drum 13. A portion of the first exhaust air duct 41 may extend horizontally in the first cabinet 10.

A space, in which the portion of the first exhaust air duct 41 and the motor 133 are received, may be formed under the first drum 13. The first exhaust air duct 41 extends horizontally under the first drum 13, and then may extend to the outside of the first cabinet 10.

The first inlet port 421 connected to the first exhaust air duct 41 may be formed at the inlet end 42A of the heat exchange channel 42. The first inlet port 421 may extend downwardly from the heat exchange channel 42. The first inlet port 421 may extend downwardly from the case 570 of the heating device 50.

The first exhaust air duct 41 may include the first duct section 411 disposed between the heating device 50 and the second cabinet 20. The first duct section 411 may be disposed between the base plate 575 and the second lower panel 204 of the second cabinet 20. The first duct section 411 may extend forwardly and rearwardly in the separation space 55.

The laundry treating apparatus B may include the first opening and closing valve 41a disposed in the first exhaust air duct 41. The first opening and closing valve 41a may control an amount of air flowing in the first exhaust air duct 41. The first opening and closing valve 41a may block an air flow in the first exhaust air duct 41. The first opening and closing valve 41a may block the air flow in the first exhaust air duct 41 when the hot air is not supplied into the first drum 13 through the first supply air duct 43.

The hot air flowing into third cabinet 30 through the second supply air duct 45 may dry the laundry placed in the third cabinet 30, and then may flow into the heat exchange channel 42 through the second exhaust air duct 44. The second supply air duct 45 may pass through the third rear panel 302 of the third cabinet 30. The second supply air duct 45 may connect the heat exchange channel 42 and the supply air hole 308. The second exhaust air duct 44 may pass through the third inner wall 306.

The second exhaust air duct 44 may connect the refresher 3 and the heat exchange channel 42. The second exhaust air duct 44 may connect the exhaust air hole 309 and the heat exchange channel 42. The second exhaust air duct 44 may extend from the lower portion of the third cabinet 30 to the lateral side to be connected to the heat exchange channel 42.

The second inlet port 422 connected to the second exhaust air duct 44 may be formed at the inlet end 42A of the heat exchange channel 42. The second inlet port 422 may extend downwardly from the heat exchange channel 42. The second inlet port 422 may extend downwardly from the first exhaust air duct 41. The second inlet port 422 may extend downwardly from the first duct section 411.

The second inlet port 422 may protrude downwardly from a lower surface of the first exhaust air duct 41. The second exhaust air duct 44 may connect the exhaust air hole 309 and the second inlet port 422. The second exhaust air duct 44 may connect the exhaust air hole 309 and the first exhaust air duct 41. The air in the second exhaust air duct 44 may join the air in the first exhaust air duct 41 to flow into the heat exchange channel 42.

The laundry treating apparatus B may include the second opening and closing valve 44a disposed in the second exhaust air duct 44. The second opening and closing valve 44a may control a flow rate of air in the second exhaust air duct 44. The second opening and closing valve 44a may block an air flow in the second exhaust air duct 44. The second opening and closing valve 44a may block the air flow in the second exhaust air duct 44 when the hot air is not supplied into the third cabinet 30 through the second supply air duct 45.

The hot air flowing into the tub 23 through the third supply air duct 48 may dry the laundry placed in the tub 23, and then may flow into the heat exchange channel 42 through the third exhaust air duct 47.

The third exhaust air duct 47 may connect the tub 23 and the heat exchange channel 42. The third exhaust air duct 47 may extend downwardly from the upper portion of the tub 23 to be connected to the heat exchange channel 42.

The third inlet port 424 connected to the third exhaust air duct 47 may be formed at the inlet end 42A of the heat exchange channel 42. The third inlet port 424 may extend upwardly from the heat exchange channel 42. The third inlet port 424 may extend upwardly from the case 570 of the heating device 50.

The laundry treating apparatus B may include the third opening and closing valve 47a disposed in the third exhaust air duct 47. The third opening and closing valve 47a may control a flow rate of air in the third exhaust air duct 47. The third opening and closing valve 47a may block an air flow in the third exhaust air duct 47. The third opening and closing valve 47a may block the air flow in the third exhaust air duct 47 when the hot air is not supplied into the tub 23 through the third supply air duct 48.

The outside air inlet port 423 may protrude forwardly from one side of the first exhaust air duct 41. The outside air inlet port 423 may allow the indoor space and the first exhaust air duct 41 to communicate with each other. The air in the outside air inlet port 423 may join the air in the first exhaust air duct 41 to flow into the heat exchange channel 42. The outside air inlet port 423 may have the outside air inlet 49 that is opened forward. The air drawn in through the outside air inlet 49 may join the air in the first exhaust air duct 41 to flow into the heat exchange channel 42.

Referring to FIG. 17, the drain pump 26 may be disposed in the heating device 50.

The drain pump 26 may be disposed in the heating device 50. The drain pump 26 may be disposed in the machine room S. The drain pump 26 may be disposed in the second cabinet 20. In the case where the machine room S is disposed in the second cabinet 20, the drain pump 26 may be disposed in the machine room S which is a portion of the inner space of the second cabinet 20. The drain pump 26 may be disposed between the cover 573 and the base plate 575. The drain pump 26 may be disposed under the tub 23 and may be disposed at the same height as the evaporator 53 or at a lower height than the evaporator 53.

The cover 573 may include a shielding wall 573a protruding downwardly. The shielding wall 573a may separate a space, in which the drain pump 26 is disposed, from the

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heat exchange channel 42. The shielding wall 573a may block the air in the heat exchange channel 42 from flowing toward the drain pump 26. The shielding wall 573a may be connected to the second side wall 572 and the front wall 574. The drain pump 26 may be disposed on one side of the shielding wall 573a, and the heat exchange channel 42 may be disposed on the other side of the shielding wall 573a. The drain pipe 532 may pass through the shielding wall 573a to be connected to the drain pump 26.

The drain pump 26 may be connected to the drain pan 531 through the drain pipe 532. The drain pump 26 may discharge the condensate, introduced through the drain pipe 532, to the outside of the laundry treating apparatus B through the second pipe 262. The drain pump 26 may discharge the water in the tub 23, introduced through the first pipe 261, to the outside of the laundry treating apparatus D through the second pipe 262.

The drain pump 26 may cool the hot air in the second supply air duct 45 by using the condensate introduced through the drain pipe 532.

The drain pump 26 may include a coolant supply pipe 265, through which water flowing into the drain pump 26 is discharged, and a coolant return pipe 268, through which the water discharged through the coolant supply pipe 265 flows into the drain pump 26.

The water in the tub 23 and the condensate in the drain pan 531 may flow into the drain pump 26 and may be extruded by the drain pump 26 to be discharged through the coolant supply pipe 265. The coolant supply pipe 265 may include a flow control valve 266 for controlling a flow amount of water discharged through the coolant supply pipe 265. The flow control valve 266 may be electrically connected to the controller 81. By adjusting an opening degree of the flow control valve 266, the controller 81 may control the flow amount of water discharged through the coolant supply pipe 265.

The coolant supply pipe 265 may include a heat exchange portion 267 disposed to heat exchange with the second supply air duct 45. The coolant supply pipe 265 may extend from the drain pump 26 toward the second supply air duct 45, to surround the second supply air duct 45. A portion of the coolant supply pipe 265 that surrounds the second supply air duct 45 may be defined as the heat exchange portion 267. The coolant supply pipe 265 may come into contact with the second supply air duct 45 or may be disposed to surround the second supply air duct 45. In addition, the coolant supply pipe 265 may be disposed adjacent to the second supply air duct 45, or a separate heat exchange member may be disposed between the coolant supply pipe 265 and the second supply air duct 45.

The heat exchange portion 267 may be disposed to surround the second hot air duct 452. After surrounding the second hot air duct 452, the heat exchange portion 267 may extend across the first hot air duct 451 toward the drain pump 26. Water flowing in the heat exchange portion 267 may be heat exchanged with hot air flowing in the second supply air duct 45 and may cool the hot air.

The water flowing through the coolant supply pipe 265 may flow into the drain pump 26 through the coolant return pipe 268. The water, heat exchanged with the hot air in the second supply air duct 45 by the heat exchange portion 267, may flow into the drain pump 26 through the coolant return pipe 268. The coolant supply pipe 265, the heat exchange portion 267, and the coolant return pipe 268 may be integrally formed with each other. A check valve 269 for preventing backflow of water may be disposed in the coolant return pipe 268. The check valve 269 may prevent the water

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in the drain pump 26 from flowing backward to the heat exchange portion 267 through the coolant return pipe 268. Accordingly, the water flowing into the drain pump 26 may be discharged to the coolant supply pipe 265 to return to the drain pump 26 through the coolant return pipe 268 and may be prevented from flowing backward.

The water flowing into the drain pump 26 through the coolant return pipe 268 may be discharged to the outside of the laundry treating apparatus B through the second pipe 262. The water used for cooling the second supply air duct 45 may be discharged to the outside of the laundry treating apparatus B by the drain pump 26.

The above description of the cooling structure using the drain pump 26 and the coolant supply pipe 265 may also be applied to a method of cooling the dehumidification duct 46. The coolant supply pipe 265 may extend from the drain pump 26 to surround the dehumidification duct 46 and may include the heat exchange portion 267. The water flowing in the heat exchange portion 267 may cool the hot air in the dehumidification duct 46. After cooling the hot air in the dehumidification duct 46, the water may return to the drain pump 26 through the coolant return pipe 268.

A detailed description of the method of cooling the dehumidification duct 46 may also be applied to a method of cooling the second supply air duct 45.

At least a portion of the second pipe 262 may be disposed adjacent to the second supply air duct 45. The second pipe 262 may extend from the drain pump 26 to the outside of the laundry treating apparatus D, and a part of the extended portion thereof may be heat exchanged with the second supply air duct 45. At least a portion of the second pipe 262 may be disposed to surround the second supply air duct 45 and may come into contact with the second supply air duct 45. The water flowing in the second pipe 262 may cool the hot air in the second supply air duct 45. At least a portion of the second pipe 262 may be disposed adjacent to the second supply air duct 45, and a separate heat transfer member may be disposed between the second pipe 262 and the second supply air duct 45.

At least a portion of the second pipe 262 may be disposed adjacent to the dehumidification duct 46. The second pipe 262 may extend from the drain pump 26 to the outside of the laundry treating apparatus D, and a part of the extended portion thereof may be disposed adjacent to the dehumidification duct 46. At least a portion of the second pipe 262 may be disposed to surround the dehumidification duct 46 and may come into contact with the dehumidification duct 46. The water flowing in the second pipe 262 may cool the hot air in the dehumidification duct 46. At least a portion of the second pipe 262 may be disposed adjacent to the dehumidification duct 46, and a separate heat transfer member may be disposed between the second pipe 262 and the dehumidification duct 46.

A method of cooling the hot air flowing in the second supply air duct 45 and the dehumidification duct 46 may selectively employ either the above method of using the coolant supply pipe 265 (hereinafter referred to as a "first method") or the above method of using the second pipe 262 (hereinafter referred to as a "second method"). However, as the method of cooling the hot air flowing in the second supply air duct 45 and the dehumidification duct 46, the laundry treating apparatus D of the present disclosure may use both the first and second methods at the same time.

The laundry treating apparatus B may include a circulation pipe 263 connected to the drain pump 26, and spraying the condensate, introduced into the drain pump 26 through the drain pipe 532, toward the evaporator 53.

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The circulation pipe 263 may extend from the drain pump 26 toward the evaporator 53. The drain pump 26 may extrude the condensate, introduced through the drain pipe 532, into the circulation pipe 263. In addition, the drain pump 26 may also extrude the water in the tub 23, which is introduced through the first pipe 261, into the circulation pipe 263.

A nozzle 264 may be disposed at one end of the circulation pipe 263. The nozzle 264 may be disposed toward the evaporator 53. The condensate flowing through the circulation pipe 263 may be sprayed by the nozzle 264 onto the evaporator 53.

The condensate sprayed onto the evaporator 53 may clean the surface of the evaporator 53, and then may be collected in the drain pan 531. The condensate collected in the drain pan 531 may flow into the drain pump 26 again through the drain pipe 532.

The laundry treating apparatus B may include a valve 263a disposed in the circulation pipe 263. The valve 263a may be electrically connected to the controller 81. The controller 81 may control an opening degree of the valve 263a.

The laundry treating apparatus B may clean the evaporator 53 by using the circulation pipe 263 and the nozzle 264. Further, the laundry treating apparatus D may readily discharge the condensate used for cleaning to the outside thereof by using the drain pan 531 and the drain pipe 532. As described above, the cleaning method using the condensate may also be applied to a method of cleaning the condenser 52.

Referring to FIG. 18, the heating device 50 may supply hot air to each of the dryer 1, the washing machine 2, and the refresher 3.

The second supply air duct 45 for circulating hot air in the refresher 3 may be disposed at a rear side of the third cabinet 30. The second exhaust air duct 44 may be disposed under the inner panel 34.

The second supply air duct 45 may be disposed behind the third rear panel 302.

The second supply air duct 45 may extend from the heating device 50 in a lateral direction to be connected to the supply air hole 308.

The second supply air duct 45 may pass through the second inner wall 206 of the washing machine 2. The second supply air duct 45 may pass through the second inner wall 206 to extend into the rear case 310.

The first hot air duct 451 may pass through the second inner wall 206 of the second cabinet 20. The first hot air duct 451 may be connected to the blowing duct 420, and air blown by the fan 60 may flow into the first hot air duct 45. The first hot air duct 451 may extend horizontally from the heating device 50.

The second hot air duct 452 may be bent upwardly from the first hot air duct 451. The second hot air duct 452 may extend upwardly from one end of the first hot air duct 451. The second hot air duct 452 may be disposed in a storage space 35 of the refresher 3 and may be disposed under the inner panel 34.

The third rear panel 302 may be disposed between the second hot air duct 452 and the first hot air duct 451.

The supply air hole 308 for supplying hot air into the inner space 33 of the refresher 3 may be disposed at the lower portion of the refresher 3. As the heating device 50 and the supply air hole 308 are disposed on the lower side of the laundry treating apparatus B, the second supply air duct 45 connecting the heating device 50 and the supply air hole 308 may be reduced in length, thereby minimizing heat loss

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occurring when the air heated by the heating device 50 flows toward the supply air hole 308. In addition, the hot air supplied by the heating device 50 may be discharged to the inner space 33 through the supply air hole 308 disposed at the lower portion of the refresher 3, such that the hot air may be distributed uniformly over the upper side of the refresher 3.

The second exhaust air duct 44 may be disposed under the inner panel 34. The second exhaust air duct 44 may be connected to the exhaust air hole 309 formed in the inner panel 34.

The second exhaust air duct 44 may extend in the left-right direction, to be connected to the heating device 50. The second exhaust air duct 44 may be connected to the first exhaust air duct 41. The second exhaust air duct 44 may be connected to the first exhaust air duct 41 by passing through the support plate 577.

The first supply air duct 43 may extend upwardly from the heating device 50. The first supply air duct 43 may be coupled to the blowing duct 420. The first supply air duct 43 may face the second rear panel 202. The first supply air duct 43 may extend upwardly from the blowing duct 420 and may be connected to the first drum 13 by passing through the first rear panel 102. The first drum 13 may be supplied with the hot air, heated by the heating device 50, through the first supply air duct 43.

The first supply air duct 43 may include a first supply air portion 431 connected to the heating device 50, a second supply air portion 432 extending upwardly from the first supply air portion 431, and a third supply air portion 433 connected to the first drum 13.

The first supply air portion 431 may be coupled to the blowing duct 420. The first supply air portion 431 may be disposed in the separation space 55.

The second supply air portion 432 may extend upwardly from an end of the first supply air portion 431. The second supply air portion 432 may face the second rear panel 202 and the first rear panel 102.

The third supply air portion 433 may extend forwardly from an upper end of the second supply air portion 432. The third supply air portion 433 may be coupled to the first drum 13 by passing through the first rear panel 102. The third supply air portion 433 may be coupled to the rear cover 132.

The first exhaust air duct 41 may extend from the first drum 13 to pass through the first inner wall 106.

The first exhaust air duct 41 may include a first exhaust air portion 41a passing through the first inner wall 106 to be connected to the first drum 13, a second exhaust air portion 41b extending downwardly from the first exhaust air portion 41a, and a third exhaust air portion 41c bent from the second exhaust air portion 41b to extend to a lateral side.

The first exhaust air portion 41a may pass through the first inner wall 106 to extend horizontally. The first exhaust air portion 41a may extend forwardly and rearwardly in the first cabinet 10, to be coupled to the front cover 131.

The second exhaust air portion 41b may extend downwardly from the first exhaust air portion 41a or may face the second inner wall 206.

The third exhaust air portion 41c may be bent from the second exhaust air portion 41b to the lateral side and may be connected to the first duct section 411. The third exhaust air portion 41c may pass through the support plate 577 to be connected to the first duct section 411. However, the third exhaust air portion 41c may be a concept that encompasses the first duct section 411.

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The second exhaust air duct **44** may pass through the support plate **577**. The second supply air duct **45** may pass through the second inner wall **206**.

The fan **60** may be disposed closer to the second inner wall **206** than to the second outer wall **205**. The fan **60** may be disposed adjacent to the refresher **3**. The fan **60** may be disposed closer to the second rear panel **202** than to the second front panel **201**. The fan **60** may be disposed behind the third rear panel **302** of the refresher **3**. The fan **60** may be disposed between the tub **23** and the second lower panel **204**. The fan **60** may be disposed between the base plate **575** and the tub **23**.

The switching device **70** for controlling the supply of air to the first supply air duct **43**, the second supply air duct **45**, and the third supply air duct **48** may be disposed at a height corresponding to the lower portion of the refresher **43**. The switching device **70** may be disposed under the tub **23**. The switching device **70** may be disposed between the tub **23** and the second lower panel **204**. The switching device **70** may be disposed between the tub **23** and the base plate **575**.

In the above arrangement structure, the length of the first supply air duct **43** and the second supply air duct **45** may be minimized. That is, the fan **60** and the switching device **70** are disposed adjacent to both the first drum **13** and the refresher **3**, such that the length of the first supply air duct **43** and the second supply air duct **45** may be minimized, thereby reducing heat loss occurring when the air heated by the heating device **50** flows through the first supply air duct **43** and the second supply air duct **45**.

The laundry treating apparatus B may include the steam generators **502** and **506**. The steam generator **502** disposed in the heating device **50** may be referred to as a "first steam generator," and the steam generator **506** disposed in the dryer **1** may be referred to as a "second steam generator."

The steam generator **502** may be connected to the tub **23** by the third steam supply pipe **502c**. The steam generator **502** may be connected to the refresher **3** by the second steam supply pipe **502b**. The steam generated by the steam generator **502** may be supplied into the second drum **24** through the third steam supply pipe **502c**. The steam generated by the steam generator **502** may be supplied into the inner space **33** of the refresher **3** through the second steam supply pipe **502b**.

The steam generator **502** may be disposed adjacent to the fan **60**. The steam generated by the steam generator **502** may be supplied into the second drum **24** and/or the inner space **33** of the refresher **3** by a blowing force of the fan **60**.

In the above arrangement structure, the length of the second steam supply pipe **502b** and the third steam supply pipe **502c** may be minimized. That is, the fan **60** and the steam generator **502** are disposed adjacent to both the second drum **24** and the refresher **3**, such that the length of the second steam supply pipe **502b** and the third steam supply pipe **502c** may be minimized, thereby reducing heat loss occurring when the steam generated by the steam generator **502** flows through the second steam supply pipe **502b** and the third steam supply pipe **502c**.

The laundry treating apparatus B may include the second steam generator **506**. The second steam generator **506** may generate steam by heating water. The second steam generator **506** may be disposed in the first cabinet **10**. The second steam generator **506** may be connected to the water supply pipes **278** and **279**. The second steam generator **506** may be supplied with water through the water supply pipes **278** and **279**.

The second steam generator **506** may be connected to the first drum **13** by the first steam supply pipe **506a**. The second

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steam generator **506** may be connected to the refresher **3** by the fourth steam supply pipe **506b**. The steam generated by the second steam generator **506** may be supplied into the first drum **13** through the first steam supply pipe **506a**. The steam generated by the second steam generator **506** may be supplied into the inner space **33** of the refresher **3** through the fourth steam supply pipe **506b**.

The fourth steam supply pipe **506b** may be connected to the upper portion of the refresher **3**. The steam flowing into the inner space **33** of the refresher **3** through the fourth steam supply pipe **506b** may be distributed uniformly to the lower side thereof.

The second steam generator **506** may be disposed closer to the first inner wall **106** than to the first outer wall **105**. The second steam generator **506** may be disposed closer to the first rear panel **102** than to the first front panel **101**.

In the above arrangement structure, the length of the first steam supply pipe **506a** and the fourth steam supply pipe **506b** may be minimized. That is, the second steam generator **506** is disposed adjacent to both the first drum **13** and the refresher **3**, such that the length of the first steam supply pipe **506a** and the fourth steam supply pipe **506b** may be minimized, thereby preventing loss of steam which occurs when the steam generated by the second steam generator **506** flows through the first steam supply pipe **506a** and the fourth steam supply pipe **506b**.

Referring to FIG. **19**, the heating device **50** may be connected to the duct system DS.

The base plate **575** of the heating device **50** may be disposed above the second lower panel **204**. The separation space **55** may be formed under the base plate **575**.

The separation space **55** may be formed between the first support plate **576** and the second support plate **577**.

The laundry treating apparatus B may include the auxiliary panel **58** and the second lower panel **204**.

The auxiliary panel **58** may shield a rear side of the separation space **55**. The auxiliary panel **58** may be connected to the second rear panel **202** and the second side panels **205** and **206**.

The auxiliary panel **58** may include a duct through hole **58a**, through which the first supply air duct **43** passes.

A portion of the first supply air duct **43** may be disposed in the separation space **55**. A portion of the first supply air portion **431** may be disposed in the separation space **55**.

The first supply air portion **431** may include a connection part **431a** connected to the blowing duct **420**, and an extension part **431b** extending rearwardly from the connection part **431a**.

The connection part **431a** may extend upwardly and downwardly. The connection part **431a** may extend downwardly from the blowing duct **420**. The connection part **431a** may extend downwardly from the first discharge port **425**.

The extension part **431b** may extend rearwardly from a lower end of the connection part **431a**. The extension part **431b** may pass through the duct through hole **58a** to extend rearwardly.

The second supply air portion **432** may extend upwardly from a lower end of the extension part **431b**. The second supply air portion **432** may extend upwardly behind the auxiliary panel **58**.

The second lower panel **204** may shield a lower side of the separation space **55**. The second lower panel **204** may be coupled to the support plates **576** and **577** and the auxiliary panel **58**.

The first support plate **576** may include a first protrusion **576a** protruding downwardly. The second support plate **577** may include a second protrusion **577a** protruding down-

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wardly. The first protrusion **576a** and the second protrusion **577a** may be referred to as “protrusions.”

The second lower panel **204** may include recesses **204a**, into which the protrusions **576a** and **577a** are inserted. The recesses **204a** may be recessed downwardly from the upper surface of the second lower panel **204**.

As the protrusions **576a** and **577a** are inserted into the recesses **204a**, the second lower panel **204** may be coupled to the heating device **50**.

The dehumidification duct **46** and the first exhaust air duct **41** may be disposed in the separation space **55**. The first exhaust air duct **41** disposed in the separation space **55** may be defined as the first duct section **411**, and the dehumidification duct **46** disposed in the separation space **55** may be defined as the second duct section **461**.

The dehumidification duct **46** may extend forwardly and rearwardly. The hot air outlet **462** may be formed on a front side of the dehumidification duct **46**. The air heated by the heating device **50** may be blown by the fan **60** to be supplied to the indoor space through the dehumidification duct **46**.

The first exhaust air duct **41** may extend forwardly and rearwardly. The outside air inlet port **423** may be formed on a front side of the first exhaust air duct **41**. The air in the indoor space may flow into the first exhaust air duct **41** through the outside air inlet port **423**.

The second exhaust air duct **44** may extend horizontally and may extend into the separation space **55**. The second exhaust air duct **44** may pass through the third inner wall **306** and the second support plate **577**. The second exhaust air duct **44** may be connected to the first exhaust air duct **41** in the separation space **55**. The air in the second exhaust air duct **44** may join the air in the first exhaust air duct **41** to flow into the heat exchange channel **42**.

Referring to FIG. 20, the heating device **50** of the laundry treating apparatus B' may be connected to the dryer **1** by the supply air duct **430** and the exhaust air duct **410**.

The first supply air duct **430** may branch off from the second supply air duct **450**. Without being directly connected to the heating device **50**, the first supply air duct **430** may branch off from the second supply air duct **450**. When the first supply air duct **430** branches off from the second supply air duct **450**, the first discharge port **425** may not be formed. The first supply air duct **430** may extend upwardly from the first hot air duct **451**. The first supply air duct **430** may branch off from the first hot air duct **451**.

The first exhaust air duct **410** may join the second exhaust air duct **440**. Without being directly connected to the heating device **50**, the first exhaust air duct **410** may join the second exhaust air duct **440**.

The first supply air duct **430** may include the first supply air portion **436** connected to the second supply air duct **450**, and the second supply air portion **437** connected to the first drum **13**.

The first supply air portion **436** may extend upwardly from the second supply air duct **450**. The first supply air portion **436** may be disposed behind the third rear panel **302**. The first supply air portion **436** may face the second side panel **206**.

The second supply air portion **437** may extend from an upper end of the first supply air portion **436** to the lateral side. The second supply air portion **437** may extend into the first cabinet **10**. The second supply air portion **437** may be coupled to the rear cover **132** of the first drum **13**.

The laundry treating apparatus B' may include a control valve **453** which is disposed at a position where the first supply air duct **430** branches off from the second supply air duct **450**.

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The control valve **453** may control a flow direction of air flowing into the second supply air duct **45**. The control valve **453** may control a direction of air flowing in the first hot air duct **451**. The control valve **453** may open and close the second hot air duct **452** and the first supply air duct **430**. The control valve **453** may open only either the second hot air duct **452** or the first supply air duct **430**. The control valve **453** may open both the second hot air duct **452** and the first supply air duct **430**. The control valve **453** may control a flow amount of air supplied to the second hot air duct **452** or the first supply air duct **430**. The control valve **453** may distribute the air, blown by the fan **60** to the second supply air duct **45**, to the second hot air duct **452** and the first supply air duct **430**.

The controller **81** may control the control valve **453**. By controlling the control valve **453**, the controller **81** may supply the hot air, supplied from the fan **60** to the second supply air duct **450**, into the inner space **33** of the refresher **3**. By controlling the control valve **453**, the controller **81** may supply the hot air, supplied from the fan **60** to the second supply air duct **45**, into the first drum **13**.

The first exhaust air duct **410** may pass through the rear side of the refresher **3** to join the second exhaust air duct **440**. That is, the first exhaust air duct **410** may extend in the rear case **310**. The first exhaust air duct **410** may pass through the third rear panel **302** to join the second exhaust air duct **440**. The air flowing out of the first drum **13** may pass through the first exhaust air duct **410** to join the air flowing in the second exhaust air duct **440**.

The first exhaust air duct **410** may include a first exhaust air portion **413** connected to the first drum **13**, a second exhaust air portion **414** extending downwardly from the first exhaust air portion **413**, and a third exhaust air portion **415** connected to the second exhaust air portion **414**.

A part of the first exhaust air portion **413** may be disposed in the first cabinet **10**. The first exhaust air portion **413** may be coupled to the first drum **13**. A part of the first exhaust air portion **413** may extend forwardly and rearwardly in the first cabinet **10**. The first exhaust air portion **413** may pass through the first inner wall **106** to extend into the rear case **310**. The first exhaust air portion **413** may extend horizontally from the rear side of the third rear panel **302**.

The second exhaust air portion **414** may extend downwardly from an end of the first exhaust air portion **413**. The second exhaust air portion **414** may face the third rear panel **302**. The second exhaust air portion **414** may be disposed in the rear case **310**.

The third exhaust air portion **415** may extend forwardly from a lower end of the second exhaust air portion **414**. The third exhaust air portion **415** may pass through the third rear panel **302** to extend forwardly. The third exhaust air portion **415** may join the second exhaust air duct **440**.

However, a structure of the supply air duct and the exhaust air duct, which are connected to the dryer **1** and the refresher **3**, is not limited to the above examples. For example, it is also possible that the supply air duct is installed in the structure as illustrated in FIG. 18, and the exhaust air duct is installed in the structure as illustrated in FIG. 20. In addition, it is also possible that the supply air duct is installed in the structure as illustrated in FIG. 20, and the exhaust air duct may be installed in the structure as illustrated in FIG. 18.

While the present disclosure has been particularly shown and described with reference to preferred embodiments thereof, it will be understood by those skilled in the art that the present disclosure is not limited to those exemplary embodiments and various changes in form and details may

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be made therein without departing from the scope and spirit of the invention as defined by the appended claims and should not be individually understood from the technical spirit or prospect of the present disclosure.

It will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the spirit or scope of the invention. Thus, it is intended that the present invention cover the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents.

Certain embodiments or other embodiments of the disclosure described above are not mutually exclusive or distinct from each other. Any or all elements of the embodiments of the disclosure described above may be combined with another or combined with each other in configuration or function.

For example, a configuration "A" described in one embodiment of the disclosure and the drawings and a configuration "B" described in another embodiment of the disclosure and the drawings may be combined with each other. Namely, although the combination between the configurations is not directly described, the combination is possible except in the case where it is described that the combination is impossible.

Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

A laundry treating apparatus may include a dryer including a first cabinet and a first drum rotatably mounted in the first cabinet; a washing machine including a second cabinet vertically disposed with respect to the first cabinet, a tub disposed in the second cabinet, and a second drum rotatably mounted in the tub; a refresher including a third cabinet disposed on one side of the first cabinet and the second cabinet, and having an inner space for receiving laundry; a heating device disposed under the tub and including a heat exchange unit for heating air supplied into the first drum and the inner space of the third cabinet; a drain pan which is disposed under the heat exchange unit, and in which condensate is stored; and a drain pump extruding water in the tub and the condensate in the drain pan.

The laundry treating apparatus may be configured as above, wherein the drain pump is disposed below the tub and is disposed in the heating device.

The laundry treating apparatus may be configured as above, wherein the heating device is disposed in the second cabinet; and the drain pump and the drain pan are disposed horizontally side by side under the tub.

The laundry treating apparatus may be configured as above, wherein the washing machine further comprises a partition wall disposed under the tub in the second cabinet, wherein the heating device and the drain pump are disposed under the partition wall in the second cabinet.

The laundry treating apparatus may be configured as above, wherein the drain pump is disposed below the tub and the drain pan.

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The laundry treating apparatus may be configured as above, wherein the heating device comprises: a heat exchange channel in which the heat exchange unit is disposed; and a cover disposed over the heat exchange channel and having a shielding wall extending horizontally, wherein the shielding wall separates a space, in which the drain pump is disposed, from a space in which the heat exchange channel is formed.

The laundry treating apparatus may be configured as above, further including a first drain pipe connecting the tub and the drain pump; a drain pipe connecting the drain pan and the drain pump; and a second drain pipe extending from the drain pump to an outside of the laundry treating apparatus.

The laundry treating apparatus may be configured as above, further including a first drain pipe connecting the tub and the drain pump; a second drain pipe extending from the drain pump to the outside of the laundry treating apparatus; and a drain pipe connecting the drain pan and the second drain pipe.

The laundry treating apparatus may be configured as above, wherein the second cabinet is disposed over the first cabinet; and the heating device and the drain pump are disposed between the first drum and the second drum.

The laundry treating apparatus may be configured as above, further including a drawer which is disposed under the drain pan, and in which the condensate generated in the heat exchange unit is stored.

The laundry treating apparatus may be configured as above, further including a drain pipe extending downwardly from the drain pan and connected to the drawer.

The laundry treating apparatus may be configured as above, further including a detergent supply device supplying detergent to the washing machine, wherein the drawer is disposed over the first drum in the first cabinet, and the detergent supply device is disposed over the first drum and is spaced apart from the drawer.

The laundry treating apparatus may be configured as above, further including a first supply air duct extending downwardly from the heating device to be connected to the first drum; a second supply air duct extending to a lateral side from the heating device to be connected to the refresher; and a third supply air duct extending upwardly from the heating device to be connected to the tub.

The laundry treating apparatus may be configured as above, wherein the first cabinet is disposed over the second cabinet, and the heating device and the drain pump are disposed below the first drum and the second drum.

The laundry treating apparatus may be configured as above, further including a supply air duct extending to the lateral side from the heating device and connected to a supply air hole formed at a lower portion of the refresher.

The laundry treating apparatus may be configured as above, further including a coolant supply pipe extending from the drain pump toward the supply air duct; a heat exchange portion connected to the coolant supply pipe and exchanging heat with the supply air duct; and a coolant return pipe connecting the heat exchange portion and the drain pump.

The laundry treating apparatus may be configured as above, further including a dehumidification duct discharging the air, heated by the heating device, to the outside of the laundry treating apparatus; a coolant supply pipe extending from the drain pump toward the dehumidification duct; a heat exchange portion connected to the coolant supply pipe

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and exchanging heat with the dehumidification duct; and a coolant return pipe connecting the heat exchange portion and the drain pump.

The laundry treating apparatus may be configured as above, further including a circulation pipe extending from the drain pump toward the heat exchange unit; and a nozzle disposed at the circulation pipe and being open toward the heat exchange unit.

A laundry treating apparatus may include a dryer including a first cabinet and a first drum rotatably mounted in the first cabinet; a washing machine including a second cabinet disposed over the first cabinet, a tub disposed in the second cabinet, and a second drum rotatably mounted in the tub; a refresher including a third cabinet disposed on one side of the first cabinet and the second cabinet, and having an inner space for receiving laundry; a heating device disposed between the first drum and the tub and including a heat exchange unit for heating air supplied into the refresher; and a drawer disposed below the heat exchange unit, and having an inner space in which water is stored.

A laundry treating apparatus may include a dryer including a first cabinet and a first drum rotatably mounted in the first cabinet; a washing machine including a second cabinet disposed under the first cabinet, a tub disposed in the second cabinet, a second drum rotatably mounted in the tub, and a drain pump for discharging water in the tub; a refresher disposed on one side of the first cabinet and the second cabinet, and having an inner space for receiving laundry; a heating device disposed under the tub and including a heat exchange unit for heating air supplied into the inner space of the refresher; a drain pan disposed under the heat exchange unit; and a drain pipe connecting the drain pan and the drain pump.

A laundry treating apparatus may include a dryer including a first cabinet and a first drum rotatably mounted in the first cabinet; a washing machine including a second cabinet disposed over the first cabinet, a tub disposed in the second cabinet, and a second drum rotatably mounted in the tub; a refresher including a third cabinet disposed on a lateral side of the dryer and the washing machine, and having an inner space for receiving laundry; a heating device disposed between the first drum and the tub and including a heat exchange unit for heating air supplied into the refresher; and a drawer disposed below the heat exchange unit, and having an inner space in which water is stored.

The laundry treating apparatus may be configured as above, wherein the heating device comprises a drain pan disposed under the heat exchange unit and above the drawer.

The laundry treating apparatus may be configured as above, wherein the drain pan has a first hole that is open downwardly, and the drawer has a second hole that is open upwardly, wherein the first hole and the second hole vertically face each other.

The laundry treating apparatus may be configured as above, further including a drain pipe extending downwardly from the drain pan, and connecting the drain pan and the drawer.

The laundry treating apparatus may be configured as above, further including a drain pump for discharging water in the tub; a first drain pipe connecting the tub and the drain pump; a first drain pipe connecting the drain pan and the drawer; and a second drain pipe connecting the drain pan and the drain pump.

The laundry treating apparatus may be configured as above, wherein the heating device comprises a base plate on which the drain pump and the drain pan are disposed, wherein the drain pump is disposed below the drain pan.

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The laundry treating apparatus may be configured as above, further including a first valve for controlling an amount of water flowing in the first drain pipe; a second valve for controlling an amount of water flowing in the first drain pipe; a third valve for controlling an amount of water flowing in the second drain pipe; and a controller electrically connected to the first valve, the second valve, and the third valve, wherein while the washing machine is in operation, the controller opens the first valve and the third valve and closes the second valve.

The laundry treating apparatus may be configured as above, further including a detergent supply device having a space in which detergent is stored, wherein the detergent supply device is disposed between the heating device and the first drum and is horizontally spaced apart from the drawer.

A laundry treating apparatus may include a dryer including a first cabinet and a first drum rotatably mounted in the first cabinet; a heating device disposed over the dryer and heating air supplied into the first drum; a refresher including a third cabinet disposed on one side of the first cabinet and the heating device, and having an inner space for receiving laundry; a supply air duct extending downwardly from the heating device and connected to a supply air hole formed at a lower portion of the refresher; and an exhaust air duct extending upwardly from the heating device and connected to an exhaust air hole formed at an upper portion of the refresher.

The laundry treating apparatus may be configured as above, wherein the heating device is disposed at a height between the supply air hole and the exhaust air hole.

The laundry treating apparatus may be configured as above, wherein the heating device comprises: a heat exchange channel in which air, introduced through the exhaust air duct, is heated; a condenser disposed in the heat exchange channel; an evaporator disposed on an upstream side above the condenser in the heat exchange channel; and a cover removably disposed over the condenser and the evaporator.

The laundry treating apparatus may be configured as above, further comprising a drawer disposed under the evaporator and having a space in which water is stored.

The laundry treating apparatus may be configured as above, wherein the drawer is disposed at an upper portion of the dryer and is disposed in the first cabinet.

The laundry treating apparatus may be configured as above, further including a washing machine including a second cabinet disposed over the first cabinet, a tub disposed in the second cabinet, and a second drum rotatably mounted in the tub, wherein the heating device is disposed between the first drum and the tub.

The laundry treating apparatus may be configured as above, wherein the heating device is disposed at a lower portion of the washing machine and is disposed in the second cabinet.

The laundry treating apparatus may be configured as above, wherein the heating device comprises: a heat exchange unit for exchanging heat the air introduced through the exhaust air duct; and a drain pan disposed under the heat exchange unit, wherein the washing machine comprises a drain pump connected to the tub and the drain pan.

The laundry treating apparatus may be configured as above, wherein a machine room, in which the heating device is disposed, is formed in the second cabinet, wherein the drain pump is disposed in the machine room.

The laundry treating apparatus may be configured as above, wherein the supply air duct comprises a first supply

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air duct extending downwardly from the heating device and connected to the first drum, a second supply air duct extending to a lateral side from the heating device and connected to the supply air hole formed at the lower portion of the refresher, and a third supply air duct extending upwardly from the heating device and connected to the tub; and the exhaust air duct comprises a first exhaust air duct extending upwardly from the first drum and connected to the heating device, a second exhaust air duct extending downwardly from the exhaust air hole and connected to the heating device, and a third exhaust air duct extending downwardly from the tub and connected to the heating device.

The laundry treating apparatus may be configured as above, wherein the second supply air duct comprises a first connection part extending to the lateral side from the heating device, a first extension part extending downwardly from the first connection part, and a second connection part extending forwardly from the first extension part; and the second exhaust air duct comprises a third connection part extending rearwardly from the exhaust air hole, a second extension part extending downwardly from the third connection part, and a fourth connection part extending to a lateral side from the second extension part and connected to the heating device.

A laundry treating apparatus may include a dryer including a first cabinet and a first drum rotatably mounted in the first cabinet; a heating device disposed over the dryer and heating air supplied into the first drum; a refresher disposed on one side of the first cabinet and the heating device, and having an inner space for receiving laundry; a supply air duct extending upwardly from the heating device and connected to a supply air hole formed at an upper portion of the refresher; and an exhaust air duct extending downwardly from the heating device and connected to an exhaust air hole formed at a lower portion of the refresher.

A laundry treating apparatus may include a dryer including a first cabinet and a first drum rotatably mounted in the first cabinet; a washing machine including a second cabinet disposed under the dryer, a tub disposed in the second cabinet, a second drum rotatably mounted in the tub, and a drain pump for discharging water in the tub; a refresher disposed on one side of the washing machine and the dryer, and having an inner space for receiving laundry; a heating device disposed under the tub and including a heat exchange unit for heating air supplied into the inner space of the refresher; a drain pan disposed under the heat exchange unit; and a drain pipe connecting the drain pan and the drain pump.

The laundry treating apparatus may be configured as above, wherein a machine room, in which the heating device is disposed, is formed in the second cabinet, wherein the drain pump is disposed in the machine room.

The laundry treating apparatus may be configured as above, wherein the heating device is disposed in the second cabinet and at a lower portion of the second cabinet.

The laundry treating apparatus may be configured as above, wherein the heating device comprises a base plate on which the heat exchange unit is mounted, wherein the drain pump is disposed over the base plate.

The laundry treating apparatus may be configured as above, wherein the heating device comprises: a heat exchange channel in which the heat exchange unit is disposed; and a shielding wall for separating a space, in which the drain pump is disposed, from the heat exchange channel.

The laundry treating apparatus may be configured as above, further including a first supply air duct disposed on a downstream side of the heat exchange unit and connecting

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the heat exchange channel and the first drum; a second supply air duct disposed on the downstream side of the heat exchange unit and connecting the heat exchange channel and the refresher; a first exhaust air duct disposed on an upstream side of the heat exchange unit and connecting the first drum and the heat exchange channel; and a second exhaust air duct disposed on the upstream side of the heat exchange unit and connecting the refresher and the heat exchange channel, wherein the heat exchange portion performs heat exchange with air flowing into the heat exchange channel through the first exhaust air duct and the second exhaust air duct.

The laundry treating apparatus may be configured as above, further including a first drain pipe connecting the tub and the drain pump; a second drain pump connected to the drain pump and discharging water in the tub to an outside of the laundry treating apparatus, wherein at least a portion of the second drain pipe is disposed to surround the second supply air duct.

The laundry treating apparatus may be configured as above, further including a coolant supply pipe connected to the drain pump; a heat exchange portion connected to the coolant supply pipe and disposed to surround the second supply air duct; and a coolant return pipe connecting the heat exchange portion and the drain pump.

The laundry treating apparatus may be configured as above, further including an outside air inlet port connected to the heat exchange channel; and a dehumidification duct connected to the heat exchange channel on the downstream side of the heat exchange unit, wherein the dehumidification duct is disposed below the drain pump.

The laundry treating apparatus may be configured as above, further including a first drain pipe connecting the tub and the drain pump; and a second drain pipe connected to the drain pump and discharging the water in the tub to the outside of the laundry treating apparatus, wherein at least a portion of the second drain pipe is disposed to surround the dehumidification duct.

The laundry treating apparatus may be configured as above, further including a coolant supply pipe connected to the drain pump; a heat exchange portion connected to the coolant supply pipe and disposed to surround the dehumidification duct; and a coolant return pipe connecting the heat exchange portion and the drain pump.

The laundry treating apparatus may be configured as above, wherein the heating device comprises a compressor connected to the heat exchange unit by a refrigerant passage, wherein the heat exchange unit comprises: a condenser for heating air in the heat exchange channel; and an evaporator disposed on an upstream side of the condenser and dehumidifying the air in the heat exchange channel, wherein the drain pan is disposed on a downstream side of the evaporator.

The laundry treating apparatus may be configured as above, further including a circulation pipe connected to the drain pump, and spraying condensate, introduced into the drain pump through the drain pipe, toward the evaporator.

The laundry treating apparatus may be configured as above, further including a partition wall disposed in the second cabinet and disposed under the tub, wherein the drain pump and the heating device are disposed under the partition wall.

The laundry treating apparatus may be configured as above, further including a drain pipe connecting the tub and the drain pump, wherein the drain pipe is connected to the drain pump by passing through the partition wall.

The laundry treating apparatus may be configured as above, wherein the heating device comprises a cover dis-

posed above the drain pump and disposed under the partition wall, wherein a buffer material is disposed between the partition wall and the cover.

A laundry treating apparatus may include a dryer having a first cabinet and a first drum rotatably disposed in the first cabinet; a washing machine having a second cabinet disposed on a lateral side of the first cabinet, a tub disposed in the second cabinet, and a second drum rotatably mounted in the tub; a refresher having a third cabinet disposed on a lateral side of the second cabinet, and having an inner space in which laundry is received; and a heating device for heating air supplied into the first drum and the inner space of the refresher.

The laundry treating apparatus may be configured as above, wherein the second cabinet is disposed between the first cabinet and the third cabinet, and the heating device is disposed in the second cabinet.

The laundry treating apparatus may be configured as above, further including a storage box having a space formed therein, wherein a height of the second cabinet is greater than a height of the first cabinet, and the storage box is disposed under the first cabinet.

The laundry treating apparatus may be configured as above, further including a second refresher disposed on a lateral side of the third cabinet and having an inner space for receiving laundry, wherein a height of the third cabinet is greater than the first cabinet and the second cabinet, and the refresher is disposed over the first cabinet and the second cabinet.

Various embodiments described herein may be implemented in a computer-readable medium using, for example, software, hardware, or some combination thereof. For example, the embodiments described herein may be implemented within one or more of Application Specific Integrated Circuits (ASICs), Digital Signal Processors (DSPs), Digital Signal Processing Devices (DSPDs), Programmable Logic Devices (PLDs), Field Programmable Gate Arrays (FPGAs), processors, controllers, micro-controllers, micro-processors, other electronic units designed to perform the functions described herein, or a selective combination thereof. In some cases, such embodiments are implemented by the controller. That is, the controller is a hardware-embedded processor executing the appropriate algorithms (e.g., flowcharts) for performing the described functions and thus has sufficient structure. Also, the embodiments such as procedures and functions may be implemented together with separate software modules each of which performs at least one of functions and operations. The software codes can be implemented with a software application written in any suitable programming language. Also, the software codes can be stored in the memory and executed by the controller, thus making the controller a type of special purpose controller specifically configured to carry out the described functions and algorithms. Thus, the components shown in the drawings have sufficient structure to implement the appropriate algorithms for performing the described functions.

The present invention being thus described, it will be obvious that the same may be varied in many ways. Such variations are not to be regarded as a departure from the spirit and scope of the present invention, and all such modifications as would be obvious to one skilled in the art are intended to be included within the scope of the following claims.

What is claimed is:

1. A laundry treating apparatus comprising:
a dryer including:

- a first cabinet; and
- a first drum rotatably disposed in the first cabinet;
- a washing machine including:
 - a second cabinet arranged vertically with respect to the first cabinet;
 - a tub disposed in the second cabinet; and
 - a second drum rotatably disposed in the tub;
- a refresher including a third cabinet disposed at a lateral side of the first cabinet and the second cabinet, the third cabinet having an inner space for receiving laundry;
- a heating device disposed under the tub, the heating device being configured to heat air supplied into the first drum and the inner space of the third cabinet;
- a drain pump configured to discharge water in the tub, wherein the heating device includes a heat exchange unit, wherein the drain pump is disposed below the tub and is disposed in the heating device.

2. The laundry treating apparatus of claim 1, further comprising a drain pan configured to store condensate, the drain pan being disposed under the heat exchange unit.

3. The laundry treating apparatus of claim 2, wherein the heating device is disposed in the second cabinet.

4. The laundry treating apparatus of claim 3, further comprising:

- a drawer disposed in the first cabinet, the drawer being located below the drain pan, the drawer being for storing water; and
- a drain pipe extending from the drain pan to the drawer.

5. The laundry treating apparatus of claim 4, further comprising a detergent supply device for supplying detergent to the washing machine,

- wherein the detergent supply device is disposed between the first drum and the heating device.

6. The laundry treating apparatus of claim 2, further comprising:

- a first pipe connecting the tub and the drain pump;
- a drain pipe connecting the drain pan and the drain pump; and
- a second pipe extending from the drain pump to an outside of the laundry treating apparatus.

7. The laundry treating apparatus of claim 1, wherein the washing machine further includes a partition wall disposed under the tub in the second cabinet, and

- wherein the heating device and the drain pump are disposed under the partition wall in the second cabinet.

8. The laundry treating apparatus of claim 1, wherein the heating device further includes:

- a heat exchange channel in which the heat exchange unit is disposed; and
- a cover disposed over the heat exchange channel, the cover having a shielding wall extending downwardly, and

wherein the shielding wall separates a first space, in which the drain pump is disposed, from a second space in which the heat exchange channel is disposed.

9. The laundry treating apparatus of claim 1, further comprising:

- a first supply air duct extending downwardly from the heating device and connected to the first drum;
- a second supply air duct extending laterally from the heating device and connected to the refresher; and
- a third supply air duct extending upwardly from the heating device and connected to the tub.

10. The laundry treating apparatus of claim 1, wherein the first cabinet is disposed over the second cabinet, and wherein the heating device and the drain pump are positioned below the first drum and the second drum.

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11. The laundry treating apparatus of claim 10, further comprising a supply air duct extending from the heating device and connected to a supply air hole formed at a lower portion of the refresher.

12. The laundry treating apparatus of claim 11, further comprising:

- a coolant supply pipe extending from the drain pump toward the supply air duct;
- a heat exchange portion connected to the coolant supply pipe, the heat exchange portion for exchanging heat with the supply air duct; and
- a coolant return pipe connecting the heat exchange portion and the drain pump.

13. The laundry treating apparatus of claim 10, further comprising:

- a dehumidification duct for discharging the air, heated by the heating device, to an outside of the laundry treating apparatus;
- a coolant supply pipe extending from the drain pump to the dehumidification duct;
- a heat exchange portion connected to the coolant supply pipe, the heat exchange portion for exchanging heat with the dehumidification duct; and
- a coolant return pipe connecting the heat exchange portion and the drain pump.

14. The laundry treating apparatus of claim 10, further comprising:

- a circulation pipe extending from the drain pump toward the heat exchange unit; and
- a nozzle disposed at the circulation pipe, the nozzle being open toward the heat exchange unit.

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